Environmental and Social Impact Assessment

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Uzbekistan: Guzar Solar and Battery Energy Storage Project

Part 4

Prepared by Juru Ltd for Masdar Clean Energy for the Asian Development Bank (ADB).

Nur-Kashkadarya Solar Project

Environmental & Social Impact Assessment: Volume III - Technical Appendices

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O‘ZBEKISTON RESPUBLIKASI EKOLOGIYA, ATROF-MUHITNI MUHOFAZA QILISH VA IQLIM O‘ZGARISHI VAZIRLIGI DAVLAT EKOLOGIK EKSPERTIZASI MARKAZI

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HUJJAT TURI Atrof-muhitga ta'sir to'g'risidagi ariza loyihasi

Davlat ekologik ekspertizasi buyurtmachisi: JURU ENERGY CONSULTING MCHJga berildi.
STIR: 303454532
Obyekt nomi: ОВОС строительства солнечной фотоэлектростанции «NUR-KASHKADARYA SOLAR PV BESS» мощностью 300 МВт, с аккумуляторной системой накопления энергии мощностью 75МВт на одной производственной площадке и сопутствующими проектными объектами в Гузарском районе Кашкадарьинской области (проект ЗВОС)
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Amal qilish muddati: 30.08.2027

Ekologik ekspertiza obyektining ekologik talablarga muvofiqligi, joylashuv nuqtalari koordinatalari, atrof-muhitni muhofaza qilish chora-tadbirlari, bajarilishi shart bo‘lgan talablar va boshqalar to‘g‘risida ilovada keltirilgan O‘zbekiston Respublikasi ekologiya, atrof-muhitni muhofaza qilish va iqlim o‘zgarishi vazirligining Davlat ekologik ekspertiza markazi va filiallarining ekspert xulosasi ushbu davlat ekologik ekspertizasi xulosasining ***ajralmas qismi hamda unda belgilangan talablar*** *bajarilishi shart hisoblanadi.*

Izoh: ***Buyurtmachi tomonidan davlat ekologik ekspertizasi xulosasida nazarda tutilgan ekologik talablarga rioya etilmaganda, davlat ekologik ekspertizasi xulosasi qonunchilikda belgilangan tartibda bekor qilinadi.***

Bosh direktor **G‘.Muxamedov**

Davlat ekologik ekspertizasining
2024-yil 30 avgustdagi 01-02/01-1504-son
xulosasiga ilova

Davlat ekologik ekspertizasi natijalari bo'yicha
Ekspert xulosasi talablari

ИП ООО «NUR-KASHKADARYA SOLAR PV» необходимо обеспечить:

В процессе строительства образуются различные строительные отходы, которые согласно требований ПКМ РУз. №40 от 28.01.2021 г. «О мерах по дальнейшему совершенствованию порядка проведения работ со строительными отходами», должны повторно использоваться, утилизироваться, перерабатываться или передаваться хозяйствующим субъектам, которые занимаются переработкой и их размещением, а также вывозятся на специализированный полигон строительных отходов, не подлежащие переработке или утилизации. Необходимо максимально утилизировать или перерабатывать строительные отходы;

не допускается в процессе строительства загрязнения почв, грунтов и подземных вод, вырубку деревьев и уничтожение травяного покрова, проведение в установленном порядке благоустройства и озеленение не менее 25% территории во исполнение общенациональных программ «Яшил макон» и «Яшил белбог» на не занятых сооружениями территориях, а также посадку широколистных пород деревьев;

аварийные риски на следует минимизировать применением современной автоматизированной системы управления и контроля за процессом приема и хранения электрической энергии, с функцией предупредительной и аварийной сигнализацией.

В целях предотвращения гибели или увечья перелетных птиц в период строительства ВЛ и эксплуатации ЛЭП предусмотреть: избегать периоды перелета птиц при организации строительно-монтажных работ; обеспечить установку различных птицезащитных устройств в виде ершей и другие; обеспечить применения различных изоляционных устройств, например яркие шары. Также, в рамках проекта предпринимать меры смягчения косвенных воздействий - избегание периодов наибольшей чувствительности для пернатых (гнездовой период) в период строительства и эксплуатации ВЛ. Предусмотреть также создание системы отпугивания птиц (размещение экспонатов, окраска опор ЛЭП специальной краской и другие).

При эксплуатации организацию работ с отходами согласно «Правил размещения и эксплуатации объектов инфраструктуры санитарной очистки и обращения с бытовыми отходами», утвержденный ПКМ РУз № 787 от 2.10.2018 г. «О мерах по дальнейшему повышению эффективности работ в области обращения с бытовыми отходами».

Не допускать смешивания разных видов отходов при их складировании и перемещении, не допускать неорганизованного накопления отходов. Предусмотреть емкости для временного хранения отходов, в том числе образующихся при строительных работах с последующим вывозом и сдачей в специализированные организации на утилизацию и переработку.

Вести систематический мониторинг и соответствующий надзор за состоянием птицезащитных устройств.

На этапе подготовки рабочего проекта провести гидрогеологические исследования по территории местоположения рассматриваемого объекта в установленном законодательством порядке.

Представить план управления земельными ресурсами с указанием конкретных мер по рекультивации нарушенных земель после проведения строительных работ.

К следующему этапу проектирования составить акт обследования данного объекта управлением экологии Кашкадарьинской области по выполнению предусмотренных и необходимых к выполнению природоохранных мероприятий по каждому пункту, план территории с нанесением мест хранения отходов, а также представить договора на вывоз твердых и жидких коммунально-бытовых отходов.

При последующем проектировании руководству предприятия необходимо уточнить номенклатуру, количество образуемых отходов, способы их утилизации, а также номенклатуру выбросов исходя из фактического установленного оборудования.

До сдачи объектов в эксплуатацию следует разработать заявление об экологических последствиях (ЗЭП), в котором разработать *уточненные экологические нормативы* предельно - допустимых выбросов загрязняющих веществ в атмосферу и нормативы образования и размещения отходов. ЗЭП представить на государственную экологическую экспертизу в установленном законодательством порядке.

За невыполнение требований нормативно-правовых актов, касательно экологии и охраны окружающей среды, а также предоставление не достоверной информации к проекту руководитель предприятия несет ответственность в соответствии с законодательством.

Заключение государственной экологической экспертизы о допустимости реализации проекта не подменяет и не отменяет необходимость получения соответствующих разрешительных документов в установленном законодательством порядке.

Управлению экологии, охраны окружающей среды и изменения климата Кашкадарьинской области необходимо взять под контроль соблюдение требований природоохранного законодательства при реализации проектных решений строительства и эксплуатации солнечной фотоэлектростанции мощностью 300 МВт на территории Гузарского района Кашкадарьинской области и выполнение требований настоящего заключения.

Особое внимание необходимо взять под строгий контроль выполнение вышеуказанных требований настоящего заключения, а именно проконтролировать:

проведение гидрогеологических исследований по территории местоположения рассматриваемого объекта в установленном законодательством порядке;

предоставление плана управления земельными ресурсами с указанием конкретных мер по рекультивации нарушенных земель после проведения строительных работ;

Не следует допускать ввода в эксплуатацию рассматриваемого объекта без положительного заключения на Заявление об экологических последствиях.

Ma’sul ekspert: KADIROV FARRUX BATIROVICH

Obyekt haqida qisqacha ma’lumot

Руководителю ИП ООО

«NUR-KASHKADARYA SOLAR PV»

копия: Управлению экологии, охраны окружающей

среды и изменения климата

Кашкадарьинской области

На государственную экологическую экспертизу представлены материалы первого этапа оценки воздействия на окружающую среду строительства солнечной фотоэлектростанции «NUR-KASHKADARYA SOLAR PV BESS» мощностью 300 МВт, с аккумуляторной системой накопления энергии мощностью 75МВт на одной производственной площадке и сопутствующими проектными объектами в Гузарском районе Кашкадарьинской области.

Реализация проекта строительства новой солнечной электростанции на территории Узбекистана осуществляется компанией «Abu Dhabi Future Energy Company PJSC» (Masdar).

«Abu Dhabi Future Energy Company PJSC» (Masdar) выиграло тендер, проводимый Министерством энергетики Республики Узбекистан на проектирование, финансирование, строительство, ввод в эксплуатацию, техническое обслуживание и передачу солнечной фотоэлектрической электростанции «Nur-Kashkadarya Solar PV» (ФЭС).

Основанием для реализации проекта строительства ФЭС являются: Постановление Президента РУз № ПҚ-125 от 14 марта 2024 года (Приложение 1); Закон Республики Узбекистан «Об инвестициях и инвестиционной деятельности», № ЗРУ-598 от 25.12.2019 г.; Закон Республики Узбекистан «Об использовании возобновляемых источников энергии» №ЗРУ-539 от 21.05.2019; Закон «О государственно-частном партнерстве» №ЗРУ-537 от 10.05.2019; Регламент подключения к единой электроэнергетической системе субъектов предпринимательства, производящих электрическую энергию, в том числе из возобновляемых источников энергии (Приложение к Постановлению Кабинета Министров от 22 июля 2019 года № 610); Концепция обеспечения Республики Узбекистан электрической энергией на 2020-2030 годы. Основной вид деятельности планируемой солнечной ФЭС заключается в прямом преобразовании солнечного излучения в электрическую энергию и распределение полученной электроэнергии непосредственно в электросети района.

Проектная мощность намечаемой солнечной ФЭС на основе возобновляемых источников энергии составляет 300 МВт. В рамках намечаемого проекта, на одной производственной площадке, также предусматривается организация Аккумуляторной системы накопления энергии (BESS) с использованием новейшей доступной технологии, имеющей более высокий КПД в обе стороны, меньшую деградацию в течение срока службы, проверенную производительность, высокую доступность.

Накопление энергии – аккумуляция энергии для ее использования в дальнейшем с использованием устройств, хранящих эту энергию. В условиях повышенного внимания к возобновляемым источникам энергии BESS компенсирует непостоянство ВИЭ, обеспечивая стабильную подачу электроэнергии. Решения BESS позволяют обеспечить резервное питание при отсутствии солнечного света. Это интеллектуальное решение включает в себя технологические модули прогнозирования спроса, прогнозирования возобновляемой генерации и оптимизации работы BESS.

В рамках намечаемого проекта срок эксплуатации ФЭС 300 МВт на отведенной территории составляет – 25 лет. Также, проектным решением намечается строительство ВЛ 220 кВ от ФЭС «Нур-Кашкадарья» до действующей ПС «Гузар» 500/220 кВ, протяженностью 1,578 км и строительство подъездной автодороги от автомагистрали М-39 до площадки ФЭС, протяженностью 1,7 км.

Общая площадь выделенного земельного участка – 7310000 м^2, из них в контуре общей площади 50000 м^2 отводится под расположение BESS, 25000 м^2 отводится под внутристанционную подстанцию.

Географические координаты объекта:

1. 38°66'58.13"СШ, 66°35'99.59"ВД;
2. 38°67'69.70"СШ, 66°36'16.84"ВД;
3. 38°69'17.43"СШ, 66°36'74.05"ВД;
4. 38°69'88.78"СШ, 66°38'37.72"ВД;
5. 38°69'13.36"СШ, 66°38'42.07"ВД;
6. 38°69'11.66"СШ, 66°39'28.09"ВД;
7. 38°68'52.22"СШ, 66°39'30.34"ВД;
8. 38°68'51.67"СШ, 66°38'04.94"ВД;
9. 38°66'97.22"СШ, 66°38'43.22"ВД;
10. 38°66'97.49"СШ, 66°39'33.06"ВД;
11. 38°66'26.81"СШ, 66°39'36.18"ВД;
12. 38°66'29.29"СШ, 66°36'66.72"ВД;
13. 38°66'58.15"СШ, 66°36'61.21"ВД.

Географические координаты площадки BESS в составе объекта:

1. 38°68'92.15"СШ, 66°36'63.59"ВД;
2. 38°69'19.43"СШ, 66°36'74.28"ВД;
3. 38°69'28.10"СШ, 66°36'93.89"ВД;
4. 38°68'91.93"СШ, 66°36'94.05"ВД.

Географические координаты внутренней подстанции ФЭС-300 МВТ объекта:

1. 38°69'05.69"СШ,66°36'79.90"ВД;
2. 38°69'07.62"СШ,66°36'80.47"ВД;
3. 38°69'06.99"СШ, 66°36'84.54"ВД;
4. 38°69'05.04"СШ, 66°36'83.98"ВД.

Территория ФЭС 300 МВт находится в окружении пустующих земель. С северо-запада – на расстоянии 670 м расположена ПС «Гузар-500», за которой на расстоянии 155м в направлении с северо-запада на юго-запад, проходит автомагистраль М39, далее параллельно автомагистрали на расстоянии 1,1 км от нее проходит железная дорога. С запада – на расстоянии 780 м ближайшие дома поселка Янгиабад. С востока – одиноко-стоящие строения, используемые пастухами только в сезон выпаса скота, расположены в 115 м от границы ФЭС. СЗЗ для ФЭС на уровне 250 м от границы объекта.

Хокимиятами Гузарского и Камашинского района 30 июля 2024 года проведены Общественные слушания материалов первого этапа оценки воздействия на окружающую среду по объекту: «Строительство солнечной фотоэлектрической станции «NUR-KASHKADARYA SOLAR PV BESS» мощностью 300 МВт, с аккумуляторной системой накопления энергии мощностью 75МВт на одной производственной площадке и сопутствующими проектными объектами в Гузарском районе Кашкадарьинской области, с участием причастных управлений хокимията, председателей и жителей МСГ Ойнокул, Янгиобод и др, инспектора по экологии районов, службы СЭБ и ОЗ, Заказчика. По итогам

Общественных слушаний было принято решение об общественной поддержке намечаемого строительства и деятельности на рассматриваемой территории.

В районе строительства крупные водотоки не имеются. Ближайшим водотоком рассматриваемого района является река Гузардарья, русло которой проходит с южной стороны от площадки ФЭС на расстоянии 13,0 км. Также, на расстояни 14 км на юг, находится Пачкамарское воохранилище, образованное слиянием рек Катта Урадарья и Кичик Урадарья.

Также проектным решением предусматривается строительство новой линии электропередач 220 кВ от ФЭС «Нур-Кашкадарья» до ПС «Гузар». Протяженность линии составляет – 1,578 км, из них 658 метров линии идет надземным маршрутом, 915 м - под землей. ВЛ 220 кВ начинается из ФЭС «Нур-Кашкадарья» как надземная, проходит над газовой трубой (газовая труба подземная), в точке 3-1 надземная часть переходит в подземную, проходит под существующими ВЛ 500 кВ и входит в существующую подстанцию «Гузар». Полоса землеотвода обеспечивает безопасную зону между высоковольтными линиями и окружающими сооружениями и растительностью, а также обеспечивает путь для наземных инспекций и доступ к опорам ЛЭП и другим компонентам линии в случае необходимости ремонта. Несоблюдение надлежащего качества полосы землеотвода может привести к опасным ситуациям, включая замыкания на землю. Расстояния, по обе стороны от проекции на землю крайних фазных проводов в направлении, перпендикулярном к ВЛЭП воздушной линии, для ВЛ 220 кВ составляют 15 метров. Общая площадь отвода земель под строительство ЛЭП 220 кВ, протяженностью 1,578 км, составляет – 27609,72 $м^2$. Маршрут ЛЭП не пересекает водные объекты и дорожную сеть. Жилых и хозяйственных построек на пути ЛЭП не имеется. Срок строительства ЛЭП 220 кВ ориентировочно 9 месяцев. Проектируемая ЛЭП 220 кВ будет находиться в зоне обслуживания «Кашкадарьинских МЭС».

Для оптимального подъезда служебного и личного автотранспорта к территории ФЭС «Нур-Кашкадарья», проектным решением предусмотрено строительство асфальтированной дороги протяженностью 1,7 км от автомагистрали М-39. Маршрут автодороги проходит с северной стороны относительно площадки ФЭС. Ширина дорожного полотна 10 м. Автодорога проходит над подземной газовой трубой и под ВЛ 500 кВ. Пересечений с другими объектами инфраструктуры не имеется.

В проекте подробно приведена климатические данные района объекта. Земли, отведенные под строительство ФЭС, являются пустующими. Древесной растительности на выделенной территории не имеется, в связи с чем вырубка растительности не намечается. На территории проекта во время полевых исследований краснокнижные млекопитающие замечены не были. Площадка не входит в охраняемые природные территории.

В гидрогеологическом отношении район проектной зоны относится к территориям с залеганием грунтовых вод около 9-11 м. Артезианские воды, используемые для питьевых целей находятся на глубине более 70 метров, с амплитудой колебаний 1,0 м. Для более детального исследования ***Заказчику при необходимости следует с привлечением*** *Министерства горно-добывающей промышленности и геологии РУз* ***выполнить гидрогеологические исследования в установленном законодательством порядке.***

Согласно проекта общая численность персонала ФЭС и BESS - 29 человек, на период строительства количество рабочих мест составит 270 человек. Строительство

ФЭС 300 МВт не связано со сносом жилых домов, в связи с чем изменения условий проживания населения не ожидается.

Проектное решение. Реализация проекта строительства новой солнечной электростанции осуществляется компанией «Abu Dhabi Future Energy Company PJSC» (Masdar). В рамках намечаемого проекта срок эксплуатации ФЭС 300 МВт на отведенной территории составляет – 25 лет. Годовой объем производства электроэнергии около 2628000 МВт ч/год.

Площадка, намеченная для строительства ФЭС Аккумуляторной системы накопления энергии, имеет неправильную ломаную конфигурацию с 14 углами Основную территорию будут занимать солнечные панели.

В рамках территории площадки ФЭС 300 МВт предполагается размещение: фотоэлектрических модулей на металлических опорных конструкциях, инверторов, контейнеров с аккумуляторными батареями с гибридными инверторами, трансформаторной подстанции в железобетонном корпусе, административно-бытового блока (в том числе диспетчерская, автоматическая телефонная станция, инженерно-бытовые помещения, жилые помещения для оперативного персонала и т.п), склада запасных частей и материалов, дизель-генератора в контейнерном исполнении со складом дизельного топлива, проходной с караульным двориком, баки запаса воды для нужд пожаротушения, ограждение, установленное по периметру площадки с охранным освещением, освещение территории фотоэлектрической станции солнечными фонарями, биотуалета (септик).

Для обеспечения рабочих горячим питанием будет организована столовая на 15 мест. Для приготовления пищи используется электрическая плита. Режим работы ФЭС 300 МВт круглосуточный, круглогодичный - 3 смены в сутки по 8 часов. Учитывая, что график работы станции вахтовый, для проживания рабочего персонала, планируется аренда жилых помещений в ближайшем населенном пункте.

На территории бетонированного склада закрытого типа предусматривается хранение необходимых для бесперебойной работы ФЭС 300 МВт материалов – элементов для ремонта оборудования (запчастей), электроды для сварочных работ и т.д. Трансформаторное масло для доливки в маслонаполненное оборудование не хранится на складе, привозится от поставщика непосредственно в период его заливки. Резервный дизель-генератор является резервным (аварийным) источником питания для подстанции и используется исключительно в аварийных ситуациях. При безаварийном режиме работы дизель-генератор включается два раза в год по одному часу в целях профилактики. Дизель-генератор, мощностью 10 кВ, вместе с надземной емкостью для хранения дизельного топлива (объемом 100 литров) устанавливается на бетонированной площадке со сливной ямой для сбора разливов под навесом.

Основное оборудование фотоэлектрической станции - фотоэлектрические панели модели N-type Bifacial со средней мощностью одной панели 570 Вт. (конфигурация расстановки солнечных панелей на площадке ФЭС 300 МВт принята – *21444 строк, в каждой строке 27 модулей. Всего модулей – 578988 ед. Размер одного модуля – 2278 мм х 1134 ммх30 мм). Солнечная панель* - объединение фотоэлектрических преобразователей (фотоэлементов) -полупроводниковых устройств, которые отвечают за улавливание солнечного излучения и преобразование его в электрическую энергию. *Контроллеры*. Интенсивность солнечного излучения в разные дни может значительно отличаться. В некоторые дни фотоэлементы работают на минимальной мощности, тогда как в другие избыток солнечного света перегружает солнечную электростанцию. Контроллер представляет собой небольшое электронное устройство, которое контролирует количество электроэнергии, поступающей от панелей, предотвращая перегрузку установки из-за избыточного солнечного излучения. *Инверторы*. Фотоэлементы генерируют постоянный электрический ток, но подавляющее большинство электроприборов работают с другим типом тока, переменным током. Инвертор - это устройство, которое преобразует постоянный ток от солнечных панелей в тип энергии, который можно использовать и передавать на большие расстояния. *Аккумуляторные батареи*, как и в любой другой системе, отвечают за хранение электрической энергии, когда источник солнечного излучения недоступен.

Фундамент сооружения должен выдерживать: Ветровые перегрузки в любом направлении. Собственный вес конструкции и поддерживаемых модулей. Снежные перегрузки на поверхности модулей (если применимо). Учитывая гидрогеологические условия района расположения проектируемой ФЭС 300 МВт, принимается забивка свай для солнечных трекеров примерно на 4 метра ниже уровня земли. Для преобразования солнечной электроэнергии в электрическую предусматриваются 38 ед. инверторных блоков. Инверторы оснащены системой управления, позволяющей полностью автоматизировать работу.

Преимуществом проекта является использование новых технологий автономной системы сухой очитки панелей Automatic Robot Cleaning System. Предусмотренная очистка является сухой, поэтому вода для этого процесса не требуется. Чистка панелей происходит сухим способом с использованием робота-уборшика (робота-пылесоса). Фотоэлектрический робот для очистки в основном включает в себя автономную систему электропитания, независимую систему ходьбы, систему управления и системы управления связью.

Проектом предусматривается организация в пределах территории ФЭС 300 МВт, внутристанционной подстанции , служащей для преобразования напряжения и выдачи электроэнергии в сеть, посредством существующей Подстанции «Гузар-500» 500/220/110/10 кВ, расположенной в западной стороны от территории ФЭС и соединенной подземным кабелем. Внутристанционная подстанция рассчитана на напряжение 220 кВ. Проектная подстанция с воздушной изоляцией (AIS) 35 кВ/220 кВ преобразует уровни выходного напряжения генерации и BESS до 220 кВ с помощью двух силовых трансформаторов для фотоэлектрической генерации и еще одного для BESS (190 МВА - 38.5кВ/230кВ, 190 МВА - 38.5кВ/230кВ, 90 МВА - 38.5кВ/230кВ) и связанных с ним электрических устройств.

Также будут установлены аккумуляторные батареи BESS для накопления электроэнергии на территории. Проектным решением принимается установка 27 батарейных контейнеров по 3,73 МВтч каждый, LFP батареи 3,2В 280 Ач в каждом модуле. В батарейных контейнерах (контейнерные блоки) расположены стеллажи с литийионными аккумуляторными батареями. Литий-ионные аккумуляторы обладают высокими показателями плотности энергии, быстрым временем отклика, долговечностью, высокой кулоновской эффективностью, быстрым временем зарядки и низкими затратами на техническое обслуживание. Для преобразования тока из переменного в постоянный и наоборот, BESS будет оснащена 54 гибридными инверторами. Для передачи энергии из сети, трансформаторы преобразуют высоковольтную энергию из сети в низковольтную энергию для хранения в аккумуляторах. Для передачи энергии в сеть трансформаторы преобразуют низковольтную энергию из аккумуляторов в высоковольтную энергию в сети. В проекте приведены подробно характеристики оборудования и управления ими.

Организационные работы на площадке ФЭС. Перед размещением солнечных панелей и контейнеров с аккумуляторными батареями предусматривается снятие плодородно-растительного слоя земли и частичная вертикальная планировка площадки. На площадке расположения зданий и сооружений предусматривается открытая система сбора и отвода дождевого и ливневого стока – водоотводные канавы и ж/б лотки. Территория вокруг административно – бытового корпуса благоустраивается и озеленяется. Так как район относится к 3 (слабой) категории селеопасности, проектом предусмотрено инженерная защита территории от ливневых и селевых потоков. Ко всем зданиям и сооружения предусмотрены технологические и противопожарные подъезды. По периметру площадки с солнечными панелями, а так же между блоками, для обеспечения технического обслуживания, предусмотрены автодороги с щебеночным покрытием. По периметру площадки предусматривается сетчатое ограждение.

На фотоэлектрической станции должны будут использованы фотоэлектрические панели с антибликовыми покрытиями**,** которые дают солнечному элементу характерный синий оттенок. Антибликовое покрытие солнечных панелей имеет антиотражающий эффект (уменьшение зеркального отражения), который также служит для **безбарьерного полета перелетных птиц.**

Для отопления помещений в зимний период и охлаждения в летний период будут применяться электронагревательные приборы и кондиционеры сплит системы «Зима-Лето». Для приготовления пищи в столовой будут установлены электроплиты - 2 штук.

Выбросы загрязняющих веществ в атмосферу. Согласно проекту ЗВОС в процессе эксплуатации предприятия состояние атмосферного воздуха не изменится. Так как применяемая технология производства электроэнергии за счет солнечных панелей является экологически чистой, отвечая концепции «Зеленой экономике и охране окружающей среды». Вредное воздействие будет происходить только на этапе строительства. В процессе эксплуатации выбросы загрязняющих веществ в атмосферу составят 0,0084 т/год, в том числе: масло минеральное (99,98%), углеводороды (0,02%). Максимальные концентрации за пределами площадки менее 0,01ПДК. При работе дизель-генератора залповые выбросы составят 0,0004 г/год, на следовых уровнях.

Водопотребление. При проведении строительных работ на площадке и эксплуатации вода используется для полива территории с целью снижения пыления, а также на хозяйственно-бытовые нужды, которая доставляется автотранспортом и сливается в специальные емкости.

Общее водопотребление в период строительства и эксплуатации солнечной ФЭС составит: 1694,785 м^3/год, из них: водопотребление для хозяйственно-бытовых нужд - 1499,785 м^3/год; водопотребление на пожаротушение - 150 м^3/год.

Планируемое водоотведение при заполнении биотуалета септика откачка осуществляется при помощи ассенизационной машины специализированных служб с последующим вывозом на ближайшие очистные сооружения. Система сбора и водоотведения ливневых и талых вод с территории ФЭС будет представлена системой лотков и отстойником, в котором будет осаждаться ил (осадок), а вода использоваться на полив территории.

На следующем этапе ОВОС необходимо определиться с приемником хозяйственно-бытовых стоков при эксплуатации солнечной ФЭС с представлением договора на вывоз стоков.

Образование отходов. В процессе эксплуатации планируемого объекта предположительно будут образовываться отходы 16-ти наименований в количестве 2822,692 т/год. Для каждого отхода даны рекомендации по их обращению и способам утилизации, а также произведена их классификация.

В результате анализа в проекте ЗВОС проведена классификация планируемого образования 16 наименований производственных отходов по классам опасности. Из 16 видов отходов 1 вид отходов отнесен 1 классу опасности (1%), 1 вид отходов отнесен ко 2 классу опасности (26%), 2 вида отхода к 3 классу опасности (0,008%), 4 вида отходов отнесены к 4 классу опасности (0,07%) и 6 видов отходов отнесены к 5 классу опасности (72,8%). Все виды отходов при выполнении условий их сбора складирования не будут оказывать вредного воздействия на окружающую среду.

В проекте ЗВОС рассмотрены возможные аварийные ситуации, связанные, в основном, с эксплуатацией оборудования на проектируемой ФЭС, с пожарами при повреждении трансформаторов и при возникновении токов короткого замыкания кабельного хозяйства. Для предотвращения и минимизации различных аварийных ситуаций в проекте представлены соответствующие рекомендации и мероприятия.

На проектируемой станции планируется использование новейших технологий по выработке солнечной энергии при проектировании данной ФЭС и внедрения полной автоматизации процесса контроля и управления, использование предупредительной и аварийной сигнализации возможность возникновения аварийных рисков на проектируемой ФЭС будет минимальна.

Вместе с тем, проект расположения солнечных панелей обеспечивает беспрепятственный доступ к подземным коммуникациям, что позволит своевременно осуществлять ремонтные работы на водопроводных трубопроводах в случае возникновения аварийных ситуаций, связанных с разрывом трубопроводов и т.п.

По результатам проведения государственной экологической экспертизы проекта строительства солнечной фотоэлектростанции мощностью 300 МВт было установлено, что при полном соблюдении природоохранных мероприятий, указанных в ПЗВОС, существенного отрицательного воздействия на окружающую среду на этапе строительства не ожидается. Воздействия на этапе эксплуатации будет носить в основном локальный характер. Незначительное минимальное воздействие, которое может иметь место на этапе эксплуатации будет предупреждено путем выполнения природоохранных мероприятий, указанных в материалах оценки воздействия на окружающую среду.

Следует отметить, что проект в целом относится к категории «зеленых технологий» и будет способствовать достижению целей и выполнению обязательств, принятых Республикой Узбекистан по Парижскому Соглашению (2021).

Анализ представленных на государственную экологическую экспертизу, материалов оценки воздействия на окружающую среду, показал их соответствие требованиям природоохранного законодательства к первому этапу оценки воздействия на окружающую среду.

Центр государственной экологической экспертизы при Министерстве экологии, охраны окружающей среды и изменения климата **согласовывает** материалы первого этапа оценки воздействия на окружающую среду строительства солнечной фотоэлектростанции «NUR-KASHKADARYA SOLAR PV BESS» мощностью 300 МВт, с аккумуляторной системой накопления энергии мощностью 75МВт на одной производственной площадке и сопутствующими проектными объектами в Гузарском районе Кашкадарьинской области.

Ma'sul ekspert: KADIROV FARRUX BATIROVICH

Technical appendix 2: Scoping Report

Nur-Kashkadarya Solar Project

Environmental & Social Impact Assessment:

Scoping Report

Consulting Firm:

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Juru's Project Manager	Nicola Davies
Juru's Project Director	Jushkinbek Ismailov

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1.0	15/10/2023	Draft for client review	Various	Marianne Lupton	Nicola Davies
2.0	21/03/2024	Updated to reflect new project layout	Various	Marianne Lupton	Nicola Davies

Disclaimer

Abbreviations

Acronym	Term
AC	Alternating Current
AoA	Analysis of Alternatives
AoI	Area of Influence
AP	Action Plan
CBD	Convention on Biological Diversity
CBO	Community Based Organisations
CHA	Critical Habitat Assessment
CHS	Community Health and Safety
CSR	Corporate Social Responsibility
DC	Direct Current
EHS	Environment, Health and Safety
EIA	Environmental Impact Assessment
EMP	Environmental Management Plan
EPC	Engineering, Procurement, and Construction
EPRP	Emergency Preparedness and Response Plan
ESIA	Environmental and Social Impact Assessment
ESMP	Environmental and Social Management Plan
GHG	Greenhouse Gas
GIIP	Good International Industry Practice
GIS	Geographical Information System
HSSE-MS	Health Safety Social Environmental Management System
HR	Human resources
IFC	International Finance Corporation
ILO	International Labour Organisation
IP	Indigenous peoples
IUCN	International Union for Conservation of Nature
LRF	Livelihood Restoration Framework
NCR	Non-compliance report
NGO	Non-Governmental Organisation
NTS	Non-Technical Summary
O&M	Operation and Maintenance
OHL	Overhead lines
OHS	Occupational Health and Safety
OPEX	Operating Expenditures
PM	Particulate Matter
POPs	Persistent organic Pollutants
PPA	Power Purchase Agreement
PPE	Personal Protective Equipment
PRs	Performance Requirements
PS	Performance Standards
PV	Photovoltaic
RAP	Resettlement Action Plan
RoW	Rights of Way
SEP	Stakeholder Engagement Plan
SIA	Social Impact Assessment
SWMP	Site Waste Management Plan
TMP	Traffic Management Plan
ToR	Terms of Reference
UN	United Nations
UNFCCC	United Nations Framework Convention on Climate Change
USTDA	United States Trade and Development Agency
WBG	World Bank Group
WHO	World Health Organisation

Table of Contents

1 Introduction

1.1 Background

Abu Dhabi Future Energy Company PJSC ("Masdar") has been awarded by the Ministry of Energy, Government of Uzbekistan, to design, build, finance, construct, commission and operate, maintain and transfer (DBFOMT) the Nur-Kashkadarya Solar photovoltaic (PV) Project[1] with a capacity of 300 MWac and a 75 MW / 75 MWh Battery Energy Storage System (BESS) ("the Project"). The Project will connect to the existing 220kV/500kV Guzar substation by combined overhead transmission line (OHTL) and underground cable (UC) to be constructed as part of the Project scope. The site and project description are elaborated in section 3.0. The Project will be designed to meet national regulations and international standards. The Project will be implemented through a long-term, i.e., 25 years power purchase agreement (a "PPA") between Nur-Kashkadarya Solar PV FE LLC Foreign Enterprise and JSC National Electric Grid of Uzbekistan ("NEGU").

The Project will support Uzbekistan to:

- Reduce energy dependence on carbon-based fuels.
- Meet renewable energy targets to deploy 5 GW of renewable energy by 2025.
- Reduce greenhouse gas emission rates.

Masdar has appointed Juru Ltd. (Juru or the ESIA Consultant) to perform an Environmental and Social Impact Assessment (ESIA) for the Project. The ESIA will be developed in accordance with the requirements of EBRD Environmental and Social Policy 2019 (ESP 2019) Performance Requirements (PRs), the International Finance Corporation (IFC) Performance Standards (PSs), Asian Development Bank (ADB) Safeguard Policy Statement (2009) and the Equator Principles.

The Project will be required to undergo a separate national environmental impact assessment (EIA) process, which Juru will perform and submit as a separate document to the Ministry of Environment for approval. The sequence of steps for the EIA and ESIA study is presented inFigure 1.. This report presents the findings of the scoping phase activities.

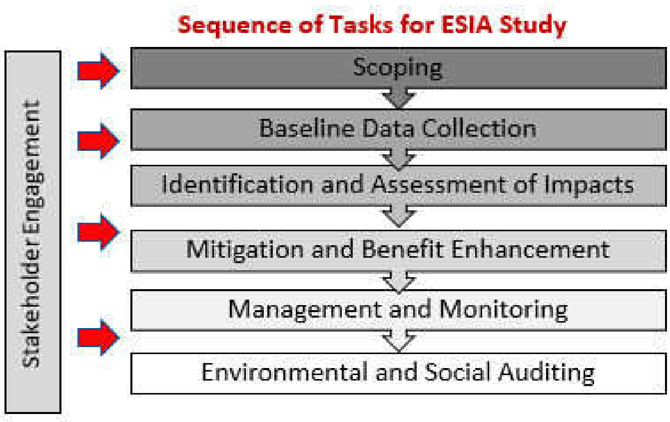


Figure 1: ESIA process - sequence of steps (source: Juru)

[1] In previous publicly disclosed documentation this Project has also been referred to as the Guzar Solar power PPP project.

1.2 Project location

The Site covers approximately 802 ha of desert landscape in the Guzar and Kamashi districts, it is approximately 55km km south-west of the city of Shahrisabz and 12 km northeast of the city of Guzar, in the Qashqadaryo region of Uzbekistan. The proposed site is located adjacent to an existing 220kV/500kV substation and a regional road (M39) runs in an east / west direction immediately north of the proposed site. The location of the Project is illustrated in Figure 2.

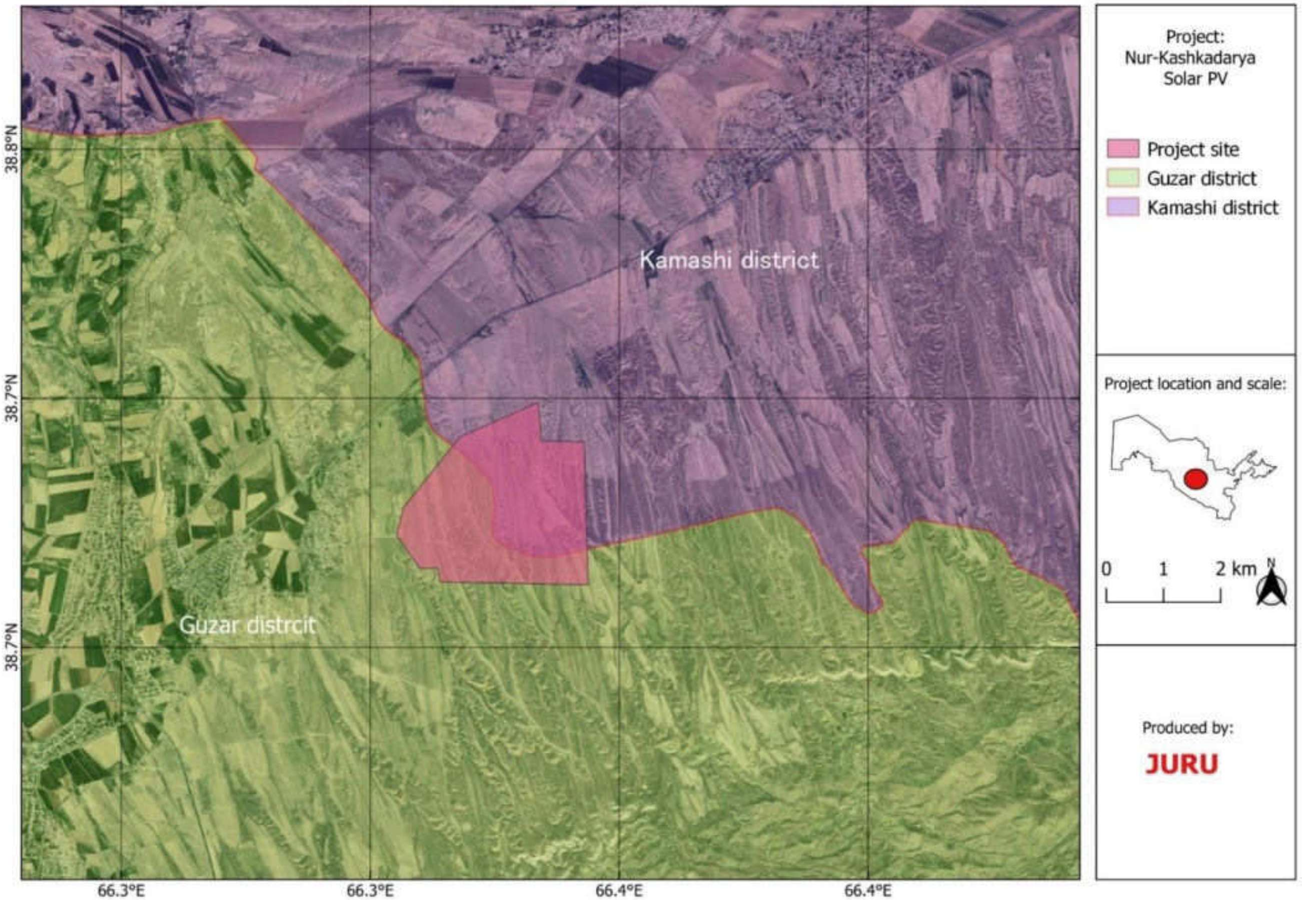


Figure 2: Project location[2]

2 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 and are still considered representative of the final layout.

1.3 Purpose of this report

The purpose of this scoping report is to identify the potential positive and negative environmental and social (E&S) issues and impacts related to the pre-construction, construction, operation and decommissioning of the Project, and to inform project stakeholders of the proposed terms of reference (TOR) for further assessment during the ESIA process, as presented in Chapter 7.

This scoping report includes a preliminary identification of the Project area of influence (AOI), a description of the E&S regulatory framework applicable to the Project and stakeholder mapping to inform and facilitate meaningful public engagement and review.

This scoping process has been informed by primary data collection and review of secondary date including:

- An initial site visit to the general Project area undertaken between 11th and 12th September 2023 and a follow up site visit on 25th January 2023;
- Consultation with representatives of the affected public, government agencies, local authorities and other organisations;
- Review of existing documentation and earlier studies for other related projects in the region; and
- Review of relevant regional and strategic environmental and social assessments or studies of relevance to the Project.

1.4 Project proponent

MASDAR

Mission "To be a global clean energy developer, with a reputation of being the partner of choice. We meet the needs of clients and communities alike"

Masdar's mission is to develop, invest in and deliver high-quality, sustainable and economically viable clean energy projects locally and globally. Masdar will develop the Nur-Kashkadarya Solar photovoltaic (PV) Project and Battery Storage. Masdar has over a decade of experience as a renewable energy developer and investor. Masdar invests in and contributes to innovative global projects, including large, utility-scale renewable energy power plants, community grid projects, and individual solar home systems. It is active across 40 countries and has developed some of the world's most significant solar and wind energy projects. Masdar has invested or committed to invest in renewable energy projects with a gross capacity of over 20 GW. In Uzbekistan, Masdar has 100 MW utility-scale PV solar plant in operation and another 1,600 MW of projects under various stages of development i.e., financial closing and construction.

1.5 ESIA team and project contact information

Juru Ltd. will perform the ESIA study. The team of specialists involved in the Project is presented in Table 1.

JURU LLC, UZBEKISTAN
Name: Viktoriya Filatova, Zilola Kazakova
Address: 10A, Chust Str., Tashkent, Uzbekistan, 100077
Email: Guzar_SolarPV_ESIA@juru.org
Phone: +998 90 515 03 92

Table 1: ESIA Team

Name	Position
Nicola Davies	Project Manager & Environmental Expert
Caleb Gordon	International Biodiversity Expert
Marianne Lupton	International Social Specialist
Viktoriya Filatova	Local Environmental Specialist
Oleg Khegay	Local Environmental Specialist
Anna Ten	Local Biodiversity Expert/Ornithologist
Zilola Kazakova	Local Social/Resettlement Specialist
Gulchekhra Nematullayeva	Social stakeholder mobilisation specialist
Timur Abduraupov	Local Herpetologist
Natalya Beshko	Local Botanist
Mariya Gritsina	Mammal expert
Eleonora Ishmuhammedova	National EIA Expert

1.6 Structure of the scoping report

The scoping report is structured as follows:

- Chapter 2: Applicable national legislation and international standards.
- Chapter 3: Description of the Project, its scope, alternatives considered and justification.
- Chapter 4: Overview of the environmental and social baseline conditions.
- Chapter 5: Assessment of the potential environmental and social impacts to be studied as well as a justification for those impacts scoped out during the scoping process.
- Chapter 6: Stakeholder mapping (as obtained during the scoping phase of stakeholder engagement).
- Chapter 7: ESIA terms of reference (TOR) and work programme.
- References
- Annex A: Scoping notification leaflet (English and Uzbek)
- Annex B: Project grievance form
- Annex C: Biodiversity survey methodology and work plan
- Annex D: Preliminary permit register
- Annex E: Information about protected areas

1.7 ESIA structure and schedule

The ESIA will contain the following volumes:

- Volume I: Non-Technical Summary (English, Uzbek)
- Volume II: ESIA Main Report (English, Russian)
- Volume III: ESIA Technical Appendices and Supporting Documents (originating language)
- Volume IV: Environmental and Social Management Plan (ESMP) (English, Russian)
- Volume V: Resettlement Plan (RP) (English, Uzbek)
- Volume VI: Stakeholder Engagement Plan (SEP) (English, Uzbek)

In addition, the following supporting documentation will be prepared:

- Critical Habitat Screening / Assessment (CHA)
- Climate Change Risk Assessment (CCRA)
- Human Rights Impact Assessment (HRIA)

Table 2 sets out the ESIA schedule.

Table 2: Project ESIA Schedule

Activity / key milestone	**Week commencing**
Scoping site visit	September 2023 / February 2024
Final scoping report and SEP and RP	March 2024
Baseline studies and desktop collection	September / October 2023 & March/April 2024
Public hearing (national EIA) and public consultation (ESIA)	August 2024
Submit National EIA to regulator	August 2024
Draft ESIA (all volumes and supporting information)	August 2024
Final ESIA report (all volumes and supporting information)	TBC
Financial Close	TBC
Notice to Proceed	TBC

2 Policy, Legislative and Institutional framework

This section sets out the Uzbek and Lender framework applicable to the Project.

2.1 Uzbekistan regulatory context

2.1.1 Relevant government ministries

Key organisations with responsibility for environmental management in Uzbekistan are:

- Cabinet of Ministers of the Republic of Uzbekistan (COM);
- The Ministry of Ecology, Environmental Protection and Climate Change (MEEPCC)[3]; and
- The Center for State Ecological Expertise, which is under the MNR.

The Cabinet of Ministers of the Republic of Uzbekistan governs the executive body in the Republic of Uzbekistan following the Constitution of the Republic of Uzbekistan (Article 98), and the Law of the Republic of Uzbekistan "On the Cabinet of Ministers of the Republic of Uzbekistan" (new edition of 2019). The COM exercises the following main functions:

- Implements measures on rational use and protection of natural resources;
- Coordinates the work of state bodies on joint conducting of natural protection events;
- Implements a large-scale ecological program of national and international importance; and
- Takes measures to eliminate the consequences of accidents and disasters as well as natural disasters.

The Ministry of Ecology, Environmental Protection and Climate Change (MEEPCC) is the main regulating body of state administration on environmental protection issues. The primary responsibilities of the MEEPCC include ensuring the implementation of a unified state policy on environmental safety, environmental protection, and the use and reproduction of natural resources; and enforcing state control over the compliance of ministries, state committees, departments, enterprises, institutions, and organisations, as well as individuals, with respect to the use and protection of land, mineral resources, water, forests, flora and fauna, and atmospheric resources. Structurally, the MEEPCC consists of the central unit (located in Tashkent), regional units (oblast) and local (district) units.

The Center for State Ecological Expertise: The Center for State Ecological Expertise's activities are directly related to the evaluation of materials for EIA and the issuance of documents determining compliance with environmental requirements for planned or executed business and other activities, as well as determining the admissibility of the implementation of the object of environmental expertise.

Due to the cross-cutting nature of sustainable development and the environment, virtually all other state bodies have some responsibility towards them. Other stakeholders that are relevant to the Project are listed below.

- Ministry of Energy of the Republic of Uzbekistan;
- Ministry for Emergency Situations of the Republic of Uzbekistan;
- Ministry of Health of the Republic of Uzbekistan;

3 The Ministry of Ecology, Environmental Protection and Climate Change (MEEPCC) was organised on a basis of State Committee of Environmental Protection of Uzbekistan by the Presidential Decree of January 25 2023 No. UP-14 "On priority organizational measures for the effective establishment of the activities of the Republican Executive Authorities"

- Ministry of Poverty Reduction and Employment;
- Ministry of Water Management of the Republic of Uzbekistan;
- Ministry of Agriculture of the Republic of Uzbekistan;
- Cadastre Agency under the Ministry of Economy and Finance of the Republic of Uzbekistan;
- Forestry Agency under the Ministry of Natural Resources of the Republic of Uzbekistan;
- Inspection of Mining, Geology and Industrial Safety Control (Kontekhnazorat) under the Ministry of Mining Industry and Geology of the Republic of Uzbekistan; and
- Ministry of Internal Affairs of the Republic of Uzbekistan.
- Ministry of Transport of the Republic of Uzbekistan
- Institute of Archaeology of Uzbekistan Academy of Sciences
- Cultural Heritage Agency under the Ministry of Tourism and Sports
- Ministry of Employment and Labour Relations of the Republic of Uzbekistan
- State Committee on Sericulture and Wool Development Industry

2.2 Green Energy Policy

2.2.1 Uzbekistan's Green Economy Strategy

At the 26th session of the United Nations Framework Convention on Climate Change (COP26), held in November 2021, under the Paris Agreement, the Republic of Uzbekistan announced the adoption of additional obligations to reduce greenhouse gas emissions per unit of gross domestic product by 2030 by 35 percent compared to 2010 levels.

Uzbekistan aims to achieve this through the implementation of the Uzbekistan's Green Economy Strategy Uzbekistan's “Strategy for transition to a green economy in the period of 2019-2030” was approved by the Resolution of the President of the Republic of Uzbekistan dated 04.10.2019 No. PP-4477 (the "Resolution").

This Resolution was adopted to ensure fulfilment of obligations under the Paris Agreement on climate change signed by Uzbekistan on April 19, 2017, as well as the implementation of the Action Strategy for five priority areas of development of the Republic of Uzbekistan in 2017-2021 as follows:

- improving energy efficiency of the economy and rational consumption of natural resources through technological modernization and development of financial mechanisms;
- inclusion in priority areas of public investment and spending of green criteria based on international best practices;
- assistance in the implementation of pilot projects in the areas of transition to Green economy through the development of mechanisms of state incentives, public-private partnership and cooperation with international financial institutions;
- development of a system of training and retraining of personnel related to labor market in green economy;
- taking measures to mitigate the negative impact of environmental disaster in the Aral Sea region;
- strengthening international cooperation in the field of "green" economy, also through the conclusion of bilateral and multilateral agreement.

Specific target indicators include:

- Reduction of emissions of greenhouse gas per unit of gross domestic product (GDP) by 10 per cent of the 2010 level;
- Doubling energy efficiency indicators and a decrease in the carbon intensity of GDP;
- Further development of renewable energy sources, with coverage of more than 25 per cent of the total volume of electricity generation;
- Providing access to modern, affordable and reliable power supply to 100% of the population and sectors of the economy

In 2022, the Decree of the President of the Republic of Uzbekistan, dated December 2, 2022 “On measures to improve the effectiveness of reforms aimed at the transition of the Republic of Uzbekistan to a "GREEN" economy until 2030” was issued to improve measures taken to ensure “green” and inclusive economic growth as part of the Strategy outlined above. This also sought to further expand the use of renewable energy sources and resource saving in all sectors of the economy. Considering the renewable energy sector specifically the goal is “increasing the production capacity of renewable energy sources up to 15 GW and bringing their share in the total volume of electricity production to more than 30 percent”.

The new law also states that from January 1, 2024, within the framework of investment projects for the construction of new solar and wind power plants with a capacity of more than 1 MW, an electric energy storage system with a capacity of at least 25 percent of the installed capacity of these stations will be introduced without fail.

2.3 Environmental Law

2.3.1 Constitution of Uzbekistan

The constitution of Uzbekistan has the following provisions relating to environmental aspects:

- Article 49: It is the duty of citizens to protect the historical, spiritual and cultural heritage of the people of Uzbekistan. Cultural monuments shall be protected by the state;
- Article 50: All citizens shall protect the environment;
- Article 53: All forms of ownership of citizens is under protection of state;
- Article 54: No property shall inflict harm to the environment; and
- Article 55: Land, subsoils, flora, fauna, and other natural resources are protected by the state and considered as resources of national wealth subject to sustainable use.

2.3.2 Law on Nature Protection, 1992 as Amended in 2021

This law is the key national environmental law for the protection of the environment and the sustainable use of resources and the right for the population to a clean healthy environment.

This law states legal, economic, and organizational basis for the conservation of the environment and the rational use of natural resources. Article 24 of this law states that the State Environmental Expertise (SEE) is a mandatory measure for environmental protection, preceded to decision making process. In addition, the law prohibits the implementation of any Project without approval from SEE.

It should be noted that Article 53 of this law confirms that if an international treaty concluded by the Republic of Uzbekistan establishes rules other than those provided for by the legislation of the Republic of Uzbekistan on nature protection, the rules of the international treaty shall be applied, except in cases where the legislation of the Republic of Uzbekistan establishes stricter requirements.

2.3.3 Law on Environmental Control, 2013 as Amended in 2022

The main objectives of this law include:

- Prevention, detection and suppression of violation of legislative requirements relating to environmental protection and rational use of natural resources.
- Monitoring the state of the environment, identifying situations that can lead to environmental pollution, irrational use of natural resources, pose a threat to the life and health of citizens.
- Determination of compliance with environmental requirements of any ongoing economic development activities.
- Ensuring compliance with the rights and legitimate interests of legal entities and individuals performing their duties in relation to environmental protection and sustainable use of natural resources.

The Article 7 of this law states that, the objects of environmental control are:

- Land, its subsoil, waters, flora and fauna, and atmospheric air;
- Natural and man-made sources of impact on the environment; and
- Activities, action or inaction that may lead to pollution of the environment and irrational use of natural resources, create a threat to the life and health of citizens.

2.4 National Environmental Impact Assessment (EIA)

According to the list of activities subject to state ecological expertise, which is established by the Resolution of Cabinet of Ministers No. 541 “On further improvement of the environmental impact assessment mechanism’ (2020), power-generating facilities are categorized as follows depending on the level of impact on the environment[4]:

- Thermal, photovoltaic, wind power and other power-generating facilities with a capacity of 300 MW or higher – Category I (high risk);
- Thermal power plants and other power-generating facilities with a capacity of 100 MW to 300 MW – Category II (medium risk);
- Thermal, photovoltaic, wind power, and other power-generating facilities with less than 100 MW capacity – Category III (low risk).

This Project will be categorized as Category I.

The national EIA will be performed by Juru in parallel to the Lender ESIA process following the process outlined below. Compliance with national requirements and a positive Environmental Approval are pre-requisites to compliance with Lender requirements. The main regulatory body for national EIA in Uzbekistan is the Ministry of Natural Resources (MNR) of the Republic of Uzbekistan.

The MNR performs its activities on the basis of the following legal acts:

- Presidential Decree of January 25 2023 No. UP-14 On priority organizational measures for the effective establishment of the activities of the Republican Executive Authorities;
- Presidential Decree of April 21, 2017 No. UP-5024 "On improving the system of public administration in the field of ecology and environmental protection";
- Resolution of the President of the Republic of Uzbekistan of April 21, 2017 No. PP-2915 “On

4 Under the Resolution of Cabinet of Ministers of Uzbekistan, no 541, all economic activities are classified into one of four categories of environmental impact: Category I (high risk), Category II (medium risk), Category III (low risk) and Category IV (local impact).

measures to ensure the organization of the activities of the State Committee of the Republic of Uzbekistan on Ecology and Environmental Protection";

- Resolution of the Cabinet of Ministers of the Republic of Uzbekistan dated January 15, 2019 No. 29 "On Approving the Provision on the State Committee of the Republic of Uzbekistan on Ecology and Environmental Protection";
- Resolution of the President of the Republic of Uzbekistan dated October 3, 2018 No. PP-3956 "On measures to ensure the organization of the activities of the State Committee of the Republic of Uzbekistan on Ecology and Environmental Protection";
- Decree of the President of the Republic of Uzbekistan, dated 30.12.2021, №-76 "On measures for environmental protection and organization of activities of state bodies in the field of environmental control";
- Resolution of the Cabinet of Ministries of the Republic of Uzbekistan dated October 7, 2020. No.541 "On measures for the further improvement of environmental impact assessment"; and
- Other laws and by-laws related to nature protection.

The national EIA procedure is regulated by:

- Law of the Republic of Uzbekistan "On Ecological Expertise" (2000); and
- Regulations "On the State Environmental Expertise", approved by the Resolution of Cabinet of Ministers No. 541 "On further improvement of the environmental impact assessment mechanism' (2020).

The Resolution specifies the legal requirements for EIA in Uzbekistan. According to the Resolution, the State Environmental Expertise (SEE) is a type of environmental examination carried out by specialized expert divisions to set up the compliance of the planned activities with the environmental requirements and determination of the permissibility of the environmental examination object implementation.

The state unitary enterprise "The Center of the State Environmental Examination" of the MEEPCC, carries out the state environmental examination of EIA of the objects of economic activity classified as categories I and II of environmental impact (high and medium risk).

The state unitary enterprise "The Center of the State Environmental Examination" of the Republic of Karakalpakstan, or the relevant regions performs the state environmental examination of EIA of the objects of economic activity classified as categories III and IV of environmental impact (low risk and local impact).

For this Project the SEE will be performed by the Centre of State Environmental Examination.

National EIAs in Uzbekistan consist of three stages to obtain the Environmental Approval:

- Stage I - Preliminary EIA report – initial and mandatory stage;
- Stage II – Statement on Environmental impacts is a non-mandatory stage and can be skipped if local regulator is satisfied with assessment provided in Preliminary EIA report.
- Stage III – Statement on Environmental Consequences - is the final stage and output that should be prepared and submitted to the regulator after completing construction/reconstruction works and before the commissioning of the project.

The intention for this project is to submit a Stage 1 Preliminary EIA report that fully meets the needs of MNR for decision by MNR as to whether a Stage II submission is required.

Alignment of ESIA and National EIA report

Stage I Preliminary EIA report - The Preliminary EIA report must contain following information:

- the state of the environment prior to the implementation of the planned activities,
- the population of the territory, land development, analysis of environmental features;

- situational plan with an indication of the geographical coordinates of the object in question,
- available recreational areas, settlements, irrigation, land-improvement facilities, farmland,
- power lines, transport, water, gas pipelines and other information about the area;
- the envisaged (planned) main and auxiliary objects, used equipment, technologies,
- the use of natural resources, materials, raw materials, fuel, analysis of their impact on the environment (both during construction and operation phases);
- expected emissions, discharges, wastes, their negative impact on the environment and ways to minimize them (both during construction and operation phases);
- storage and disposal of waste (both during construction and operation phases);
- analysis of alternatives to the planned or ongoing activities and technological solutions from the standpoint of nature conservation, taking into account the achievements of science, technology and best practices;
- organizational, technical, technological solutions and measures that exclude negative environmental consequences and reduce the impact of the object of examination on the environment;
- analysis of emergency situations (with an assessment of the likelihood and scenario of preventing their negative consequences);
- forecast of environmental changes and environmental consequences as a result of the implementation of the object of examination;
- environmental measures to prevent the negative effects of the implementation of the object of examination; and
- results of public hearings.

It is necessary to highlight, that based on changes in local regulation, public hearings must be conducted in accordance with the procedure indicated in the law, represent all environmental impact assessments (to be justified by calculations) for construction and operation phases (if applicable).

Stage II Statement on Environmental impacts – during this phase additional information is provided in relation to key issues e.g., where specific modelling or impact assessment has been required. It is possible the outputs of the finalized EIA process could be communicated at Stage I which may negate the need for additional information to be provided (under Stage II) thus streamlining the approval process and the issue of permits for construction.

The Statement on Environmental impacts should include:

- Supporting materials (extract from PPA agreement; Presidential decree);
- Situation plan signed by the ecologist from the local ecology department and developer;
- Baseline conditions;
- Assessment of environmental problems of the selected site based on the results of engineering and geological surveys, models and other necessary studies;
- Environmental analysis of technology in relation to identified problems of the site;
- Results of public hearings (MoMs);
- Reasoned studies of environmental measures to prevent the negative effects of the implementation of the object of examination
- Hydrogeological report;
- Approvals and non-objections of the relevant authorities;

- Letter from Sanitary Epidemiological Wellbeing Department on the size of Sanitary Protection Zone;
- Environmental management plan;
- Environmental monitoring plan.

Stage III Statement on Environmental Consequences is the final stage of the SEE process and shall be carried out prior to the commissioning of the project. The report describes in detail the changes in the project made as a result of the analysis of the SEE during the first two stages of the EIA process, the comments received during public hearings, the environmental standards applicable to the project in relation to waste generation, water discharge, air emissions, and the environmental monitoring requirements related to the project, as well as the main conclusions.

2.5 Applicable E&S Legislation and Standards

2.5.1 Environment

The following Laws are relevant to the Project:

- The Law of the Republic of Uzbekistan “On Water and Water Use” (1993) as amended in 2022;
- The Law of the Republic of Uzbekistan “On Ecological Expertise” (2000) as amended in 2021;
- The Law of the Republic of Uzbekistan “On Atmospheric Air Protection” (1996, amended on 21.04.2021);
- The Law of the Republic of Uzbekistan “On Protection and Use of Vegetation” (1997) as amended in 2016;
- The Law of the Republic of Uzbekistan "On Protection and Use of the Wildlife" (1997) as amended in 2016;
- The Law of the Republic of Uzbekistan “On Protected Natural Reserves” (2004) as amended in 2022;
- The Law of the Republic of Uzbekistan “On Wastes” (2002) as amended in 2021;
- The Law “On the sanitary and epidemiological well-being of the population” (2015) as amended in 2021;
- The Resolution of the Cabinet of Ministries of the Republic of Uzbekistan №541 “On further improvement of the environmental impact assessment mechanism’ (2020) as amended in 2022;
- The Resolution of Cabinet of Ministries of the republic of Uzbekistan №820 “On measures to further improve the economic mechanisms for ensuring nature” (2018) as amended in 2021;
- The Resolution of the Cabinet of Ministers of the Republic of Uzbekistan No 14. “On approval of the regulation on the procedure for the development and agreement of projects with environmental standards” (2014) as amended in 2022; and
- Resolution of Cabinet of Ministers of Republic of Uzbekistan No.95 “On approval of general technical regulations of environmental safety” (2020) as amended in 2022.

2.6 Applicable National Environmental Standards

Uzbekistan has a large set of specific standards that refer to emissions, effluent discharge, and noise standards, as well as standard to handle and dispose specific wastes ranging from sewage to hazardous wastes. The following summarizes these laws and standards along with other international best practice standards. The ESIA will compare all standards and recommend “project standards” to be followed.

2.6.1 Air Quality

National Standards – Air quality in Uzbekistan in measured against Maximum Permissible Concentrations (MPC) and Maximum Permissible Emissions (MPE). Ambient Air Quality Standards, or MPCs, are established by SanPiN 0293-11 (May 16, 2011). According to the United Nations Environment Program (UNEP), Uzbek national ambient air quality standards meet World Health Organization (WHO) standards.[5] The MPCs relevant to the Project are show in Table 3**.**

Table 3: National Air Quality MPCs

Pollutant	Uzbekistan MPC (mg/m^3)			
	30 min	24 Hour	Monthly	Annually
Nitrogen Dioxide (NO_2)	0.085	0.06	0.05	0.05
Nitrogen Oxide (NO)	0.6	0.25	0.12	0.06
Sulphur Dioxide (SO_2)	0.5	0.2	0.1	0.05
Dust	0.15	0.1	0.08	0.05
Carbon Monoxide (CO)	5.0	4.0	3.5	3.0

Emission standards are stipulated by The Resolution of the Cabinet of Ministers of the Republic of Uzbekistan No. 14 of January 21, 2014 “On Approval of the Regulation on the Procedure for Developing and Coordinating Environmental Draft Projects”. It states that the main criterion for establishing MPE are quotas for pollutants. WBG and EU Standards – The Asian Development Bank (ADB) and the International Finance Corporation (IFC) adhere to the guidelines set forth by the World Health Organization (WHO). The European Bank for Reconstruction and Development (EBRD) follows the regulations established by the European Commission Directive 2008/50/EC on ambient air quality. This directive sets the framework for managing and improving air quality across Europe. (Table 4). As part of the ESIA an assessment against the most stringent guidelines will be made and used as “project standards”.

Table 4: EU Ambient Air Quality Guidelines

Pollutant	Averaging Period	Objective	Concentration	Comment
$PM_{2.5}$	Annual	Limit value	25 µg/m^3	
PM_{10}	24 hours	Limit value	50 µg/m^3	Not to be exceeded more than 35 days per year
PM_{10}	Annual	Limit value	40 µg/m^3	
NO_2	Hourly	Limit value	200 µg/m^3	Not to be exceeded more than 18 hours per year
NO_2	Annual	Limit value	40 µg/m^3	

5 https://wedocs.unep.org/bitstream/handle/20.500.11822/17141/Uzbekistan.pdf?sequence=1&isAllowed=y

Pollutant	Averaging Period	Objective	Concentration	Comment
SO_2	Hourly	Limit value	350 µg/m^3	Not to be exceeded more than 24 hours per year
SO_2	24 hours	Limit value	125 µg/m^3	Not to be exceeded more than 3 days per year
CO	Maximum daily 8 hour mean	Limit value	10 mg/m^3	

Table 5: WHO Ambient Air Quality Guidelines

Pollutant	Averaging Period	Concentration
$PM_{2.5}$	24 hours	15 µg/m^3
$PM_{2.5}$	Annual	5 µg/m^3
PM_{10}	24 Hour	45 µg/m^3
PM_{10}	Annual	15 µg/m^3
NO_2	Hourly	200 µg/m^3
NO_2	Annual	10 µg/m^3
NO_2	24 hours	25
SO_2	24 hours	40 µg/m^3
CO	Maximum daily 8 hour mean	10 mg/m^3

2.6.2 Noise

National noise standards are set out in SanPIN No. 0331-16. Admissible noise level into the living area, both inside and outside the buildings is used to ensure the rules of acceptable noise levels for residential areas in Uzbekistan. These rules and regulations establish permissible noise parameters in residential, public buildings and residential buildings of populated areas created by external and internal sources, as well as general requirements for measurements, measurement methods and hygienic noise assessment at research sites. Evaluation of the sound level at the calculation point is performed for the day and night period of the day (from 7:00 to 23:00 hours and from 23:00 to 7:00 hours) and considers the maximum intensity of the sound source level during the half-hour period. Table 6 presents the permissible noise levels for the premises most relevant for the project.

Table 6: Noise limits from SanPiN No. 0331-16

Purpose of premises or territories	Time	SanPiN No. 0267-09
Territories adjacent to homes, clinics, dispensaries, rest homes, boarding houses, nursing homes, child care facilities, schools and other educational institutions, libraries.	From 7 am to 11 pm	55 dB(D)
	From 11 pm to 7 am	45 dB(A)

To meet WBG guideline requirements noise impacts should not exceed the levels presented in Table 7 or result in a maximum increase in background levels of 3 dB at the nearest receptor location off site.

Table 7: WBG Noise Level Guidelines

Receptor	One-hour L_{aeq} [6](dBA)	
	Daytime 07.00-22.00	Night-time 22.00 – 07.00
Residential; institutional; educational	55	45
Industrial; commercial	70	70

The levels are almost identical to WBG noise level guidelines which are based on the standards of the World Health organisation (WHO) with the exception of the periods where WBG standards are slightly more stringent defining night-time noise as applicable at 22:00 instead of 23:00 under national standards. As such IFC guideline limits will be used for the Project.

Workplace Noise - In order to protect the health of staff in the workplace Uzbekistan, utilizes the law (SanPiN) No. 0325-16. Sanitary Standards for Permissible Noise Levels at the Workplace. This standard provides acceptable noise levels for various types of work, the most relevant of which are listed in Table 8**.** In addition, the WBG provides noise limits for various working environments, which are also illustrated in Table 8.

Table 8: Working environment Noise Limits

Type of work, workplace	SanPiN No. 0325-16	General EHS Guidelines of WBG
Performance of all types of work at permanent workplaces in industrial premises and at enterprises operated since March 12, 1985	80 dB (A)	
Heavy industry		85 Equivalent Level Laeq, 8h
Light industry		50-65 Equivalent Level Laeq, 8h

2.6.3 Water Quality

Water resource management, allocation and use in Uzbekistan falls under the Ministry of Agriculture and Water Resources (MAWR), which oversees national authorities I.e., provisional and district departments of agriculture and water resources, and inter provincial and inter district canal management authority.

According to SanPin № 0244-07 from 29.12.2007 Sanitary protection of water pipelines is provided by organizing a sanitary protection strip (zone). The width of the sanitary protection strip should be taken on both sides of the extreme lines of the conduit:

a) in the absence of groundwater - not less than 10 m with a diameter of water pipes up to 1000 mm and not less than 20 m with a diameter of water pipes more than 1000 mm;

No temporary or permanent discharges to surface water bodies are envisaged. The site is subject to works adjacent to ephemeral water bodies (streams) that cross the site and for completeness it is proposed to verify the baseline water quality pre-construction. Water quality standards are given as set out in Table 9.

[6] Laeq- equivalent average sound pressure level

It is not anticipated that any foul water discharges to surface water courses in relation to the construction or operation works, however for completeness, these are provided in Table 9 as per the following standards.

- Sanitary requirements for development and approval of maximum allowed discharges (MAD) of pollutants discharged into the water bodies with waste waters (SanPin No. 0202-06)
- SanPiN No 0255-08 which provides the criteria for hygienic assessment of the level water bodies

Table 9: Sanitary discharge standards (MAD and WBG guidelines for sanitary discharges)

№	Name of pollutants	Concentration (mg/l) (MAD)
1	Ammonium nitrogen	1
	Nitrate nitrogen	9,1
2	Nitrite nitrogen	0,2
3	Acetone	17,16
4	Acetonitrile	1,5
5	Beryllium	0,002
6	BOD20, BOD 5	15
7	Suspended substances	150,0
8	Diethyl ether	2,0
9	Fats	1,0
10	Potassium	50,0
11	Calcium	180,0
12	Caprolactam	10,73
13	Polyaniline	1,0
14	Xylene	1,0
15	Magnesium	40,0
16	Tall oil	1,0
17	Diesel fuel	1,0
18	Methanol	1,0
19	Molybdenum	1,0
20	Carbamide	80,0
21	Sodium	120,0
22	Sodium thiosulphate	16
23	Ammonium nitrate	10
24	Free fatty acids (FFAs)	3,9
25	Rhodanidae	1,0
26	Sulphur	10,0
27	Sulphur carbon	2,0
28	Turpentine	0,5
29	Synthetic surfactants	20,0

№	Name of pollutants	Concentration (mg/l) (MAD)
30	Styrene	0,56
31	Sulphates	100,0
32	Mercury (II) chloride (Sulema)	0,0001
33	Antimony	0,2
34	Dry residue (mineralisation)	2000,0
35	Tellurium (to Background)	0,05
36	Titanium	0,1
37	Toluene	2,8
38	Phosphates	2,5
39	Total phosphorus	3,0
40	Fluorides (fluorine ion)	1,5
41	Chlorides	350,0
42	Cyclohexane	0,1

2.6.4 Waste Management

Waste management laws relevant to the Project are listed below and key requirements described in the following sections:

- Law of the Republic of Uzbekistan “On wastes” (2002) amended in 2029; and
- SanPin № 0127-02 “Sanitary Procedures for inventory, classification, storage and disposal of industrial waste”

The Law of the Republic of Uzbekistan “On wastes” (2002) amended in 2029
The principal objective of this law is to prevent the negative impacts of solid wastes on human lives and health as well as the environment, reduce waste generation and encourage rational use of waste reduction techniques.

Article 19 Provided generated waste is subject to export and import operations, or hazardous waste is subject to transportation, an environmental certification procedure shall be completed by the Project to confirm compliance with sanitary and environmental norms and standards associated with waste management.

Article 20 states that transportation of hazardous waste shall be in specially designated types of vehicles with a waste certificate and permit. The responsibility for safe transportation of hazardous waste shall be with the transporting organisation.

Article 22 of the Law on Wastes specifies the general requirements for waste storage and disposal. Waste disposal of recyclable waste is prohibited in Uzbekistan. In addition, storage and disposal of waste in the environment including in nature conservation and protected areas, settlements, health and recreational areas or historical and cultural facilities is prohibited.

SanPin № 0127-02 “Sanitary Procedures for inventory, classification, storage and disposal of industrial waste”

This regulation and norm ensure optimal hygienic accounting and inventory of industrial wastes, determination of toxicity index and classification of industrial waste by hazard classes with optimal selection of ways to neutralise and utilise them.

SanPiN of the Republic of Uzbekistan dated 29/7/2002 No 0128-02 – "Hygienic classifier of toxic industrial wastes in the Republic of Uzbekistan. Hazardous waste is classified into four groups known as "hazard classes". Waste hazards are assessed based on this law. Hygienic classifier of industrial hazardous waste and SanPiN No 0127-02-Sanitary procedures for industrial waste inventory, classification, storage and disposal. Waste hazard classes include:

- Class I: Extremely hazardous waste;
- Class II: Highly hazardous waste;
- Class III: Moderately hazardous waste;
- Class IV: Low hazardous waste.

Other relevant regulations and standards are listed below and their requirements will be incorporated into the ESIA assessment:

- SanPiN № 0157-04 "Sanitary requirements to the storage and neutralization of solid domestic waste on special grounds in Uzbekistan"
- SanPiN of the Republic of Uzbekistan dated 16/11/2011 No 0300-11 "Sanitary Rules and Standards for managing collection, inventory, classification, treatment, storage and disposal of industrial waste in the context of Uzbekistan
- Regulation "On the Procedure for the Disposal, Collection, Pay Settlement, Storage and Removal of Waste Industrial Oils" annexed to the Decree of the Cabinet of Ministers dated 04/09/2012 No.258
- Regulation on the Procedure for Handling Coloured and Black Metal Scrap" annexed to the Decree of Cabinet of Ministers dated 06/06/2018 No. 425
- SanPiN No. 0158-04 - Sanitarian Rules and Norms on collection, transportation and disposal of wastes containing asbestos in Uzbekistan.

2.6.5 Land rights, acquisition and resettlement

The following land Laws are relevant to the Project:

- Civil Code of the Republic of Uzbekistan (1997) as amended on 8.11.2022.
- Land Code of the Republic of Uzbekistan (1998) as amended on 1.10.2022.
- Law of the Republic of Uzbekistan on State Land Cadastre No.666-I of 28.08.1998.
- Presidential Decree № UP-5495. Decree "On measures on cardinal improvement of investment climate in the republic of Uzbekistan".
- Appendix No. 2 to the Resolution of the Cabinet of Ministers № 146 (2011), regulation "On the Procedure for Compensation for Losses of Land Owners, Users, Tenants and Owners, as well as Losses of Agricultural and Forestry Production".
- Resolution № 911 of the Cabinet of Ministers (2019) "On the Procedure for withdrawal of land plots and compensation to owners of immovable property located on the land plot.
- Law No 781 "On procedures for the withdrawal of land plots for public needs with compensation" October 1st, 2022.[7]

Law No. 781 specifies cases when the land plots can be acquired for public need among which construction (reconstruction) of roads and railways of national and local significance is also specified. Law No. 781 also prescribes procedures of land acquisition, communication with project affected

7 Law 911 and Law 781 work alongside each other to address land withdrawal and compensation matters.

people (PAPs), compensation calculation, and demolition of affected assets. From October 1st, 2022, all projects that require the acquisition of land for public needs should be managed following this Law.

2.6.6 Archaeology and Cultural Heritage

The principal legislation applicable to the archaeology and cultural heritage study comprise the

- Constitution of the Republic of Uzbekistan, the Criminal Code of the Republic of Uzbekistan[8],
- Law No. ZRU-229 "On protection and use of the objects of archaeological heritage" (13 October 2009)[9],
- Law No. 269-II "On the Protection and Use of Cultural Heritage Sites (30 August 2001, as amended)[10],
- Presidential Decree No. R-5181 "On improving the protection and use of objects of tangible cultural and archaeological heritage" (16 January 2018) [11]

Presidential Decree no. PP-4068 "Regarding the strengthening of the protection, management and enhancement of tangible and intangible cultural heritage" (19 December 2018)[12]. The relevance of these requirements will be determined during the ESIA process.

2.6.7 Right of way and land acquisition process in Uzbekistan

Procedures to establish a right of way (ROW) in Uzbekistan are the same for legal entities and individuals. ROW or limited use of a land plot is determined in the Land Code of Uzbekistan, Civil Code (under the term servitude), and the Resolution of Cabinet of Ministries No.911 dated 16.11.2019.

The grid interconnection works is expected to require a new 200kV underground cable. There is no right if way required to be established for the unground cable between the site and the existing substation, however a safety zone of one meter either side of the cable is required to be set up as described below.

Article 30 of the Land Code (LC) determines engineering, electrical power and other lines (including underground cables) and constructions as a reason for receiving the right to servitude. Following Article 30 of the Land Code, Article 173 of the Civil Code (CC), and Article 30 of Annex A of the Resolution of Cabinet of Ministers No. 1060 dated December 29, 2018, servitude is established by agreement between persons demanding the establishment of servitude and the owner, user, lessee, proprietor of the land plot. If they do not achieve consent, the servitude shall be established by a court decision at the user's claim. The agreement on servitude shall be subject to state registration and preserved when the land plot is transferred to another person. Servitude agreements can be terminated in cases of the cessation of the reason according to which it was established.

Article 173 of CC also states that the burdening of a land parcel by servitude does not deprive the owner of the parcel of the rights of possession, use, and disposition of this parcel.

8 Criminal Code of the Republic of Uzbekistan of September 22, 1994 No. 2012-XII (as amended on 03-12-2019) Available at: https://www.lex.uz/acts/111457

9 Law of the Republic of Uzbekistan dated August 30, 2001 No. 269-II "On the Protection and Use of Cultural Heritage Sites". Available at: https://www.lex.uz/acts/10375#1526009

10 Law of the Republic of Uzbekistan dated 13 October 2009 No. ZRU-229 "On protection and use of the objects of archaeological heritage". Available at https://lex.uz/docs/1526179

11 Presidential Decree No. R-5181 of 16 January 2018 "On improving the protection and use of objects of tangible cultural and archaeological heritage". Available at: https://www.lex.uz/docs/3506339

12 Presidential Decree No. PP-4068 of 19 December 2018 "Regarding the strengthening of the protection, management and enhancement of tangible and intangible cultural heritage". Available at: https://lex.uz/ru/docs/4113474/.

Calculation and compensation of losses due to servitude agreement are performed following Law No 781 “On procedures for the withdrawal of land plots for public needs with compensation” (if it is a project for public needs) the Resolutions of Cabinet of Ministers No.146 from 25 May 2011 “On measures to improve the procedure for granting land plots for urban development activities and other non-agricultural purposes” and No. 911 from 16 November 2019 “On additional measures for enhancing modalities of providing compensation on withdrawal and allocation of land plots and safeguard the property rights legal and physical entities”.

Article 86 of the LC states that losses caused to the owners of land parcels, landowners, land users and lessees are liable to be fully refunded (including the lost profit) in the case of limitation of their rights in connection with land acquisition. Refunding of losses is carried out at the expense of the resources of the corresponding centralized funds for compensation of losses to individuals and legal entities in connection with the seizure of land plots from them for public needs and by enterprises, establishments and organizations the activity of which causes limitation of rights of land parcel owners, landowners, land users and lessees or worsening the quality of the neighbouring lands in the order established by legislation.

Article 173 of the CC states that the parcel owner burdened with the servitude has the right unless otherwise provided by a Law, to demand from the person in whose interests the servitude is established proportional payment for the use of the parcel.

2.6.8 National norms and standards for underground transmission cables

Resolution of the Cabinet of ministers of the Republic of Uzbekistan No.1050 “On Approval of Rules for Protection of Electric Grid Facilities” (2018) establishes the following security zone for underground cables:

- along underground cable power lines - in the form of a part of the surface of a plot of land located underneath the subsoil plot (to a depth corresponding to the depth of laying cable power lines), limited by parallel vertical planes spaced on both sides of the power line from the outer cables at a distance of 1 meter, and when cable lines pass in cities under sidewalks - 0.6 meters towards buildings and structures and 1 meter towards the roadway.

2.7 Labour and employment

The labour policy in Uzbekistan is applied at the national government level and is reflected in the following relevant laws, regulations, and national social programmes.

- Labour Code of the Republic of Uzbekistan 1996 as amended on 18.05.2022;
- Law “On the employment of the population” No. 642 of 20.10.2020;
- Resolution of the Ministry of Labour and Social Protection of the Population, Ministry of Health of the Republic of Uzbekistan, registered on 29.07.2009, reg. number 1990 “About the approval of the list of occupations with unfavourable conditions, in which the use of the labour of persons under 18 years of age is prohibited”
- Decree No. 133 of 11 March 1997 to approve normative acts necessary for the realization of the Labour Code of the Republic of Uzbekistan;
- Decree of the Cabinet of the Ministers No. 1011 of 22 December 2017 "On Perfection of the Methodology of Definition of Number of People in Need of Job Placement, including the Methodology for Observing Households with Regard to Employment Issues, also for the Development of Balance of Labour Resources, Employment and Job Placement of Population";
- Decree of the Cabinet of the Ministers No. 965 of 5 December 2017 "On the Measures of

Further Perfection of the Procedure of Establishment and Reservation of Minimum Number of Job Places for the Job Placement of Persons who are in need of Social Protection and Face Difficulties in Searching Employment and Incapable of Competing in Labour Market with Equal Conditions"; and

- Decree No. 964 of 5 December 2017 "On the Measures for Perfection of the Activity of Self-Government Bodies Aimed at Ensuring Employment, firstly for the Youth and Women"

As a member of the International Labour Organization (ILO) since 1992, Uzbekistan has ratified 17 ILO conventions, including the eight fundamental conventions (bold) set out in Table 10.

Table 10: Labour Conventions ratified by Uzbekistan

Convention	Date
Universal Declaration of Human Rights (1948)	1991
CCPR - International Covenant on Civil and Political Rights (1966)	28-Sep-95
Convention on the Elimination of All Forms of Intolerance and of Discrimination Based on Religion or Belief (1981)	30-Aug-97
EU Partnership and Cooperation Agreement (1996)	21-Jun -96
C029 - Forced Labour Convention, 1930 (No. 29)	13-Jul-92
P029 - Protocol of 2014 to the Forced Labour Convention, 1930	16-Sep-19
C087 - Freedom of Association and Protection of the Right to Organise Convention, 1948 (No. 87)	12-Dec-16
C098 - Right to Organise and Collective Bargaining Convention, 1949 (No. 98)	13-Jul-92
C100 - Equal Remuneration Convention, 1951 (No. 100)	13-Jul-92
C105 - Abolition of Forced Labour Convention, 1957 (No. 105)	15-Dec-97
C111 - Discrimination (Employment and Occupation) Convention, 1958 (No. 111)	13-Jul-92
C138 - Minimum Age Convention, 1973 (No. 138) Minimum age specified: 15 years	06-Mar-09
C182 - Worst Forms of Child Labour Convention, 1999 (No. 182)	24-Jun-08
C081 - Labour Inspection Convention, 1947 (No. 81)	19-Nov-19
C122 - Employment Policy Convention, 1964 (No. 122)	13-Jul-92
C129 - Labour Inspection (Agriculture) Convention, 1969 (No. 129)	19-Nov-19
C144 - Tripartite Consultation (International Labour Standards) Convention, 1976 (No. 144)	13-Aug-19
C047 - Forty-Hour Week Convention, 1935 (No. 47)	13-Jul-92
C052 - Holidays with Pay Convention, 1936 (No. 52)	13-Jul-92
C103 - Maternity Protection Convention (Revised), 1952 (No. 103)	13-Jul-92
C135 - Workers' Representatives Convention, 1971 (No. 135)	15-Dec-97
C154 - Collective Bargaining Convention, 1981 (No. 154)	15-Dec-97
CEDAW - Convention on the Elimination of All Forms of Discrimination against Women	19-Jul-95
C187 - Promotional Framework for Occupational Safety and Health Convention, 2006 (No. 187)	14-Sep-21

Measures have been enacted via a national action plan to implement these conventions into national law, including a legal and institutional framework to prevent forced labour. The legislation of the Republic of Uzbekistan (Constitution, Labour Code, Law on Employment) prohibited the use of child and forced labour. Article 7 of the Labour Code stipulates that forced labour, namely compulsion to perform work under the threat of some form of punishment (including as a means of labour discipline) is prohibited.

2.8 Permits and licenses

Required permits and licenses known at this time are listed in Annex D. A permit register will be maintained for the Project duration and reviewed and updated regularly.

2.9 Lender requirements

The following Lender requirements will be considered to provide maximum flexibility to the Project financing. The Project will principally set out to comply with the requirements of ADB, EBRD and IFC.

2.9.1 EBRD Policy

The ESIA will consider the E&S requirements of the EBRD as set out in the following Performance Requirements (PRs).

- The European Bank for Reconstruction and Development (EBRD) Environmental and Social Policy 2019 (ESP 2019);
- EBRD PRs:
  - PR1 – Assessment and Management of Environmental and Social Risks and Impacts;
  - PR2 – Labour and Working Conditions;
  - PR3 – Resource Efficiency and Pollution Prevention and Control;
  - PR4 – Health, Safety and Security;
  - PR5 – Land Acquisition, Restrictions on Land Use and Involuntary Resettlement;
  - PR6 – Biodiversity Conservation and Sustainable Management of Living Natural Resources;
  - PR7 – Indigenous Peoples;
  - PR8 – Cultural Heritage;
  - PR10 – Information Disclosure and Stakeholder Engagement.

With reference to EBRD ESP 2012, Appendix 2, this project is proposed to be categorised as Category “B”. A project is categorised B when its potential environmental and/or social impacts are typically site specific, and/or readily identified and addressed through effective mitigation measures. EBRD will finally determine the scope of environmental and social appraisal on a case-by-case basis. Category B projects do not require EBRD disclosure of key ESIA documents over and above the requirements of PR10.

2.9.2 International Finance Corporation (IFC)

The Project will also consider the requirements of the IFC Performance Standards 2012 (IFC PSs), including:

- Performance Standard 1: Assessment and Management of Environmental and Social Risks and Impacts.
- Performance Standard 2: Labour and Working Conditions
- Performance Standard 3: Resource Efficiency and Pollution Prevention.
- Performance Standard 4: Community Health, Safety, and Security.
- Performance Standard 5: Land Acquisition and Involuntary Resettlement.

- Performance Standard 6: Biodiversity Conservation and Sustainable Management of Living Natural Resources.
- Performance Standard 7: Indigenous Peoples.
- Performance Standard 8: Cultural Heritage.

IFC PS1 establishes the importance of: (i) integrated assessment to identify the social and environmental impacts, risks, and opportunities of projects; (ii) effective community engagement through disclosure of project-related information and consultation with local communities on matters that directly affect them; and (iii) management of social and environmental performance throughout the life of the project.

IFC PS2 through IFC PS8 establish requirements to avoid, reduce, mitigate or compensate for impacts on people and the environment and to improve conditions where appropriate. While all relevant social and environmental risks and potential impacts should be considered as part of the assessment, IFC PS2 through IFC PS8 describe potential social and environmental impacts that require particular attention in emerging economies and sensitive and critical natural and human environments. Where social or environmental impacts are anticipated, they are to be managed through an Health Safety Social Environmental Management System (HSSE-MS) consistent with the requirements of IFC PS1.

IFC PS3 refers to the World Bank Group EHS Guidelines. These guidelines are the technical reference documents for environmental protection and set out specific examples of good international industry practice (GIIP). The General EHS Guidelines contain information on crosscutting issues applicable to projects in all industry sectors, including geothermal. They provide guidance on performance levels and measurements considered achievable at reasonable cost by new or existing projects using existing technologies and practices. Projects are expected to comply with standards and guidelines identified in the General EHS Guidelines where host country requirements are less stringent or do not exist.

This project is considered a Category “B”: *“Business activities with potential limited adverse environmental or social risks and/or impacts that are few in number, generally site-specific, largely reversible, and readily addressed through mitigation measures”.*

2.9.3 World Bank Group Guidelines

The EBRD PRs and IFC PSs refer to the World Bank Group (WBG) Environment, Health and Safety (EHS) Guidelines as general guidance for implementing GIIP. The EHS Guidelines applicable to the Project include the following:

- WBG General EHS Guidelines (April 2007) - cover the four areas of the environment; occupational health & safety (OHS); community health & safety (CHS); construction and decommissioning; and
- WBG EHS Guidelines Electric Power Transmission and Distribution (April 2007).

2.9.4 Asian Development Bank

ADB SPS 2009 sets out three key safeguard policies as follows:

- Safeguard Requirements 1: Environment;
- Safeguard Requirements 2: Involuntary Resettlement); and
- Safeguard Requirements 3: Indigenous Peoples (not expected to be triggered for this Project).

The requirements of these safeguards are broadly aligned with the requirements of EBRD PRs and IFC PSs (as outlined above) relating to identification of impacts, the requirement to develop and implement plans to avoid, minimize, mitigate, or compensate for the potential adverse impacts; and to ensure affected people are informed and consulted during project preparation and implementation. SPS 2009 requires projects to reflect internationally recognized standards such as the World Bank Group's Environmental, Health and Safety Guidelines (refer to section 4.7.2 above).

Broadly speaking, ADB SPS 2009 is also aligned on the topics of climate change, gender and biodiversity, however key ADB policy documentation relevant to this Project includes:

- ADB Social Protection Strategy (2001);
- ADB Access to Information Policy (2018); and
- ADB Gender and Development Policy (1998).

An Initial Environmental Examination (IEE)[13] including the supporting documentation listed below was performed in September 2022 in accordance with ADB SPS 2009. Supporting documentation/assessments included:

- Project environmental management plan (EMP)
- Biodiversity Management and Evaluation Plan
- COVID health & safety risk mitigation measures
- Critical habitat assessment[14]
- Site Suitability report[15]
- Social Safeguard Due Diligence Report[16]

This was supported by consultations with the Project community in December 2021 and January 2022 as reported in the IEE. The Project is categorised as category B project.

2.9.5 Equator Principles

The Equator Principles (EPs) are voluntary standards signed up by various financing institutions to serve as a common baseline and risk assessment framework. EP4 includes ten principles covering:

- Review and categorisation
- E&S Assessment
- Applicable E&S Standards
- E&S Management System and EP Action Plan
- Stakeholder Engagement
- Grievance Mechanism
- Independent Review
- Covenants
- Independent Monitoring and Reporting
- Reporting and Transparency

This project is likely to be categorised as "Category B" under EPIV. As such, it must apply the requirements of the applicable IFC PSs and WBG EHS Guidelines as defined above. The project must also implement effective stakeholder engagement, grievance management and an Health Safety Social Environmental Management System (HSSE-MS). EP4 also sets out specific requirements relating to

[13] Initial Environmental Examination Report *for Utility-Scale Solar Photovoltaic and BESS PPP Project in Guzar District, Qashqadaryo Region of the Republic of Uzbekistan* Project no.: 49407-005 Client: Ministry of Energy, Uzbekistan & ADB – Asian Development Bank, September 2022 (Draft 2), Uzbekistan: Solar Public-Private Partnership Investment Program (Tranche 2), Dornier Suntrace GmbH.

[14] Guzar Solar PV Critical Habitat Assessment for Utility-Scale Solar Photovoltaic PPP Project in Guzar District, Qashqadaryo Region of the Republic of Uzbekistan, MOE, ADB, September 2022, Suntrace GmbH.

[15] Guzar Solar PV Site Suitability Report (Pre-FS) for Utility-Scale Solar Photovoltaic PPP Project in Guzar District, Qashqadaryo Region of the Republic of Uzbekistan, MOE, ADB, October 2021, Suntrace GmbH.

[16] Social Safeguard Due Diligence Report for Utility-Scale Solar Photovoltaic PPP Project in Guzar District, Qashqadaryo Region of the Republic of Uzbekistan, MOE, ADB, July 2022, Suntrace GmbH

Human Rights in line with the United Nations Guiding Principles on Business and Human Rights (UNGPs) by requiring a human rights due diligence (HRDD) and improving the availability of climate-related information, such as the Recommendations of the Task Force on Climate-related Financial Disclosures (TCFD) to incorporate a Climate Change Risk Assessment (CCRA) for the assessment of potential transition and physical risks of Projects.

2.9.6 Good International Industry Practice (GIIP)

The Project will also follow GIIP, including, but not limited to:

- Voluntary Principles on Security and Human Rights (est. 2000); (http://www.voluntaryprinciples.org/);
- United Nations Guiding Principles for “Protect, Respect and Remedy” Human Rights Framework (2011); (https://www.business-humanrights.org/en/un-secretary-generals-special-representative-on-business-human-rights/un-protect-respect-and-remedy-framework-and-guiding-principles);
- United Nations Code of Conduct for Law Enforcement Officials; and (https://www.un.org/ruleoflaw/blog/document/code-of-conduct-for-law-enforcement-officials/);
- United Nations Basic Principles on the Use of Force and Firearms by Law;
- Use of Security Forces: Assessing and Managing Risks and Impacts (February 2017);
- Worker's Accommodation: Processes and Standards (Guidance Note by IFC and EBRD, 2009), and
- Stakeholder Engagement: A Good Practice Handbook for Companies Doing Business in Emerging Markets, 2007.
- World Bank (2016a). Managing the Risks of Adverse Impacts on Communities from Temporary Project Induced Labor Influx. The World Bank, Washington, D.C.
- IFC/EBRD Guidelines for workers accommodations (2009).
- IFC Good Practice Note on Non-Discrimination and Equal Opportunity.
- IFC Stakeholder Engagement- Good Practice Handbook for Companies Doing Business in Emerging Countries;
- IFC Good Practice Note – Addressing Grievances from Project-Affected Communities;
- IFC Good Practice Manual – Doing Better Business Through Effective Public Consultation and Disclosure;
- IFC Handbook for Labour and Working Conditions - Measure & Improve Your Labour Standards Performance.
- IFC Good Practice Note Managing Contractors’ Environmental and Social Performance (2017).

2.10 International conventions and agreements

Fundamental conventions and agreements (in addition to the ILO conventions mentioned in Table 10 signed and ratified by Uzbekistan that are relevant to the Project are provided in Table 11.

Table 11: Conventions relevant to the Project that have been ratified by Uzbekistan

Convention name
ENVIRONMENT / CLIMATE CHANGE

United Nations Framework Convention on Climate Change (UNFCC) (New York, 1992) (Official Gazette of RM no. 61/97) including Paris Agreement (joined April 2017)
United Nations Convention on Biological Diversity (Official Gazette of RM no. 54/97)
United Nations Convention to Combat Desertification (UNCCD) (26/12/2006)
Convention on the Prohibition of Military or Any Other Hostile Use of Environmental Modification Techniques (05/26/1993)
Basel Convention on the Control of Transboundary Movements of Hazardous Wastes and their Disposal (12/22/1995)
The Convention on the Protection and Use of Transboundary Watercourses and International Lakes
Convention Concerning the Protection of the World's Cultural and Natural Heritage (ratified 1993)
Convention for the Safeguarding of the Intangible Cultural Heritage. Paris (ratified 2008)
Convention on International Trade in Endangered Species of Wild Fauna and Flora (07/01/1997)
Convention on the Conservation of the Migratory Species of Wild Animals (Bonn Convention) (05/01/1998)
Convention on Wetlands of International Importance, especially the Water Fowl Habitats of Aquatic Birds (Ramsar Convention) (1975) (ratified 2001)
Vienna Convention for the Protection of the Ozone Layer (1985).
Montreal Protocol to Protect the Ozone Layer (including 1990 and 1999 amendments)
Convention on Environmental Impact Assessment in a Transboundary Context (Espoo, 1991) - the 'Espoo (EIA) Convention'
Convention on Access to Information, Public Participation in Decision Making and Access to Justice in Environmental Matters (Aarhus Convention) (Official Gazette of RM no. 40/99

2.10.1 Climate Policy

Multi-lateral development banks have developed their own climate related policies for alignment with the Paris Agreement. These can be summarised as follows[17]:

EBRD:

- From end-2022, all activities to be Paris aligned.
- Increasing green financing to >50% by 2025.
- Aims to achieve cumulative net GHG emissions reductions of 25- 40 million tons annually.

IFC:

- Committed to 85% alignment starting 01 July 2023
- 100% alignment starting 01 July 2025
- Growing its climate- related investments to an annual average of 35 percent of its own-account long- term commitment volume between 2021 and 2025

ADB:

In 2018, ADB committed to ensuring at least 75% of its operations support climate action and to reaching a cumulative $80 billion in climate finance by 2030. Today's announcement elevates this ambition to 100$ billion. The additional $20 billion will support the climate agenda in five areas:

[17] IFC Technical Guidance for Financial Institutions — Assessment of Greenhouse gas emissions (2023)

- Climate Mitigation: Including energy storage, energy efficiency, and low-carbon transport. Cumulative mitigation finance is expected to reach $66 billion.
- Transformative Adaptation Projects: Focusing on sectors like urban, agriculture, and water. Cumulative adaptation finance is expected to reach $34 billion.
- Private Sector Operations: Creating commercially viable projects, targeting $12 billion in private sector climate finance from ADB's resources and anticipating an additional $18-30 billion from private investors.
- Green Recovery from COVID-19: Utilizing platforms like the ASEAN Catalytic Green Finance Facility to leverage funds for low-carbon infrastructure.
- Policy-Based Lending: Supporting reforms in DMCs for enhanced climate resilience and mitigation.

3 Project Description

3.1 Summary of needs case

Uzbekistan has abundant renewable energy resources (solar and wind) and renewable generation potential, largely untapped. This potential has remained virtually untapped due to a lack of incentives, experience, historically subsidised natural gas prices for the country’s gas-fired thermal power plants that constitute over 80% of the total power generation capacity, and low tariffs.

The Government of Uzbekistan (GOU) aims to increase its power supply and has adopted the 2030 Energy Strategy, which defines several objectives and directions for electricity supply between 2020-2030. One of the Energy Strategy objectives includes developing and expanding renewable energy use and its integration into the unified power system. To fulfil this objective, the Government of Uzbekistan intends to “Ensure diversification in power and heat energy sectors through the increased share of renewable energy sources and creation of renewable energy investment project mechanism utilising PPP [public-private partnership] approaches, enhancement of government policies related to the development of renewable energy sources, demonstration of renewable projects”.

In May 2019, the laws of the Republic of Uzbekistan, "On the use of renewable energy sources" and "On public-private partnership", were adopted. Thus, a regulatory and legal framework has been created to accelerate the implementation of renewable energy projects such as this one. Uzbekistan plans to increase the share of renewable energy sources to 25% by 2030.

The Nur-Kashkadarya PV and BESS Project is part of the Ministry of Energy program which aims to enable the rapid roll-out of competitively priced, utility-scale solar PV power in Uzbekistan. The approach is a largely standardised solution based on a templated Public Private Partnership (PPP) transaction.

The Project aligns with the Uzbekistan Energy Sector Strategy (BDS18-237(F)), the Green Economy Transition approach (BDS15-196(F)) aimed at supporting cleaner production and distribution of energy through greater energy and resource efficiency. The Project is also part of a more comprehensive program to help the broader integration of renewables into the national grid.

The Project benefits can be summarised as follows:

- reduce energy dependence on carbon-based fuels
- providing clean energy to households in the Kashkadarya region
- meet RES targets
- reduce greenhouse gas emission rates
- creating job opportunities for the local labor force during the construction and operation stages, along with the indirect impact of consumed materials and services in the supply chain
- establishment of a new private sector and associated businesses focused on the solar power generation sub-sector reducing state’s share in the power sector.

4 Analysis of Alternatives

A robust site selection and technology assessment has been performed to identify the 802 ha site and grid connection approach. The ESIA report will document the information on the decision making relating to the following:

- No project alternative
- Site location

- Site boundary and layout
- Grid connection alternatives
- Technology alternatives
    - solar versus other energy generation technologies)
    - battery technology selection

4.1 Project location and setting

The PV Project site is 802 ha site immediately adjacent to the 500kV Guzar substation in the Guzar and Kamashi districts (55 km south-west of Shahrisabz and 12 km northeast of Guzar cities in Qashqadaryo regions) of Uzbekistan. Project location indicated in Figure 3. Parallel to the northern boundary of the site is the regional road M39 that runs between Shahrisabz and Guzar. To the west is the existing 220kV/500kV Guzar substation. The nearest communities are Yangiabad, Khalqabad, and Batosh from Guzar district and Aynarkul from Kamashi district. The location of communities in relation to the Project is indicated in Figure 4.

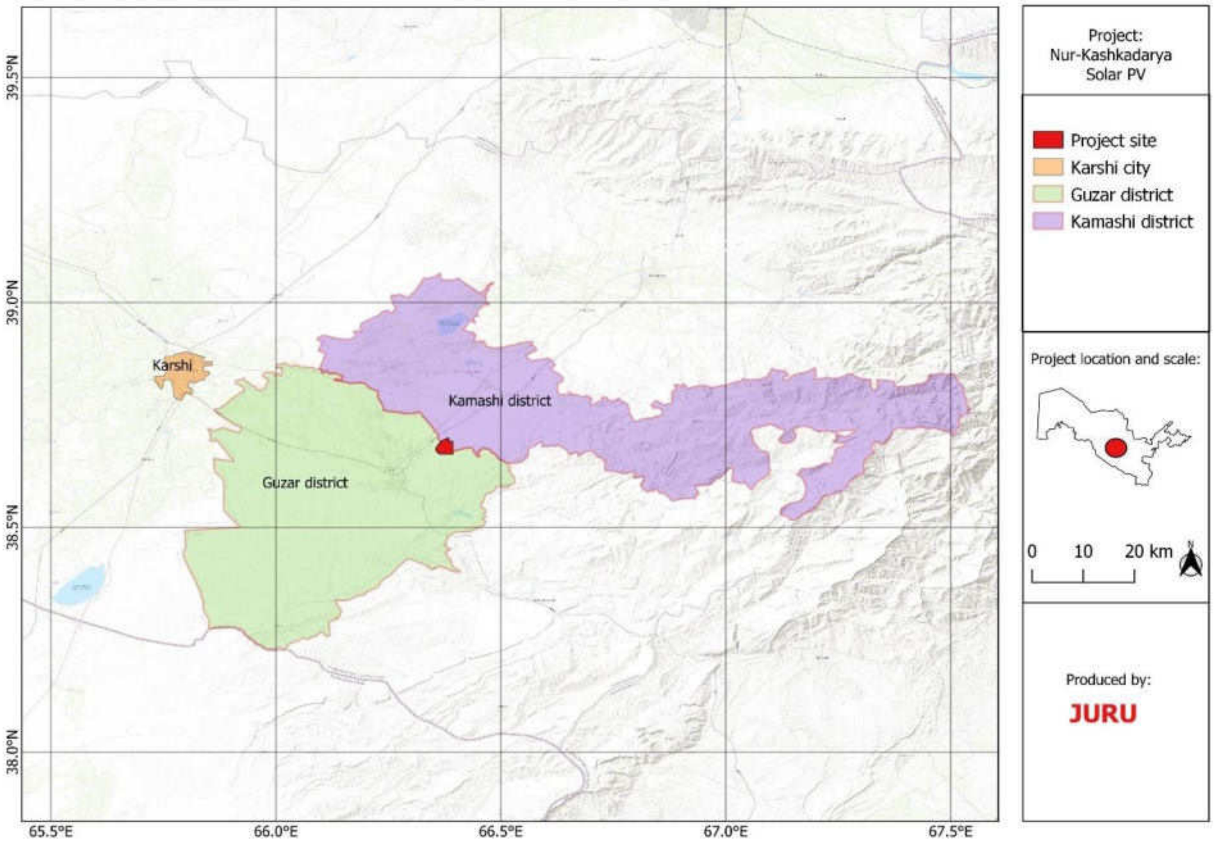


Figure 3 Location of the Project site (preliminary borders)[18]

18 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 3Figure 2 and are still considered representative of the final layout.

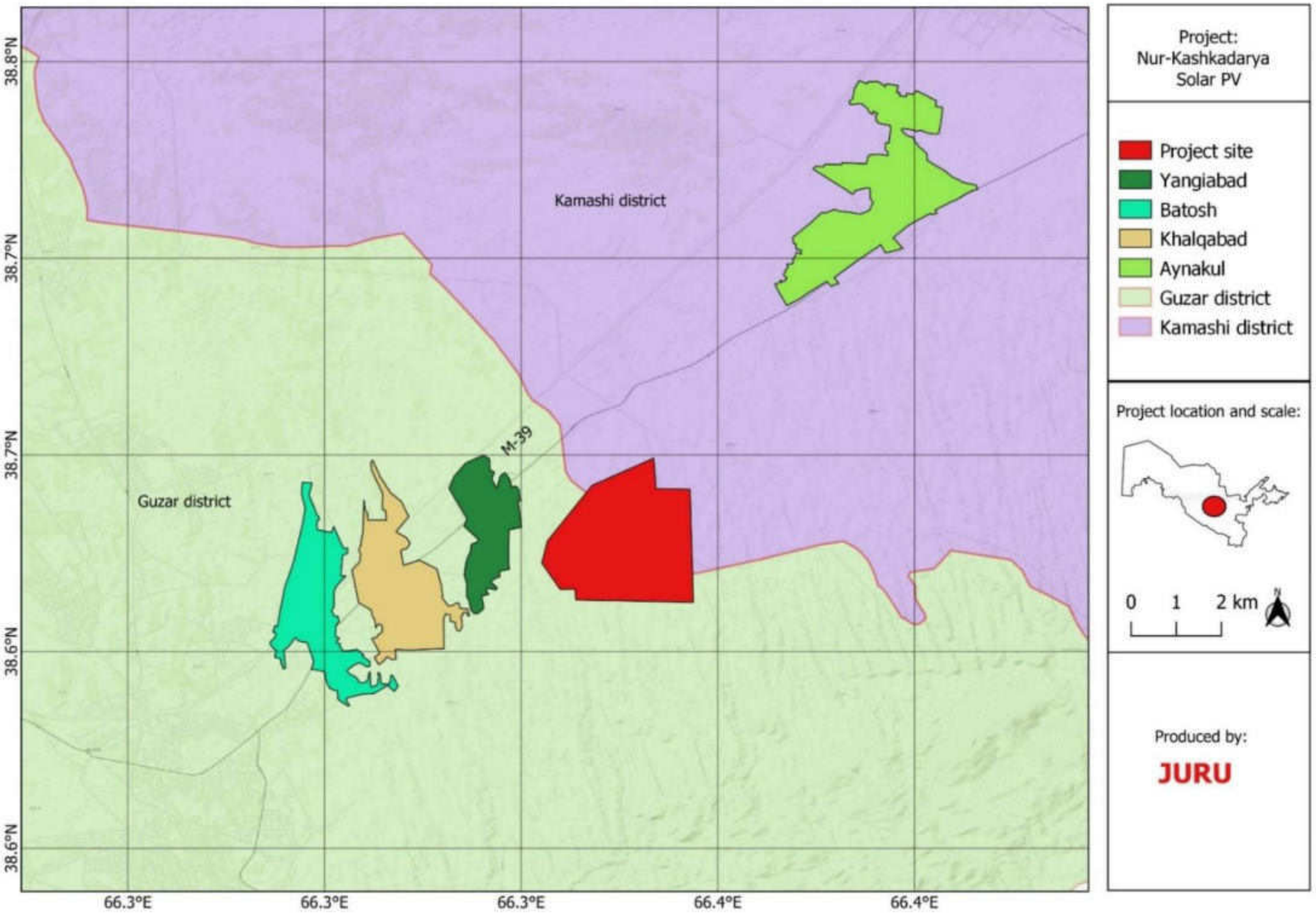


Figure 4 Location of communities near the Project site (preliminary borders)

This Site topography is undulating and covered by rain fed fields and short grass. The site is boarded to the north and west by existing 500kV transmission line (see figures below). There are no permanent waterbodies or rivers on the site, however there are a number of ephemeral channels running from south-southeast and from a north-northwest direction. The site is predominately used for agricultural purposes (refer to section 5.4 for further detailed land use description).

Key features of the site location and area are illustrated in the figures below. A full description of the environmental and social baseline is provided in Chapter 4. The reference Universal Transverse Mercator (UTM) coordinates (Zone 41) are represented in Table 12 and Figure 5 below.

Table 12: Site coordinates

Point No	Northing	Easting
Pb01	38.699187°	66.383718°
Pb02	38.692248°	66.368003°
Pb03	38.678126°	66.356682°
Pb04	38.672517°	66.355245°
Pb05	38.665820°	66.359916°
Pb06	38.665887°	66.363660°
Pb07	38.663099°	66.363911°
Pb08	38.662543°	66.393775°
Pb09	38.691258°	66.392985°
Pb10	38.691442°	66.384288°

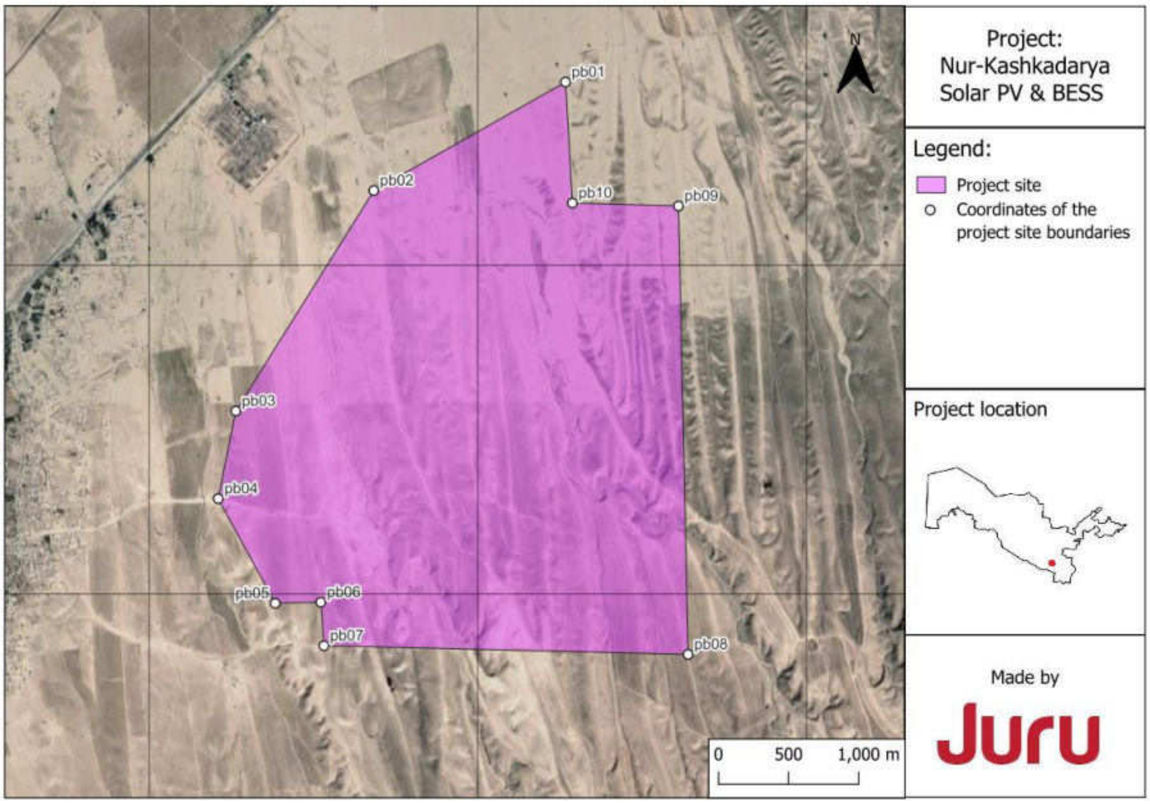


Figure 5: Project site boundaries and coordinates[19]

Figure 6: Rain fed fields on the Project site

Figure 7: Camel thorn (Alhagi pseudalhagi) on the north part of the Project site

19 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 5 and are still considered representative of the final layout.

Figure 8: Unpaved Road across the Project site

Figure 9: Existing OHTL to the north of the proposed site

Figure 10: Piles of garbage and existing OHTL along the western border of the Project site

Figure 11: Herder's hut on the Project site

Figure 12: Guzar Substation (view from the Project site)

Figure 13: Farm plot near the existing 500kV substation (may be crossed by grid connection)

Figure 14: The view on the Guzar SS where the final connection will be made

Figure 15: Boundary of farm plot near the existing 500kV substation (may be crossed by grid connection)

Figure 16: Deep ravines within the Project site

Figure 17: House outside the Project boundaries

Figure 18: Leakage from the underground water pipeline (outside the Project boundaries)

Figure 19: Grazing animals near Guzar SS and site

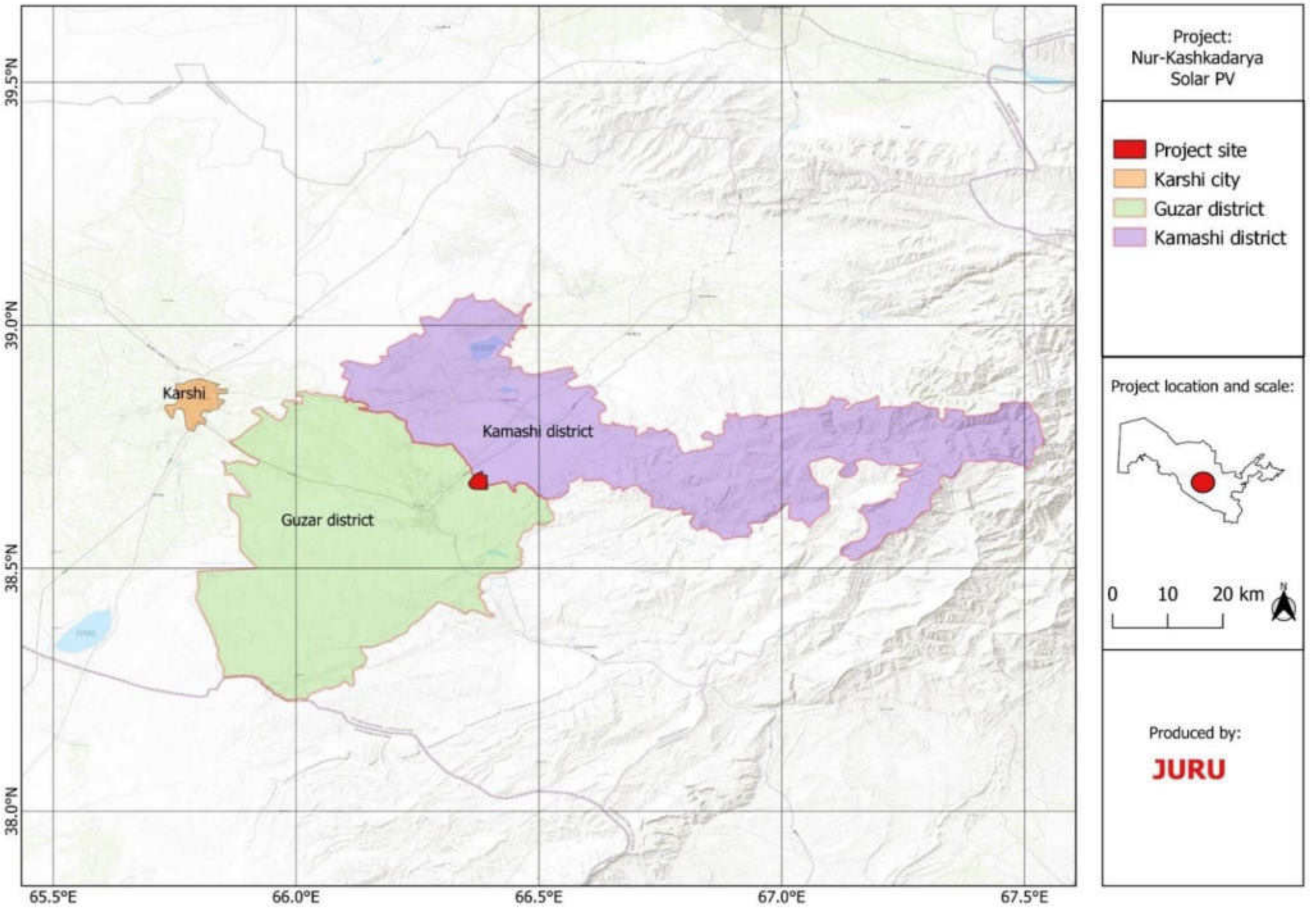


Figure 20: Wider Project area[20]

20 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 20 and are still considered representative of the final layout.

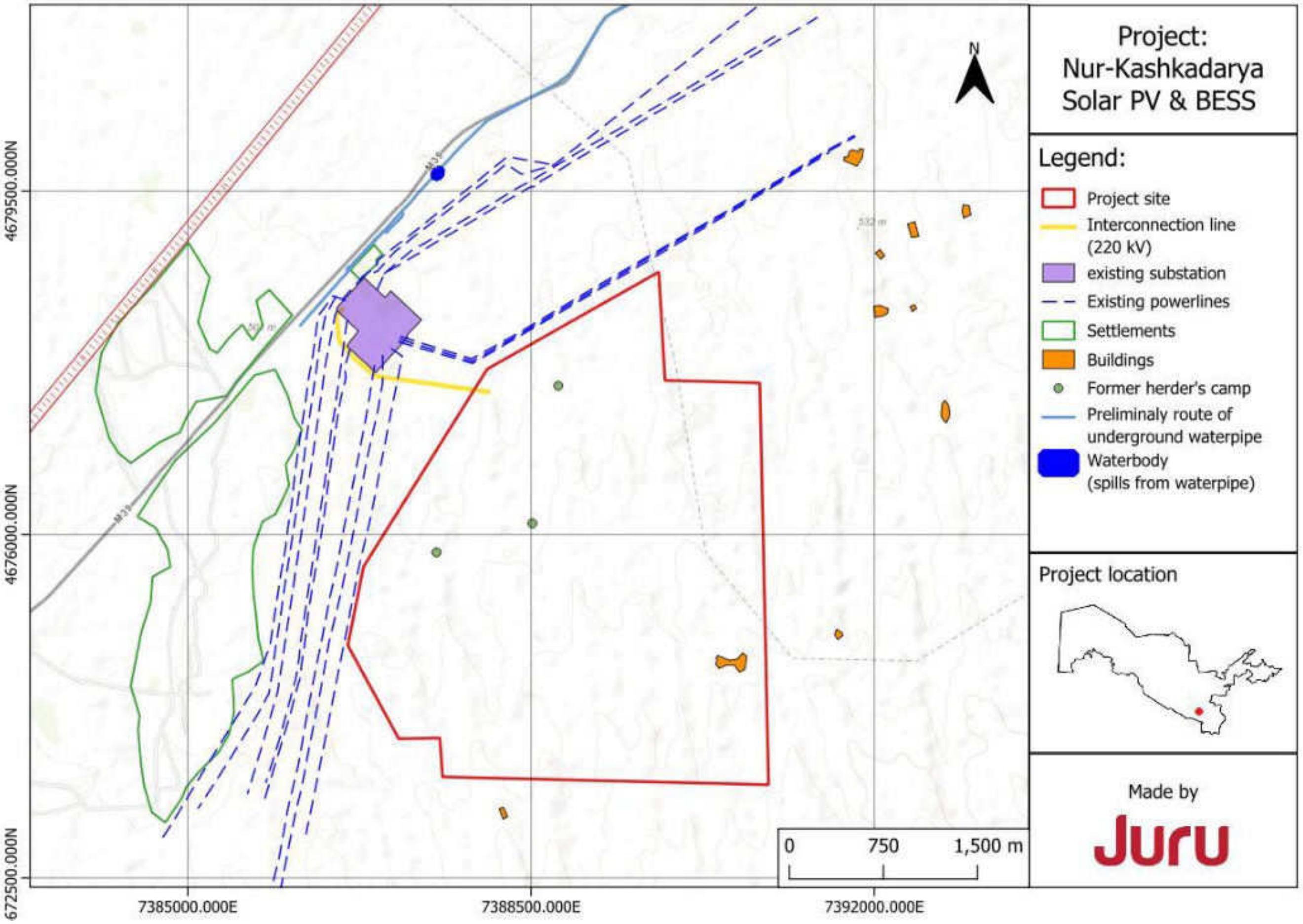


Figure 21: Project setting (map view)[21]

21 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 21 and are still considered representative of the final layout.

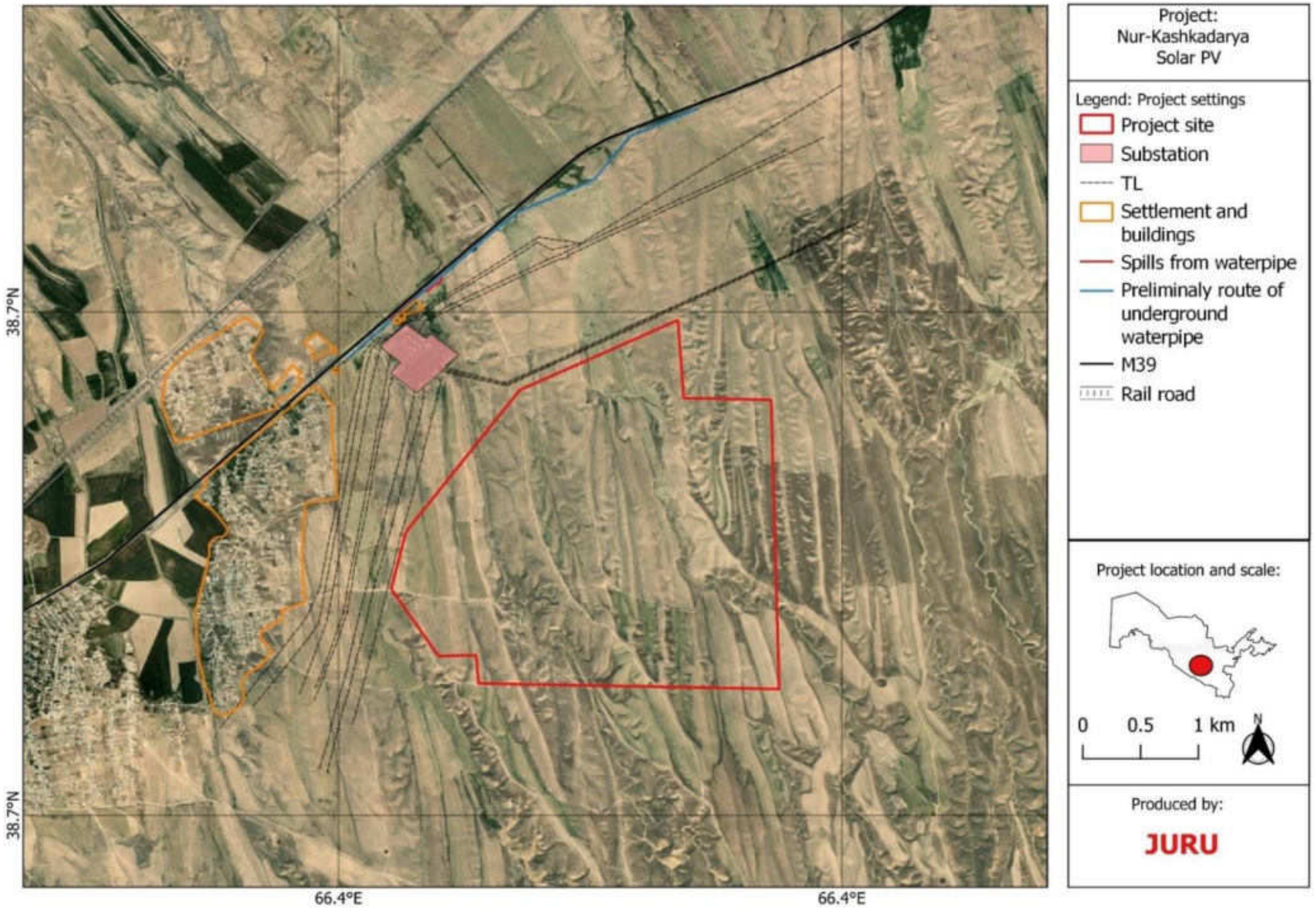


Figure 22 Project setting (aerial photo view)[22]

22 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 22 and are still considered representative of the final layout.

4.2 Project receptors

There are a total of four communities that are located near to the Project site. In the Guzar district, there are three communities; namely, Yangiabad, Khalqabad, and Batosh. In the Kamashi district, there is one community, Aynakul. In addition, there are scattered households which are located to the north-east, approximately 1 km from the Project boundary. During the scoping site visit, residents from these households stated that they informally use the Project site to graze livestock seasonally. The total number of households that use the land will be determined during future site visits and surveys.

During scoping site visits, it was identified that there are five temporary herder camps and five stables for livestock located within the Project site. There are also some temporary structures such as herder camps and stables just outside of the Project site. The structures were constructed by members of the local community and are used only during certain periods of the year.

There is a water pipeline along M39 main road, 1.3 km from the northern borders of the Project area and water has accumulated in certain areas due to leakages from the pipe. Local herders use this surface water for watering their livestock.

The Guzar substation is located to the northwest of the PV Project site. There are several existing OHTLs that are located along the western boundary and the northern boundary. Three residential properties belonging to one household are located in a compound adjacent to the Guzar substation.

The Project site belongs to two landowners (SWID and Kamashi State Forestry) and seven leaseholders. Detailed information regarding the land use has been provided in the Land ownership section (section 4.10).

4.3 Project components

4.3.1 Overview

Photovoltaic (PV) power uses solar panels to convert sunlight into electricity by converting solar radiation into DC electricity. PV inverters convert the direct current into alternating current, and the transformers (located in the Power Stations) will raise the voltage from Low Voltage (LV) to Medium Voltage (MV). Then, the energy generated will be conducted through an underground medium voltage (MV) network of 35 kV to the 35/220 kV Substation. An overview of the process is illustrated in Figure 23.

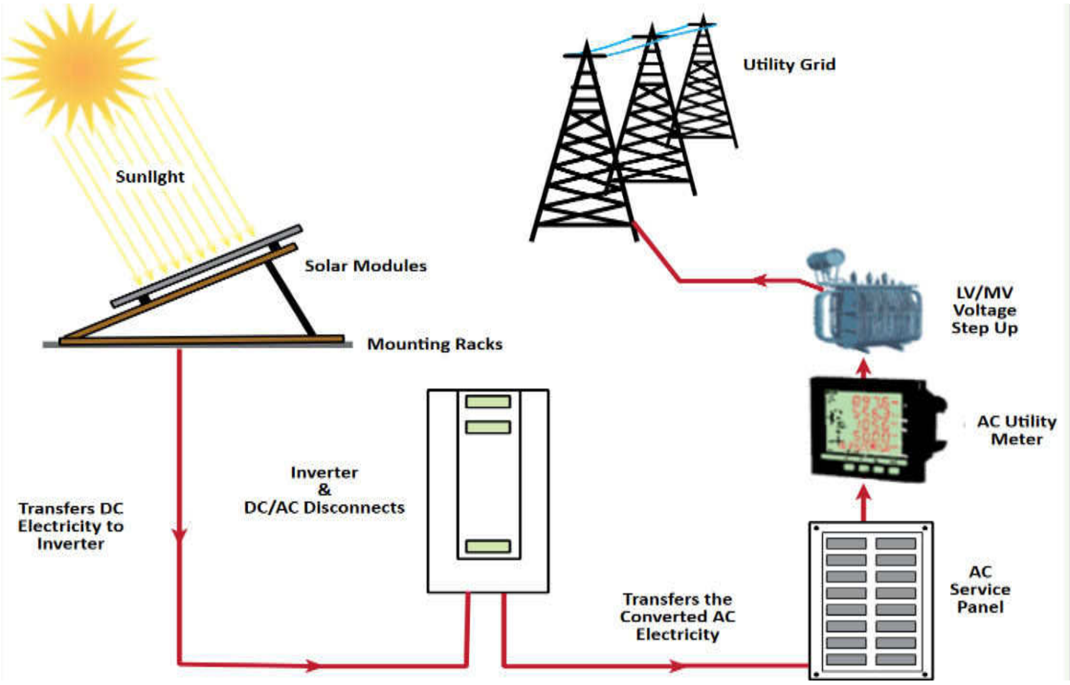


Figure 23: Overview of the PV process (compiled from IFC, 2015)

4.3.2 Primary component - PV Plant

A summary description of the primary components of the PV Plant is provided in Table 13.

Table 13: Description of PV Plant components

Project components	Summary description
PV panels and tracking system	Photovoltaic modules based on monocrystalline silicon technology are planned to be used. Modules will be divided into strings across the site (exact layout is not determined at this time).
Trackers	The modules will be set on a horizontal (north-south) axis to track the sun's position. The structure of the trackers will be installed employing direct driving to the ground whenever possible to a minimum depth determined by the geotechnical studies. Only if direct driving is not possible will a pre-drilling method be used.
LV electrical installation	The Low Voltage (LV) electrical installation is considered to refer to the downstream of the LV/MV transformers located in each of the Power Stations of the Solar PV Plant. It is the connection between PV modules on each string, between strings and string inverters and the inverters and the low voltage panel of the power station.
Invertors	Inverters convert the DC electricity produced by PV arrays into AC electricity compatible with utility grids. In addition, PV inverters often provide system protection and data communications.
Medium voltage (MV) (35kV) internal cabling	The MV power station comprises the MV cells and the power transformer, which is responsible for raising the output voltage of the inverters (800 V) to 35 kV. An MV ungrounded power network connects the MV power stations with the 35/220 kV Substation. The MV cables will run in trenches (directly buried or under a tube, depending on the section).
Control System	The PV Plant's monitoring and control system will be based on open products on the market and will include the SCADA and the Plant's control system, as well as all the necessary equipment to communicate with the rest of the Facility's systems

Figure 24: PV Panels (example)

Figure 25: MVS 8850-LV “Power Station (example)

4.3.3 Primary components - Battery Energy Storage System

Battery energy storage systems (BESS) can help address the intermittency of solar power and enable a power system to respond rapidly to large fluctuations in demand, making the grid more responsive and reducing the need to build back up power plants.

A BESS is a set of energy accumulators that, through an electrochemical process, can store electrical energy. The BESS consists mainly of batteries and a battery control and monitoring system (BMS). As depicted in Figure 26, the smallest, indivisible battery unit is called a cell, within which chemical reactions occur. Cells are connected within modules. These modules are equipped with voltage, current and temperature sensors to monitor the state of the cells. The modules, in turn, are connected inside cabinets, commonly called battery racks, until the desired system DC voltage level is reached at the design level. The battery racks also contain additional control and protection module. This BMS monitors the primary variables, such as voltages, currents and temperatures, at the level of the modules included in the rack and the cell. The stored power is transformed when needed by the power conversion system in the form of DC by the batteries into AC and vice versa by executing the appropriate current control to discharge and charge the batteries.

The BESS shall consist of [23]:

- A power conversion system (PCS) suitable for outdoor installation on a user-furnished concrete pad or the user-furnished box pad.
- Lithium-ion battery with a design life expectancy rating of 10 years under site operating conditions, suitable for outdoor installation, and a battery management system (BMS). Any replacement / modification required to meet Warranty requirement or design life shall be included in scope.
- An Energy Storage Unit as per type tested design and SOC
- DC Cables.
- Instrumentation and communication cables.
- Interfacing to meet statutory requirements and comply with Local Grid Code. All required hardware (AI & DI Cards).
- Fire Detection & protection system.
- HVAC system for BESS.

23 Guzar feasibility Study report, MOE, ADB, September 2022, Dornier Group.

The BESS, and associated equipment, shall be provided in self-contained National Electrical Manufacturers Association (NEMA) enclosure(s) rated for the site conditions with adequate ventilation and multiple layers of protection. Hazard and fire protection includes gas extinguishing method using NOVEC1230 and water or liquid Fire Fighting (Suppression) System) (FFS) along with the detection of the combustible gas like hydrogen or carbon monoxide and their exhaust system. The FFS include smoke detector, control panel, alarm device, exhaust pipe and bump head. It uses clean fire suppression gas to minimize the second loss.

The exact configuration of the BESS is not confirmed at this time, but is likely to include between four and five 40 ft containers. Thermal conditioning systems shall maintain ambient temperature within warranty requirements to minimise the chance of fire spread or thermal runaway. The BESS components and associated ancillary equipment shall have working space clearances required by local code, and electrical circuitry shall be within weatherproof enclosures marked with the environmental rating suitable for the type of environment in compliance with NFPA standards. All BESS containers will be locked to prevent unauthorised access. The BESS area will be segregated from the PV Site and shall have its own separate transformer for connection to the 220kV substation and a separate meter to measure the incoming and outgoing energy.

The areas between and around the equipment will be finished with gravel and kept free of vegetation or other material that could spread a fire. The lifetime of the BESS is approximately 10 years (>6000 cycles) after this time the batteries will be replaced.

A B C

Figure 26 : A: Typical battery system configuration - Battery cell (175×27×200mm), B: Battery Pack (contains 40 battery cells), C: Battery Cluster (contains 18 battery packs).

Typically, there are 15 battery clusters in one battery container as show in Figure 27 below.

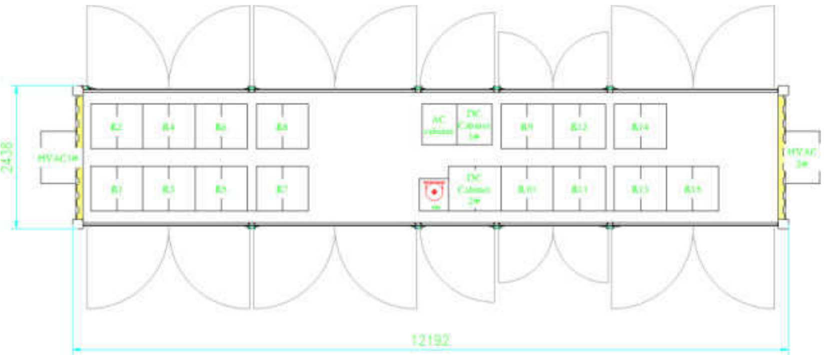


Figure 27: Typical BESS container

4.4 Substation and grid interconnection

The final location for the project substation and the option for connecting the Project substation to the existing "Guzar 500kV substation" are not confirmed at this time. The ESIA will provide further information about the options analysis and the final preferred solution.

The three options for the interconnection under consideration are:

1. Underground cable from the Project substation to the existing "Guzar 500kV substation.
2. Overhead transmission line (OHTL) from the Project substation to the existing "Guzar 500kV substation.
3. Combination of OHTL and underground cable from the Project substation to the existing "Guzar 500kV substation.

A new ROW (estimated to be 60m) is required to be established for an overhead grid connection and compensation and setback requirements as per national law will be required[24] A smaller protection area is required for an underground connection. Considerations for the final connection will include existing OHTL's and gas pipelines around the proposed site, current land use and biodiversity sensitivity.

The Project 35kV / 220kV air-insulated substation (AIS) will transform the generation and BESS output voltage levels to 220 kV through two power transformers for the PV generation and another for BESS and the associated electrical devices. The AIS switchyard shall have the following systems:

- 220 kV MV switchgear
- Substation control building
- Protection and control panels.
- AC/DC auxiliary power supply panels.
- DC battery banks (in a separate room) and chargers.
- Two MV/LV transformer for supplementary services.
- Telecom panels.
- Other rooms (offices, warehouse, toilets, etc.).
- Secure entrance and gate post building.
- Two firefighting tanks.
- Oil collector with treatment facilities for oiled effluents.

24 Livelihood impacts and compensation requirements will be outlined in the project Livelihood Restoration Plan (LRP).

All end-user work will all be within the existing substation boundary in the northwest corner of the site in an available area of approximately 30 x 90 meters. In addition, some work is required to extend the existing 500/220kV control room. Relay equipment will be installed in the existing 110kV control room. The interconnection system will incorporate protection, control and communication equipment meeting national and international standards.

4.5 Access road and internal roads

The Project site is located adjacent to the main road (M-39), between Shahrisabz and Guzar. The M-39 road is directly connected to Tashkent and the nearest logistics hubs are Karshi and Samarkand. The distance from the Project Site to Samarkand along the M-39 road is approximately 130km and to Karshi is approximately 50km. The town of Shahrisabaz is not considered as a logistics hub due to the fact that it is difficult for large vehicles to cross the Takhta Karacha Pass. The distance from the site to Shahrisabz by road is approximately 55km and about 10km to Guzar. In order to get to the Project site, a short access road from the main road (M-39) will be constructed. The smallest distance from the M-39 to the project site is about 1000m and is likely to be the main access point. The main road (M-39) is a double lane road between the two cities of Shahrisabz and Yakkabog, and a single lane between Yakkabog and Guzar. Viaducts and overhead lines/structures and some road damage (potholes, cut-outs, etc.) were observed on the road from Shahrisabz to the site.

Figure 28: Existing Road near the site

.

4.6 Water supply and treatment

The Project will require water for construction works (e.g. foundations, cement manufacture) and domestic use operations. Typical expected water use volumes are described in Table 14. Based on the projected water consumption volumes, the Project does not require long-term significant water storage or a dedicated water abstraction borehole. The following key assumptions regarding water use are highlighted:

- Cement batching will be undertaken on-site using a temporary facility dedicated for the Project.
- During operation, dry cleaning will be used for module cleaning,
- Septic tanks of 2,500 litres will treat sewage water during construction and operation.

Table 14: Typical water requirements

Phase	Task / Sub-task		Water quantity	Unit	Water quantity (m3/ construction phase)
Construction	Human consumption	Potable water	3	L / person / day	296
		Non-potable water / Sewage water	27	L / person / day	2,661
	Construction activities	Dust mitigation	3,000	L / km road / day	10,496
		Soil compaction	15	L / m road	263
		Concrete curing	250	L / m³ concrete	120
		Cleaning machinery	500	L / construction machine / day	1,916
Total Construction Phase (m3/construction phase)					**15,752**
Phase	**Task / Sub-task**		**Water quantity**	**Unit**	**Water quantity (m3/ construction phase)**
Operation (O&M)	Human consumption	Potable water	3	L / person / day	11
		Non-potable water / Sewage water	27	L / person / day	99
	PV Module cleaning	Dry cleaning	N/A	L / MWp / cleaning	N/A
Total Operation & Maintenance Phase (m3/construction phase)					**110**
Assumptions Construction Phase:					
Installed Peak Power (MWp)					300
Construction duration (months)					19
Number of daily workers (average)					270
Assumptions O&M Phase:					
Number of daily workers (average)					**29**

4.7 Associated facilities

Under EBRD PR1 and IFC PS1, associated facilities are defined as facilities that are not funded as part of the project and that would not have been constructed or expanded if the project did not exist and without which the project would not be viable. The scope of the project will include the design, supply, installation, testing and commissioning of all electrical equipment at the Project and the finally determined interconnection option. As such the interconnection and end-user works in the Guzar substation are considered part of the Project scope.

No associated facilities are identified in connection with this Project. No further consideration of associated facilities will be included in the ESIA.

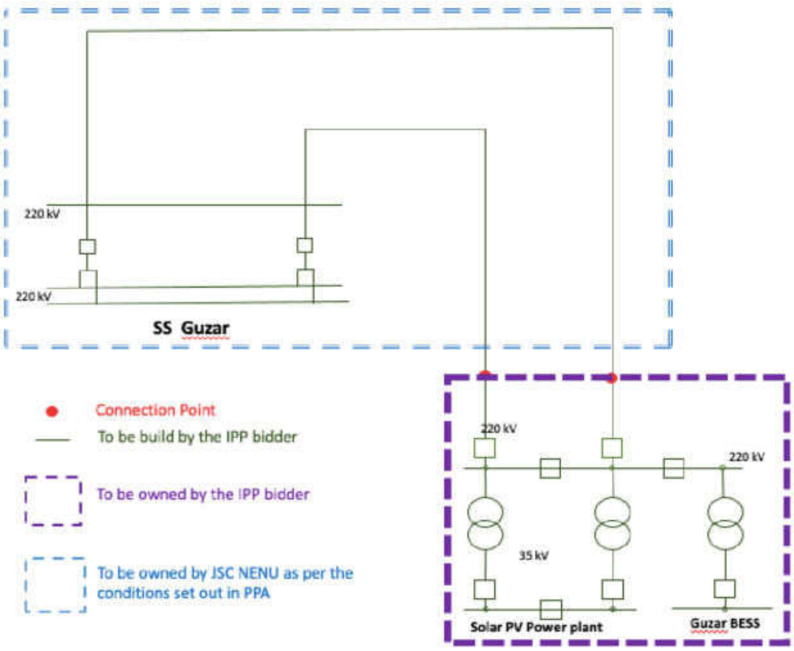


Figure 29: Construction and ownership arrangements

4.8 Supporting infrastructure

Supporting infrastructure for the construction and operation phase includes:

- Temporary batching plant
- on-site buildings (operational control centre, office, and welfare facilities)
- diesel generator for emergency power supply
- guard house
- laydown area (temporary)
- new drainage system
- emergency response (fire suppression, hydrants, water storage tanks).

The PV Plant will have a storage drainage system that allows the evacuation of rainwater outside the PV Plant and maintain drainage patterns within the site related to the ephemeral gulleys that cross the site. The drainage system will consist of a perimeter drainage network and another ditch-shaped internal drainage network on the side of the interior roads where uncontaminated surface water runoff is planned to be collected and routed to the existing drainage network. The area will be kept dry by designing the buildings to enable runoff away from structures and to ensure drainage for both subsurface and surface water to protect foundations. Drainage design will be sized for 1:100-year storms and consider future climate change predictions in sizing.

4.9 Security system

The Project security system will have the following features:

- perimeter security system
- access control system
- closed-circuit television (CCTV) system
- monitored and alarms in the access doors to the MV Power Stations or any other building of the installation.

The system itself will be responsible for automatically managing the alarm signals, first checking if it is an unwanted alarm. At the point of intrusion, the system will send a warning signal to the security centre and the person responsible will verify the alarm notify as necessary third-party security forces, firefighters, etc. During construction, security guards will employ additional security measures using permanent surveillance.

4.10 Land ownership status

There will be land acquisition required for the Project which will then be leased to the Private Partner through the SPV. The Project will involve permanent land take of 802 ha for the solar Project site. There is also land acquisition associated with the construction of the final proposed interconnection option and access to the right of way for construction purposes.

Land requirements for the access road (length as yet undetermined) are also required to be included in the land compensation calculations. Facilities such as construction and erection staging (unloading, site storage, workshop) areas, construction (workers) camps, spoil disposal sites, underground cable construction and access tracks established for the new 220 kV may also take up additional land temporarily during the construction stage. The temporary land take areas will be estimated during detailed engineering design.

According to the information provided by Cadastral Departments of Guzar and Kamashi districts, it was identified that on the Project site there are total of two landowners namely the Sericulture and Wool Industry Development (SWID) Committee in Guzar district site and Kamashi State Forestry Department in Kamashi district and a total of six lease holders which are indicated in the table below. Additionally, part of the territory is in reserve. Further surveys will be performed as part of the ESIA process to confirm the leaseholders and to gather information about land use. Indicative land ownership and land lease statuses are provided in Figure 30. For further information on land use, see section 5.4.

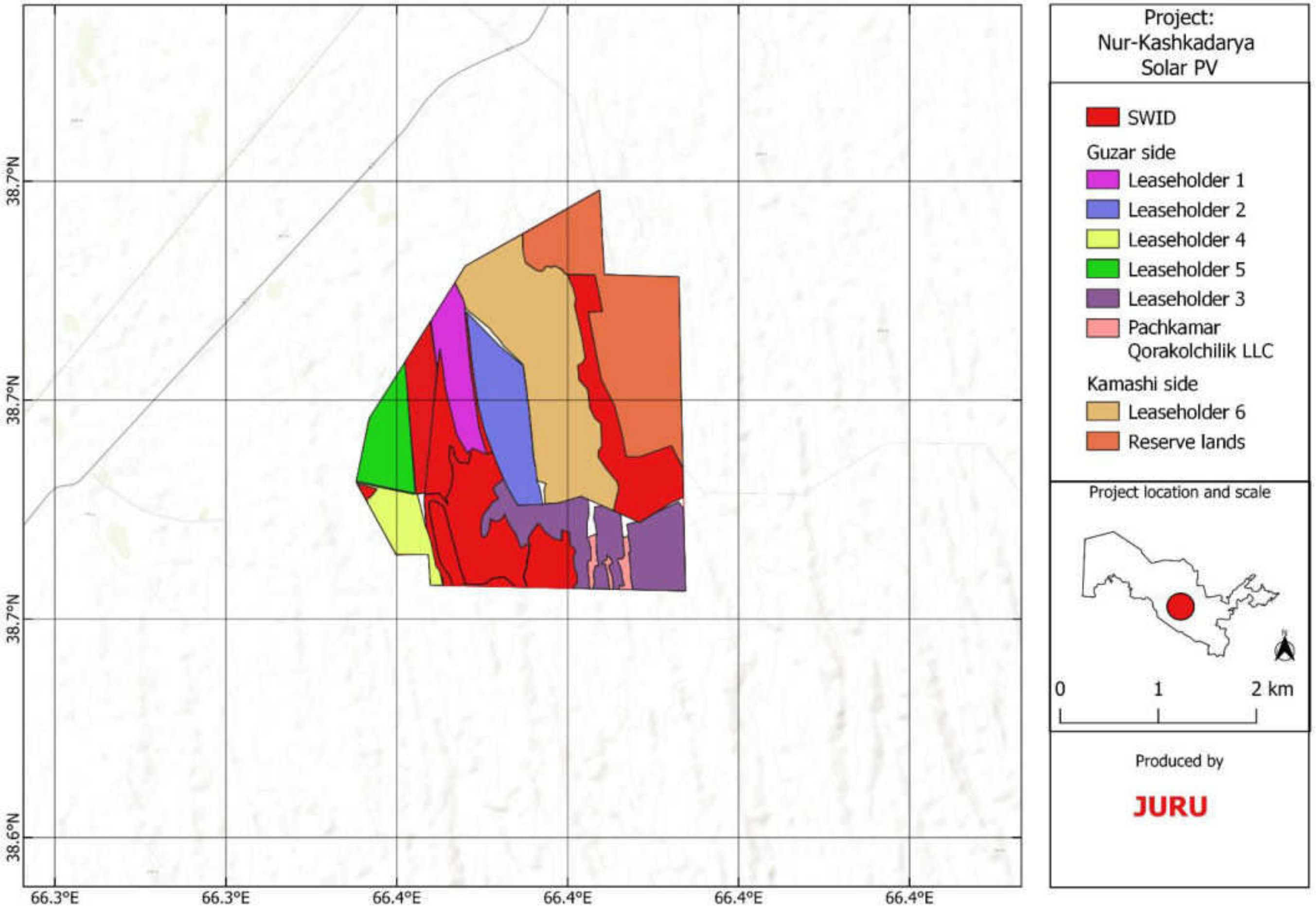


Figure 30 Land use[25]

25 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 30 and are still considered representative of the final layout.

4.11 Project accommodation

Some temporary construction accommodation is planned to be located at the Site. The location for this is undetermined, however it is likely to be alongside the substation area and within the planned site boundary. The accommodation will be designed and operated following the "Worker's Accommodation Processes and Standards": Guidance Note by IFC and EBRD[26]. In addition, the Project will allow Contractors to employ offsite accommodation provided it meets the guidance requirements of the above and adheres to the management and measures to be stipulated in the ESIA. These requirements will be defined in a Worker accommodation management plan by the Project Company and elaborated further by the EPC in their Accommodation Management Plan.

4.12 Project workforce

The PV Project construction phase will last up to 18 months. The following graph depicts the approximate number of workers at each stage of the construction (civil work, electrical, mechanical installation and commissioning).

The works will be led by an engineering, procurement and construction (EPC) contractor who will have a number of contracts with suppliers and sub-contractors for the following activities:

- Transportation contractor
- Civil works /balance of plant contractor
- Electrical contractor (PV site) (low voltage)
- Electrical contractor (medium and high voltage)
- Service suppliers (accommodation, geotechnical surveys, road construction, security, catering)

Figure 30 depicts the approximate number of workers (conservative) for each activity of the construction (civil work, electrical, mechanical installation, and commissioning). The total workforce required during the peak construction period could reach approximately 600 workers (made up of 40% skilled and 60% non-skilled). The EPC Contractor will aim for a breakdown of the workforce of 60% Uzbek workers and 40% expatriate workers.

26 Workers' accommodation: processes and standards (ebrd.com)

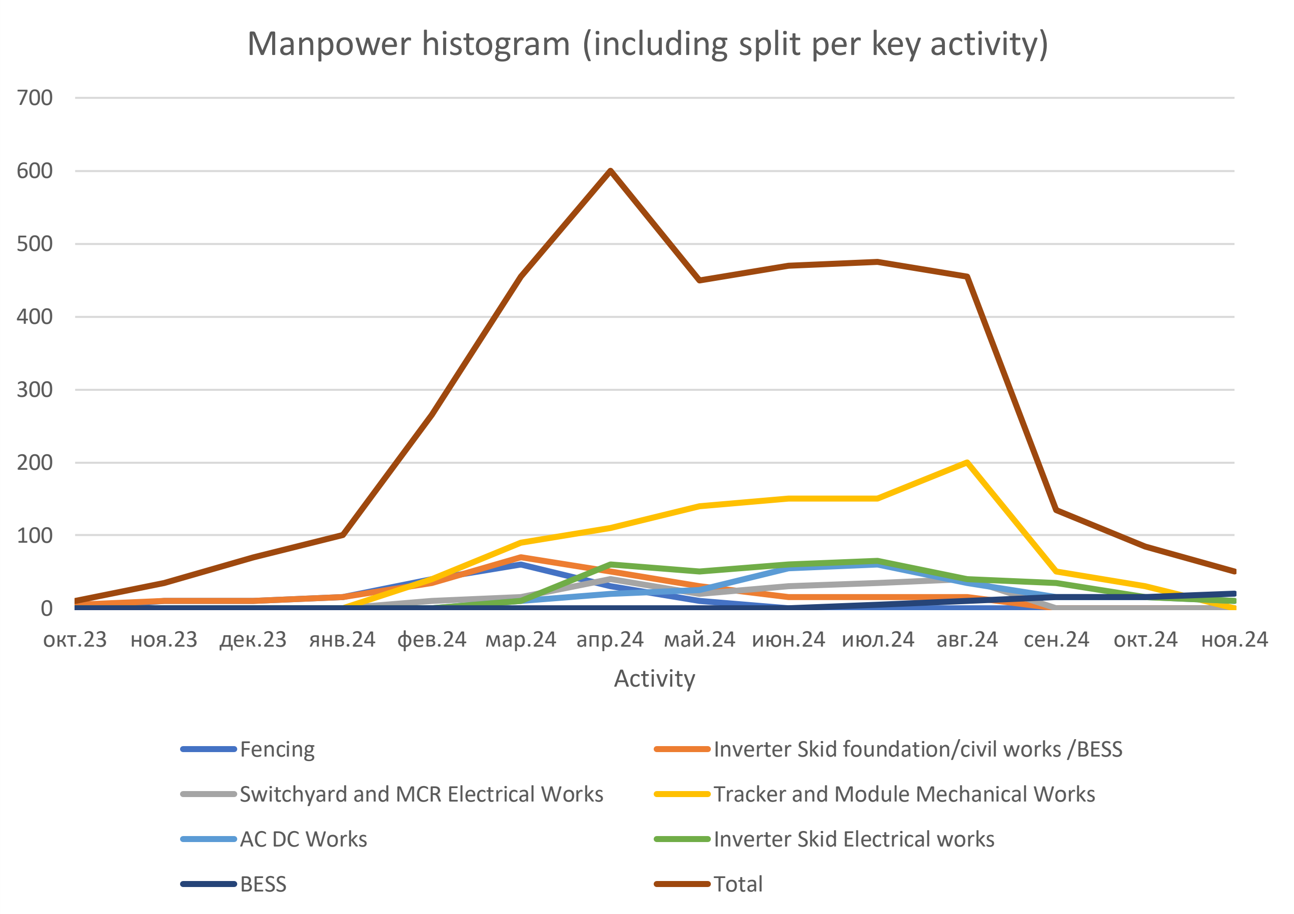


Figure 31: Typical manpower plan

4.13 Project development process

The Project development process includes the following stages:

- Site selection/screening (including developing the project concept) (finalised);
- Development (including assessment of technology and supplier options, contracting strategy, technical feasibility, E&S assessment, permitting and financing);
- Mobilisation (including detailed design, project implementation activities, local permit requirements, procurement and contracting);
- Construction (including mobilisation, site preparation, testing and commissioning);
- Operation; and
- Decommissioning.

An overview of the project activities for each phase is presented below.

- Mobilisation phase
  - Transportation of civil construction materials to site
  - Storing of materials
  - Recruitment of local workforce / services
  - Identification of local materials
- Site set up
  - Preparation of accommodation facility
  - Procurement

  - Site fencing
  - Site office construction
  - Construction phase - civil works
  - Secure site
  - Construct access road
- Site clearance / excavations
  - Foundation works (including delivery of cement)
  - Cabling excavations (220kV underground cable and internal low voltage cable trenches)
  - Transportation of abnormal loads materials to site
  - OHTL tower excavations (tower foundation)
  - Construction of operations building, stores and maintenance yard
  - Enabling work
- Construction phase - mechanical and electrical works
  - PV/BESS infrastructure installation
  - Excavation for placement of tracking system
  - Construction of SS
  - Installation of SS equipment
  - Installation of OHTL towers
  - Commissioning
- Operation phase
  - Operation of PV / BESS project.
  - Day to day maintenance
  - Periodic / planned maintenance
  - Monitoring
- Decommissioning phase (construction)
  - Reinstatement of excavated areas
  - Removal of construction materials
  - Rehabilitation of temporary storage and accommodation areas

4.14 Transportation of components

The PV and BESS system transportation from the factory will be a combination of sea and land freight. The main and alternative routing solutions are given hereunder:

- Direct Rail Route traditional route from China origins via Kazakhstan up to Tashkent hub or nearby logistics hubs of Karshi and Samarkand;
- Sea Rail Route via Turkey alternative route, by sea to Mersin port in Turkey and further by rail to destination; or
- Sea Rail Route via Baltic Sea alternative route, by deep sea to Baltic ports, and thereafter by rail to destination.

4.15 Maintenance

The PV and BESS Plant will be maintained and operated by skilled personnel, ensuring that the system is in optimal condition and that all parts are fully serviced and functional. Routine maintenance will likely be undertaken on the PV and BESS equipment twice a year. This typically consists of a major maintenance period and a minor maintenance period. The major maintenance is relatively non-intrusive and involves checking connections and inspections. This will encompass all PV and BESS equipment, including the fire system. Minor maintenance is typically a visual inspection and rectification of any accumulated noncritical defects.

4.16 Operation

During operation, all works on the Site will be controlled under safe work systems. This means all work is risk assessed to protect personnel and equipment. The PV Plant and BESS operation will be managed following the OESMP and a Health Safety Social Environmental Management System (HSSE-MS)

4.17 End-of-life disposal/decommissioning

With regards to the decommissioning of the PV and BESS components, the requirements will be determined during the ESIA for application during the procurement contract stage, these will set certain obligations in respect of PV components/battery return and disposal as well as other requirements.

4.18 Project schedule

The planned implementation schedule is set out in Table 15.

Table 15: Current project implementation schedule

Activity	Date
Scoping	September 2023
Consultation on national EIA	~~Late~~ October 2023- July 2024
Submission of national EIA	August 2024
Submission of draft ESIA	August 2024
Lender disclosure period	30-Sep-2024 till 15-November-2024
Finalise ESIA (including public consultation comments and ongoing studies)	27-Sep-2024
Finalise ROW and land agreements / compensation	November/December 2024
Financial close and Notice to Proceed	February 2025
Construction starts (interconnection works)	Full NTP in March 2025 although early works (site leveling and grading would start in November)
Construction Start (civil works) – main PV site and BESS	TBC
1st Module Assembly at Site	TBC
BESS container installation	TBC
Early generation	TBC
Commercial Operation Date	July 2025

Activity	Date
Expected Lifetime	25 years

5 Baseline Conditions

5.1 Area of influence

The direct area of influence (AOI) is defined by where Project impacts may be felt or observed, e.g. the zone of visual impact or the distance from the working area where noise or air quality impacts may be identified. The indirect AOI area is defined as where secondary or induced benefits or impacts may be realized, including employment impacts or impacts from an influx of workers. The ESIA will further define the direct and indirect AOI for each respective topic under consideration. For the scoping phase, a potential AOI of approximately 50 km from the Project site has been defined for scoping potential receptors and considering the potential avifauna risks and potential impacts and benefits on the wider communities in the AOI (Figure 32 below).

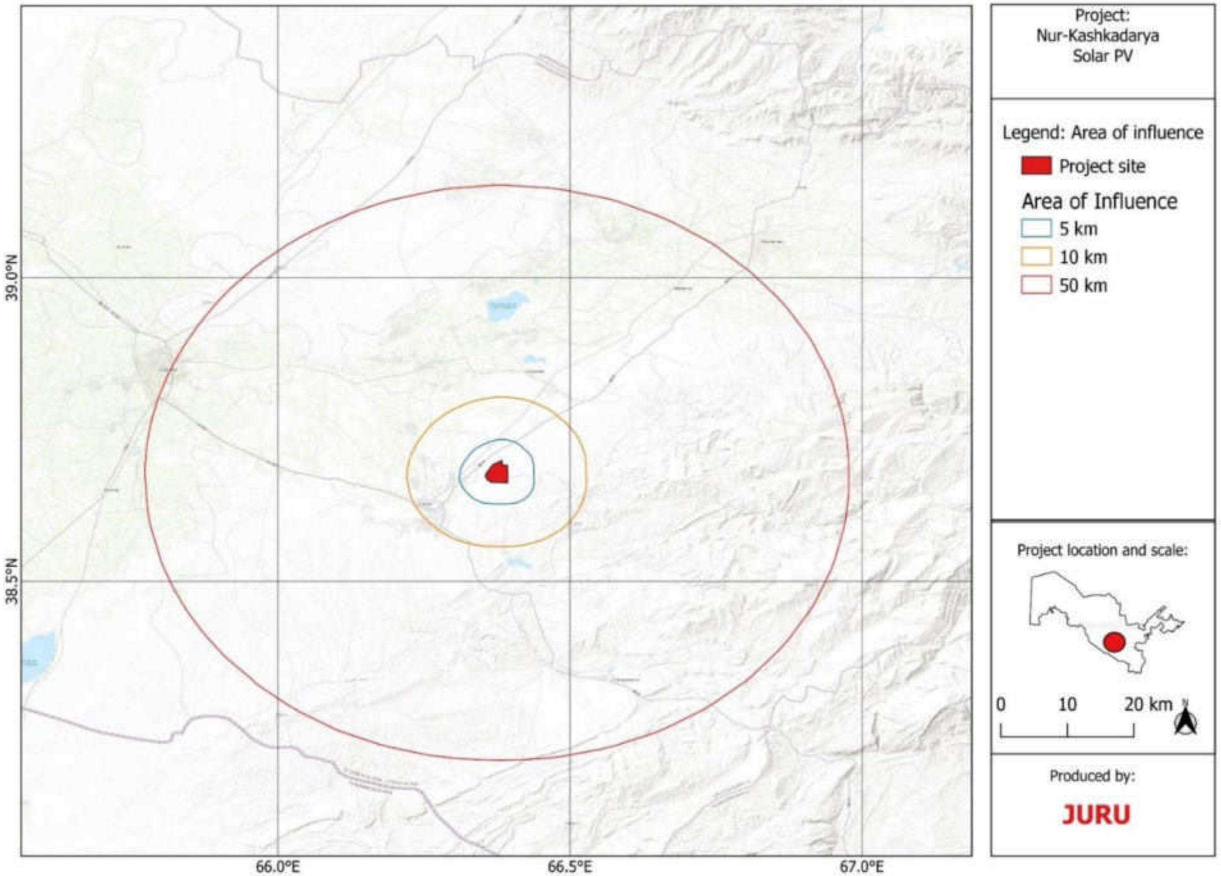


Figure 32: Project area and AOI[27]

27 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 32 and are still considered representative of the final layout.

5.2 Baseline data collection, scoping site visit and receptor mapping

Baseline data collection to inform the scoping assessment has been obtained from secondary source information, including:

1. Desk-based review of laws, policies, reports from the relevant governmental and non-governmental institutions and existing national and international publicly available information data from websites.
2. Review of the ADB Initial Environmental Examination Report – Uzbekistan: Solar Public-Private Partnership Investment Program (Tranche 2) prepared by Dornier Group from September 2022
3. Review of the ADB Social Safeguard Due Diligence Report for Utility-Scale Solar Photovoltaic PPP Project in Guzar District, Qashqadaryo Region of the Republic of Uzbekistan prepared by Dornier Group from July 2022
4. Review of the ADB Site Suitability Report (Pre-FS) for Utility-Scale Solar Photovoltaic PPP Project in Guzar District, Qashqadaryo Region of the Republic of Uzbekistan prepared by Suntrace from October 2021
5. Review of the Critical habitat assessment Report for Utility-Scale Solar Photovoltaic PPP Project in Guzar District, Qashqadaryo Region of the Republic of Uzbekistan prepared by Suntrace from January 2022
6. IBAT PS6 & ESS6 Report. Generated under licence 1781-26131 from the Integrated Biodiversity Assessment Tool on 17 January 2022 (GMT). www.ibat-alliance.org
7. Site visit to identify physical, biological and socio-economic features on the Project site and 10 km AoI, in particular, to identify any:
8. wells, boreholes on the Project site
9. infrastructure utilities such as network cables, OHTLs, gas pipelines, water pipelines etc
10. waste municipal landfills, disposals in the vicinity of the Project site
11. existing concrete batching plant in the area
12. existing OHTLs in the general Project area
13. soil contamination and air emissions sources (anthropogenic) cases.
14. Ground truth baseline locations for air, noise, soil and water baseline monitoring locations.
15. Understand the existing road quality
16. Check location of the structures on site indicated in the previous studies
17. Perform terrestrial scoping ecology walk-over in the area of the project site and underground cable
18. Consultations with local stakeholders and scoping meetings with local khokimiyats
19. Distribute Project leaflet with GRM information

The site visit was performed on 11 and 12 September by the following specialists from Juru:

- Viktoria Filatova (Senior environmental consultant)
- Anna Ten (Senior biodiversity expert)
- Oleg Khegay (Environmental consultant)
- Asad Nabiev (Social/resettlement consultant)

Figure 33 below presents the route track undertaken during the scoping site visit.

A further site visit was performed on 25th January with the representatives of Masdar. After the site visit to the Project area, meetings were held with the Cadastral departments of Guzar and Kamashi districts in terms of getting information about land owners and land users in the area.

For the ESIA report, a comprehensive desktop review of available information will be undertaken to develop further the biophysical baseline, which will include governmental and non-governmental reports, scientific publications, available aerial imagery and maps of the area.

Further fieldwork will be undertaken to obtain additional primary data regarding the biophysical characteristics and cultural heritage around the Project area through observations and surveys by qualified professionals. The scope of this work is defined in Annex C.

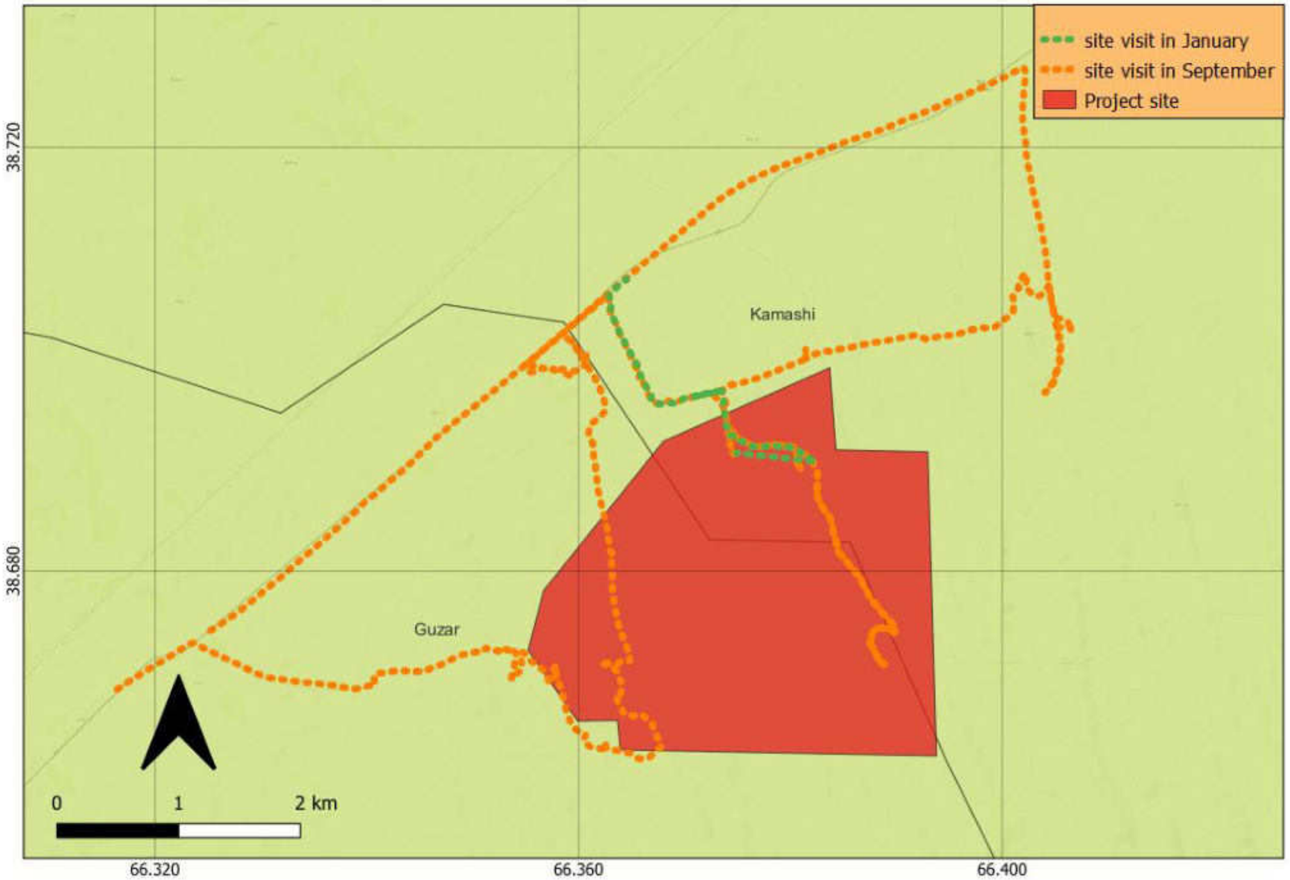


Figure 33: Track route of the scoping site visit (September 11-12)[28]

28 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 33 and are still considered representative of the final layout.

5.3 Physical overview

5.3.1 Climate

Uzbekistan has an arid and continental climate characterised by large variations in temperature within days and between seasons. Large parts of the country (79% by area) feature flat topography either in the form of semi-desert steppes or desert zones, including desert areas in the far west that have formed due to the drying of the Aral Sea.

The majority of the territory of Uzbekistan is attributed to a moderate climate zone. According to the criteria of the UNESCO world map of desertification and the UN convention to combat desertification, the country has an aridity index from 0.03 to 0.20 which categorises it as an arid region, subject to intensive desertification and droughts. The southern part of the country is located in the arid subtropical climate zone. In the west, the climate is sharply continental with dry, hot summers and relatively cold snowless winters and a moisture deficit, with a significant excess of evaporation over precipitation. The remaining south-eastern areas have a continental climate, including the area covering the largest cities of Tashkent and Samarkand, and contain high mountains forming part of the Tien-Shan and Gissar-Alai Ranges[29].

The Kashkadarya region is located in the basin of the Kashkadarya River on the western slope of the Pamir-Alai Mountains. In the mountainous east, the climate is moderately humid, while in the plains the weather is colder in winter and hotter and drier in summer. The project area in Guzar district exhibits a hot desert climate (Köppen classification BWh). High temperatures, scarce rainfall, abundant sunshine, and low humidity characterize the arid conditions.

Table 16: Average monthly statistics of air temperature and precipitation

	January	February	March	April	May	June	July	August	September	October	November	December
Avg. Temperature °C (°F)	2.9 °C (37.2) °F	4.7 °C (40.4) °F	10.3 °C (50.5) °F	15.5 °C (59.9) °F	21.3 °C (70.4) °F	26.1 °C (78.9) °F	27.7 °C (81.9) °F	26 °C (78.9) °F	21.2 °C (70.1) °F	15.1 °C (59.1) °F	8.5 °C (47.3) °F	4 °C (39.1) °F
Min. Temperature °C (°F)	-1.9 °C (28.5) °F	-0.9 °C (30.4) °F	4.1 °C (39.4) °F	8.5 °C (47.4) °F	13.4 °C (56.1) °F	17.3 °C (63.1) °F	19.2 °C (66.5) °F	17.8 °C (64) °F	13.4 °C (56) °F	8.3 °C (46.9) °F	3.1 °C (37.6) °F	-0.8 °C (30.5) °F
Max. Temperature °C (°F)	8.4 °C (47) °F	10.4 °C (50.6) °F	16 °C (60.9) °F	21.5 °C (70.7) °F	27.8 °C (82) °F	32.9 °C (91.3) °F	34.5 °C (94.1) °F	33.1 °C (91.5) °F	28.5 °C (83.3) °F	22.1 °C (71.8) °F	14.6 °C (58.3) °F	9.8 °C (49.6) °F
Precipitation / Rainfall mm (in)	54 (2)	68 (2)	77 (3)	65 (2)	33 (1)	5 (0)	1 (0)	0 (0)	2 (0)	11 (0)	36 (1)	43 (1)
Humidity(%)	63%	62%	60%	58%	46%	33%	34%	38%	42%	48%	59%	62%
Rainy days (d)	7	8	8	6	4	1	0	0	0	2	5	5
avg. Sun hours (hours)	6.9	7.5	8.6	10.4	12.4	13.3	13.1	12.3	11.2	9.7	7.8	7.1

29]https://climateknowledgeportal.worldbank.org/sites/default/files/2021-09/15838-Uzbekistan%20Country%20Profile-WEB.pdf

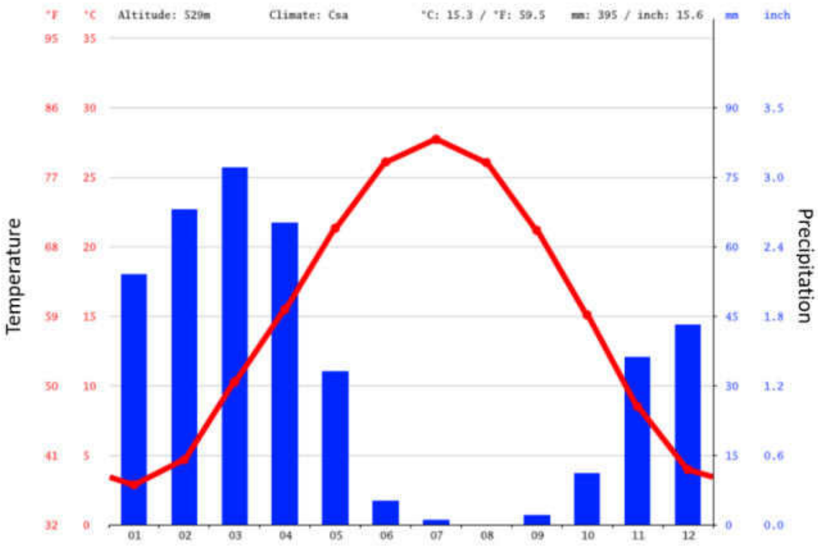


Figure 34: Average monthly precipitation and temperature variability at the project area

Precipitation in the Guzar district displays a marked seasonal variation, with the driest conditions occurring in August. During this month, rainfall levels plunge to an extreme low of 0 mm, indicative of an exceptionally arid climate. Conversely, the wettest period transpires in March, when precipitation averages 77 mm, accounting for the majority of the annual rainfall.

5.3.2 Climate projections

Uzbekistan is exposed to a range of weather-related extreme events, including dust storms, mudflows, floods, drought, and avalanches and is significantly threatened by climate change, with serious risks already in evidence.[30] Historical trends indicate an increasing average temperature of 0.13°C per decade between 1901 and 2013, rising more steeply (0.51 °C per decade since 1983) with a temperature increase greatest at low altitudes and more prevalent during the winter months. The average number of days with a maximum of 40°C in the central part of the Kyzylkum desert has increased from 10 days in the 1950s to more than 20 days in 2016[31] [32]. Uzbekistan climate projections, based on the Projections of the World Bank Climate Change Knowledge Portal, are as follows[33]:

- Increase in annual mean temperature of 1.3 to 2.1°C by 2030, 1.8 to 3.3°C by 2050, and 2.0 to 5.4°C by 2085.
- Increase in annual maximum temperature of 2.1 to 6.3°C and increase in minimum temperature of 2.2 to 5.6°C by 2085.
- There will be an increase in long-lasting heat waves from three to nine days by 2030, between four and 17 days by 2050, and between six and 43 days by 2085.
- Anticipated change in total annual precipitation ranges from a decrease of three per cent to an increase of 12 % by 2030 and a decrease of 6 % to an increase of 18 % by 2085, with most projections showing an increase.

30 World Bank's Overview of Climate Change Activities in Uzbekistan (October 2013) https://openknowledge.worldbank.org/bitstream/handle/10986/17550/855660WP0Uzbek0Box382161B00PUBLIC0.pdf?sequence=1andisAllowed=y

31 Climate Service Center Germany. 2016. Climate-Fact-Sheet: Uzbekistan.

32 http://www.un-gsp.org/sites/default/files/documents/tnc_of_uzbekistan_under_unfccc_english_n.pdf The 3rd National communication of the Republic of Uzbekistan under the UN Framework Convention on Climate Change. Tashkent 2016.

33 https://www.climatelinks.org/resources/climate-risk-profile-uzbekistan Uzbekistan Climate risk profile, (17/02/2022)

- Likely increased precipitation between November and April, with precipitation in other months remaining stable or decreasing slightly.
- Dry spells are expected to grow longer by up to four days by 2085.
- An overall increase in arid conditions due to changing precipitation patterns and increased temperatures.
- Increased in the intensity of heavy rain events by 3 to 11 % and frequency by 7 to 36 % by 2030, and in intensity by 7 to 23 % and frequency by 12 to 74 % by 2085. [34,35]
- The number of hot days in Uzbekistan to increase by 28.6 days by 2040-2059, under an RCP 8.5 scenario.
- The number of tropical nights (minimum temperature above 20°C) is projected to increase over 31 days by 2040-2059, under an RCP 8.5 scenario[36].

5.3.3 Air quality

There are no anthropogenic sources of air pollution near the Project site except transport emissions from the adjacent main road. Most soil conditions of the territory are sandy soil, making the site prone to dust generation. During the scoping site visit, no major generation of dust which significantly decreased the visibility was observed.

5.3.4 Soils and topography

The Project area is situated in the northern part of the Kashkadarya valley, near the foothills of the Kitab-Shakhrisabz basin. The specific study area within Guzar district consists of a relatively flat plain at elevations between 490-530m (Figure 35). The plain itself comprises flat or gently sloping terrain, with little relief or prominent elevations. Overall, the study area features a flat, low-lying plain in contrast to the steeper topography nearby. The variety of soil is presented in Figure 36.

34 Climate Service Center Germany. 2016 (https://www.climate-service-center.de/products_and_publications/fact_sheets/climate_fact_sheets/index.php.en)

35 Tashkent. 2016. Third National Communication of the Republic of Uzbekistan Under the UN Framework Convention on Climate Change.

36 RCP 8.5 is a high emission scenario also referred to as «business as usual».

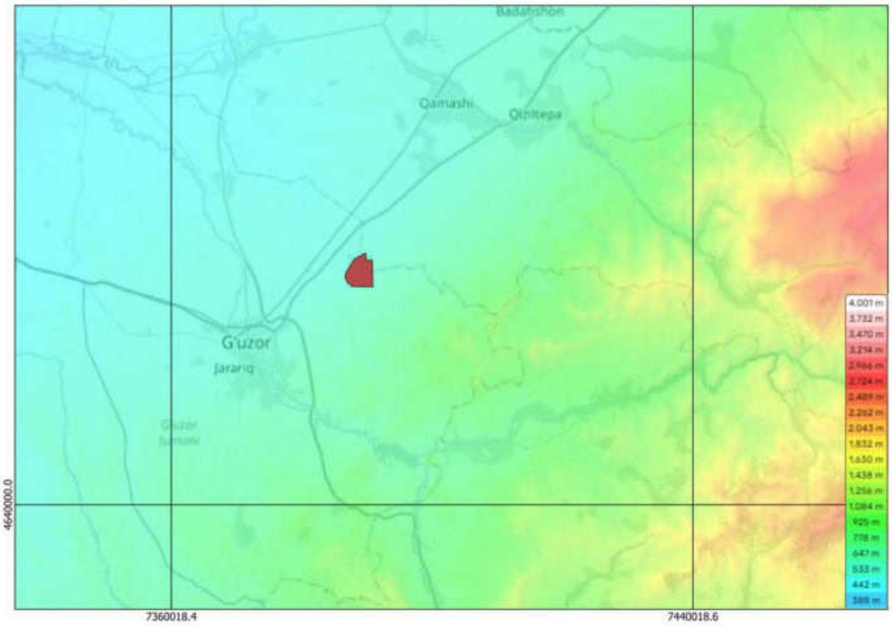


Figure 35: Topographic map and surrounding area

The foothills in Guzar district are covered mostly by loess loam soils. On the undulating plains, the loess loam overlies deposits of gravel. Overall, the soils in Guzar are predominantly loamy, overlying layers of sand and gravel.

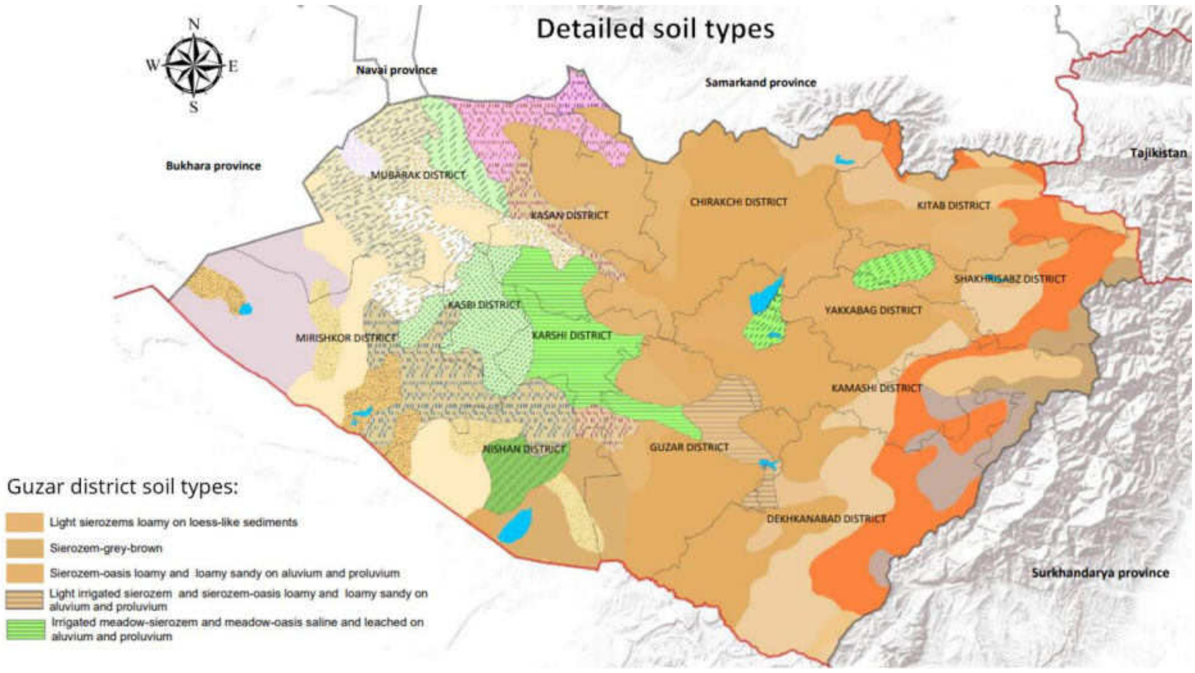


Figure 36 Soil profile for the project area

Source: Extract from *Geodatabase and Diagno abase and Diagnostic Atlas: Kashkadarya Province, Uzbekistan Zafar Gafurov, Sarvarbek Eltazarov, Bekzod Akramov, Kakhramon Djumaboev, Oyture Anarbekov and Umida Solieva.*

5.3.5 Geology and seismicity

The geological structure of the Kashkadarya District is varied across the territory. The mountainous part, having risen during the Hercynian Mountain building epoch in the Paleozoic era, is composed of crystalline rocks such as shale, limestone, marble and granite. The flat plains of the district occupy the far eastern part of the Turan tectonic plate. This area is blanketed by sedimentary deposits such as sand, clay, and conglomerates.

Uzbekistan's territory is related to the active tectonic structure of the lithosphere of Western Tian-Shan, the development of which results in the formation of deep fault networks.[37] A considerable part of the territory of Uzbekistan belongs to the zone of seismic intensity VII (very strong) (MSK scale)[38]. Several seismically active zones exist in Uzbekistan, which align with significant tectonic deformation strike lines. These internal zones are capable of producing earthquakes with magnitudes of M ≥ 5. The Kashkadarya region faces a moderate risk from earthquakes due to its proximity to seismically active fault zones in Central Asia. The regional seismicity stems largely from the active collision between the Indian and Eurasian plates. This creates fault networks that accumulate strain before periodically rupturing to generate earthquakes. Major active faults near Kashkadarya include the Gissar-Kokshaal fault zone to the northeast and the Karakul-Chardara fault to the south. These structures and others in the area are capable of producing earthquakes with estimated magnitudes of 6 to 7.

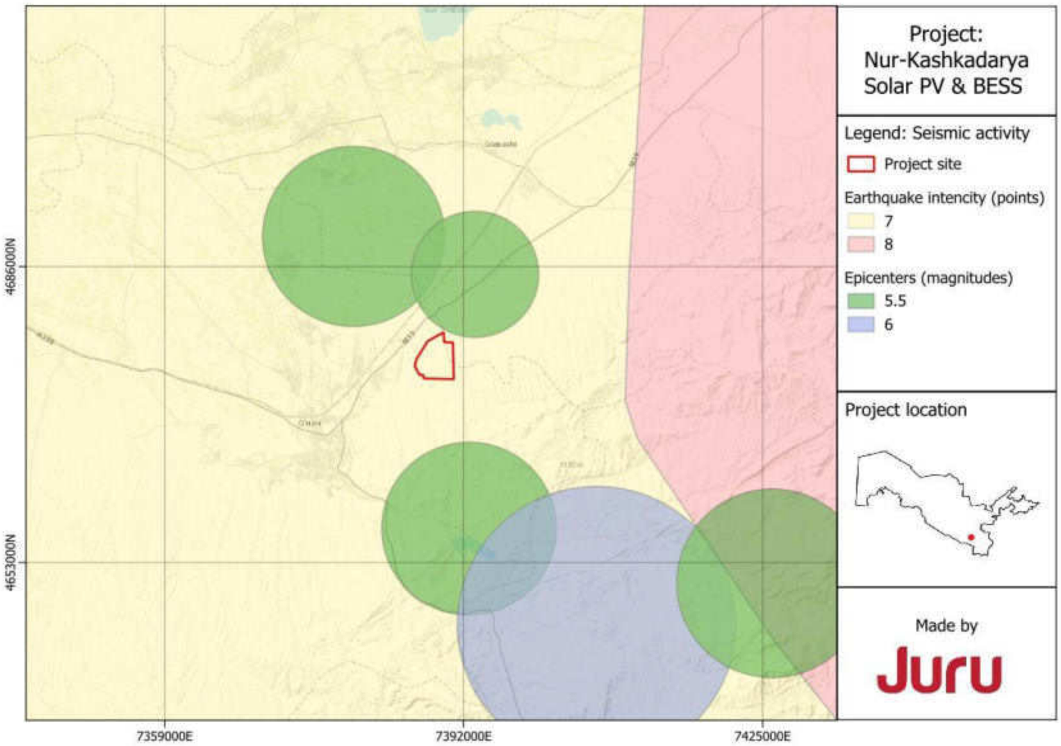


Figure 37: Seismic zoning near the project area[39]

37 Mavlyanova N. et al (2004): Seismic code of Uzbekistan. Proceeding's 13th World Conference on Earthquake Engineering Vancouver, B.C., Canada. August 1-6, 2004. Paper No. 1611

38 Medvedev-Sponheuer-Karnik (MSK) scale. This is similar to the Modified Mercalli Intensity scale used in the United States and Europe.

39 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 32 and are still considered representative of the final layout.

Source: "Seismological Atlas of Uzbekistan." G.O. Mavlonov Institute of Seismology, 2021.

5.3.6 Hydrology (surface water and groundwater)

In the Kashkadarya region, rivers are of the greatest importance among surface waters. There are 33 rivers in the area with a length of more than 20 km[40]. No signs of permanent surface water features were observed within the site during the site visit. No wells or boreholes were identified. Offsite there are pools of water (refer to Figure 18) generated by leaks from the adjacent water pipeline. On site there are several ephemeral gulleys (small channels) (refer back to Figure 16) crossing a site, they are dry and can concentrated with water in rain. Dominant orientation of gullies is south-southeast to north-northwest directions. The Project site is located approximately 12 km from the Guzardaryo river.

The Kashkadarya region is rich in groundwater. Groundwater found in quaternary sediments is suitable for drinking and is used by the population for household needs and in animal husbandry. Additionally, healing mineral thermal waters have been discovered in Cretaceous and Paleogene deposits. Several reservoirs have been constructed in the Kashkadarya region, and their waters are used for irrigation. These reservoirs include Chimkurgan on the Kashkadarya River, Pachkamarskoye on the Guzardarya River, and Talimardzhanskoye, which was built along the Karshi main canal.

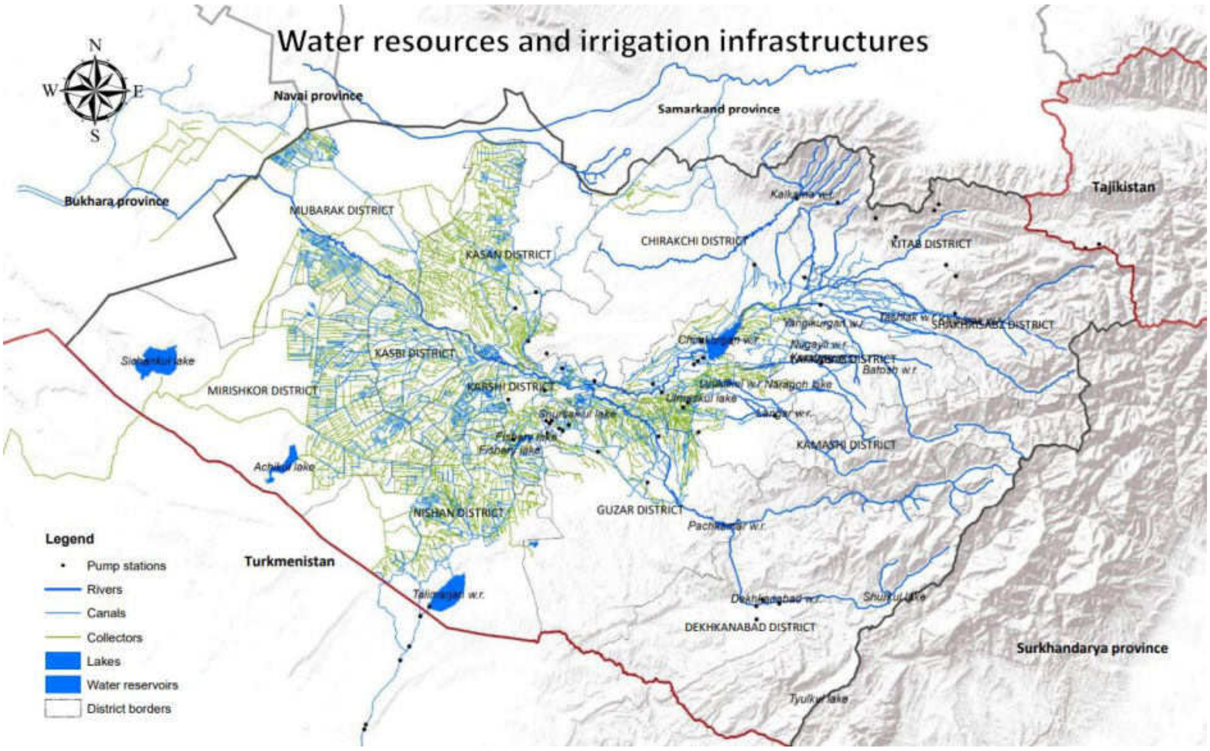


Figure 38: Water resources of Kashkadarya region

5.3.7 Physical Infrastructure

__Waste__

40 Ochilova B.A. Geographical location of Kashkadaya

Solid waste collection services are available in major cities like Karshi but more limited in rural areas. Recent reforms in Uzbekistan's waste management system have led to the establishment of 13 state-owned enterprises under the managment of Toza Hudud with 174 regional and city branches across the country. As part of the Toza Hudud network, a new Kashkadarya Service Center site is planned for development in the provincial capital of Karshi, which will aim to improve waste management infrastructure and operations in the region. No additional information and waste disposal location is available at this time.

Roads

There is the main road M39 situated close to the project site, with the total length 725km. The M39 is a major national highway that connects the cities of Termez, Guzar, Karshi, and Shakhrisabz. There are also unpaved roads along the Project site.

Airfields

The closest airfields to the Project site are: The Karshi Airport (IATA: KSQ, ICAO: UTSK). No public airfields were identified within 10km of the project site.

Communication network

Mobile networks in Uzbekistan are available only in villages, and their working zones are not distributed in unpopulated areas such as the Project Site. It is possible to use 3G and 4G in most parts of the Guzar district; Figure 39 below represents the coverage areas for Uzmobile as the provider with the best coverage in the AOI.

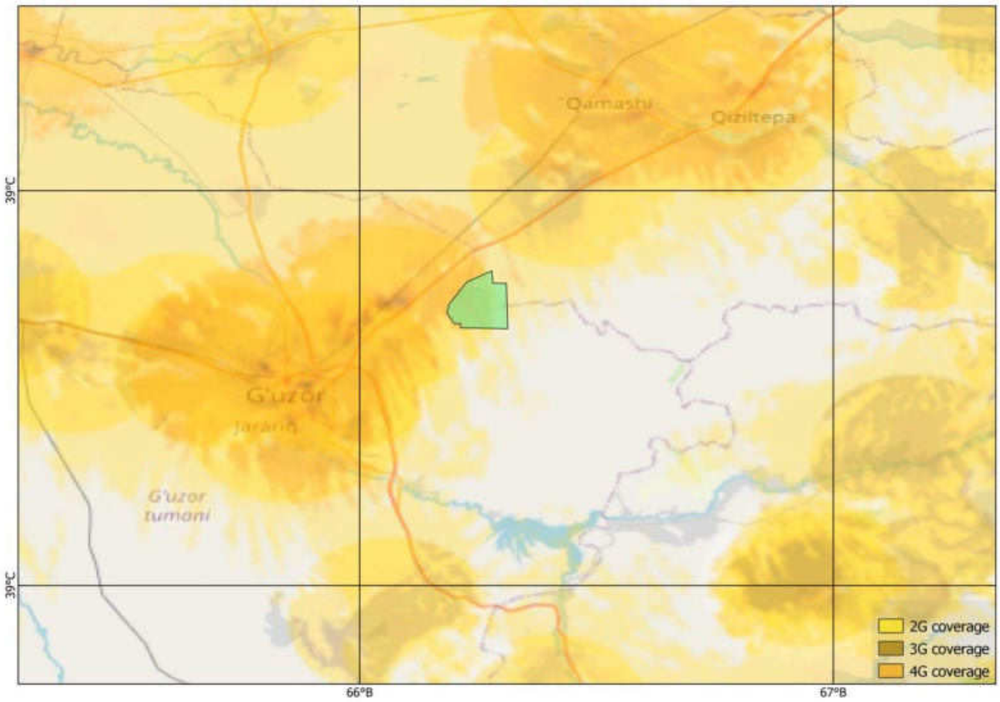


Figure 39: Mobile network coverage in the Guzar region
(Source: https://beeline.uz/ru/coverage-and-offices/)

5.4 Land use

Scoping site visit observations revealed that the majority of the Project site is designated as farmland, with a small part allocated for grazing and livestock activities. The land is not irrigated and is farmed using rainfall only. People consulted during the scoping site visit stated that there had been a severe drought in 2023 and many of the crops did not survive, therefore the land was predominantly being used for grazing livestock at the time of the scoping report. Previously land has been used to grow wheat, chickpeas, rye and livestock feed.

A total of ten official land users, including; two land owners Guzar district department of SWID committee and Kamashi district department of SWID committee and eight lease holders were identified as being impacted by the Project, three from Kamashi district and seven from Guzar district.

See the Land ownership section (section 4.11) for further details.

During the scoping site visits, many of the households located just outside the Project site (particularly to the northwest) stated that they informally (i.e., without a lease agreement) use the Project land for grazing their livestock seasonally. The total number of households that use of the land will be determined during future site visits.

Some local community members have constructed temporary structures such as herder camps and stables that are only used during certain months of the year. These are outside of the Project footprint, except for five temporary herder camps and five stables for livestock which the Project will impact. The livelihood impact assessment and the livelihood restoration plan will consider the impact on those structures further as part of the ESIA process.

The following figures are photographs taken during the scoping site visit.

Figure 40: Photo of the herder's hut within Project footprint (from the outside)

Figure 41 Photo of the herder's hut within Project footprint (interior)

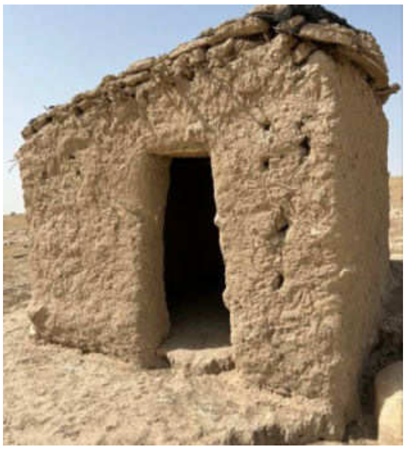


Figure 42: Structure just outside the Project boundary (south-east)

Figure 43: Example of the household (to the northwest of the Project boundary)

There are a large number of existing overhead transmission lines (OHTL), to the west of the Project site. A letter has been sent to NEGU to determine if it owns all of the OHTLs and what sanitary protection zone (buffer zone) is required for the different voltages of OHTLs. These transmission lines link to the Guzar substation. One existing transmission line will cross through the Project site, and a buffer zone has been identified between the OHTL and the solar panels.

Figure 44: Existing OHTLs

Figure 45: Existing OHTLs outside the Project boundaries

5.5 Socio-economic overview

5.5.1 General characteristics

Administratively the Project site is located in two districts -the Kamashi and Guzar Districts of the Kashkadarya region (refer to Figure 2). It is located 8 km northeast of Guzar city. The site is accessible from the M39 roadway. The site has an undulating topography; there are no permanent surface water bodies or rivers on the site.

5.5.2 Description of region and district

Kashkadarya region

Kashkadarya region is situated in the south-eastern part of Uzbekistan in the basin of the river Kashkadarya and on the western slopes of the Pamir-Alay mountains. It borders the Samarkand region, Bukhara region, and Surkhandarya region as well as the Republics of Tajikistan and Turkmenistan. The territory of Kashkadarya region is 28,570 km². As of 1 January 2023, the permanent population in Kashkadarya region was 3,482,300 people, among them, the urban population totalled 1,491,600 people (42.8% of the population) and the rural totalled 1,990,700 people (57.2% of the population). The administrative centre of the region is Karshi city, located in the southwestern part of Kashkadarya region.

The gross regional product of Kashkadaraya region is made up of agricultural, forestry and fishery sectors 35.9%, industry 18.8%, construction 6.8%, retail, consumer and food services 7.1%, transportation and storage, information and communication 5.5% and other sectors 25.9%.

As of January 2023, there were 8,869 active enterprises and organizations in the Kashkadarya region. The largest number of enterprises were in the trade (3,189) and agriculture, forestry and fishery (1,850) sectors of the region.

At the end of 2022, there were 557 pre-school educational organisations in the region and for the academic year 2022-2023 there were 1,220 general education institutions, and nine higher educational institutions in the region. In 2022, there were a total of 91 hospitals and 543 polyclinics that provided general medical services to the population.

The average monthly salary of a person living in Kashkadarya region amounted to 3,199,659 UZS (or approximately USD $263) as per statistics provided for the second quarter of 2023.

Guzar District

According to the information provided by the Statistics Agency under the President of the Republic of Uzbekistan, Guzar District comprises an area of 2,660 km² and consist of one city, five towns and 80 rural settlements. The population of Guzar District as of 1 January 2023 was 217,300 people, among them, the urban population totalled 51,000 people (23.5% of the population) and the rural totalled 166,300 people (76.5% of the population). The ethnic composition of which are predominantly Uzbeks (93.2%), with the remainder of the district coming from the following ethnic groups Tajiks (5.5%), Russians (0.2%) and representatives of other nationalities (1.1%).

Kamashi District

The Kamashi District comprises an area of 2,660 km² and consist of one city, five towns and 115 rural settlements as per the information provided by the Statistics Agency under the President of the Republic of Uzbekistan. The population of Kamashi District as of 1 January 2023 was 286,800 people, among them, the urban population totalled 68,100 people (23.7% of the population) and the rural totalled 218,700 people (76.3% of the population). The proportion of men and women is very similar. Men comprised to 145,800 people (50.8%) while the number of women reached 141,000 people (49.2%).

5.5.3 Access to services and connectivity

Guzar District

There are 188 pre-school educational organizations, 78 schools, and 87 libraries in Guzar district. With regards to healthcare, there are 45 healthcare organizations, among them 25 are private organizations. The district is equipped with natural gas, potable water and electricity supply, however only a small portion of the population has access to these services. A total of 30,2% of the population of the district have access to gas supply (liquefied gas is 13,3%) while 26,1% of the population has access to potable water supply. The total length of the district's transmission lines is 2,181 km and the total number of substations is 12. Roads of local, international and state importance also cross the district.

Kamashi District

In Kamashi district there are 24 kindergartens, 31 pre-school educational organisations, 95 schools, and one public library. With regards to healthcare, there are two hospitals and 28 polyclinics that provided general medical services. Also, there are four cultural centres and one amusement park.

5.5.4 Economy, employment and livelihoods

Guzar District

The economy of the Guzar district is specialized in the fields of industry, animal husbandry and building materials. There are 13 joint ventures, 350 small industrial enterprises and one large industrial enterprise ("Gissarneftegaz" LLC JV, "Shurtangaz Chemical Complex" LLC," Kitob Ip Yigiruv"JSC) in the district.

The number of economically active people in Guzar district comprised 83,100 people in 2022, among them 75,400 people are employed and the rest 7,700 are unemployed. Women make up a higher proportion of the unemployed, with 4,200 women registered as unemployed, compared to 3,500 men. The unemployment rate of Guzar district amounts to 9,3%. The number of people who migrated to work abroad in 2022 was 7,797 people.

Kamashi District

There are 3,017 enterprises in Kamashi District, of which 1,136 are in agriculture, 368 in industry, 171 in construction, 713 in trade, the remaining 629 enterprises belong to other sectors. The number of small business enterprises is 2,705, among them active enterprises are 2,594. The main sectors in which the small business operate are industry (63.2%), agriculture (99.7%), investment (64.3%), construction (100%), and trade (98.9%).

The number of economically active people in Kamashi district in 2021 comprises 160,200 people, however only 94,500 of people were reported as being employed.

5.5.5 Community profile

There are four communities namely Yangiabad, Khalqabad, Batosh, and Aynakul located relatively close to the Project site. Yangiabad, Khalqabad, Batosh belong to Guzar district, while Aynarkul is the part of Kamashi district Location of these communities are indicated in Figure 4.

Yangiabad community is located approximately 1.5 km away from the western boundaries of the Project site. The total area of Yangiabad community is 217.5 ha, and with a total of 1,151 households. According to the information provided by the khokimiyat, the total number of residents is 6,510. The ethnic groups located in the community include Uzbeks (6,471), Tajiks (38) and Azerbaijanis (one). Regarding vulnerable groups, in 2022 there were 109 disabled community members, 16 households without breadwinners, two single elderly people, 630 low-income households and 77 households with large numbers of dependent children (4 or more) as provided by the khokimiyat. There are two local schools with 789 pupils and seven pre-school educational facilities with 265 pupils.

Khalqabad community is located approximately 2.6 km west of the Project site. The total area of Khalqabad community is 184 ha, and the total number of households is 934. According to the information provided by the khokimiyat, the total number of residents is 4,936. The ethnic breakdown of the community is Uzbeks (4,858), Tajiks (77) and Russians (one). Regarding the vulnerable groups, in 2022 there were 106 disabled community members, 16 households without breadwinners, two single elderly people, two low-income households and 77 households with large numbers of dependent children, as provided by the khokimiyat. There are two local schools with 927 pupils and six pre-school educational facilities with 275 pupils.

Batosh community is located approximately 5km west of the Project site. The total area of Batosh community is 148.4 ha, and the total number of households is1,022. According to the information provided by the khokimiyat, the total number of residents is 4,907. The ethnic breakdown of the community is Uzbeks (4,907), Tajiks (three) and Turkmens (one). Regarding the vulnerable groups, in 2023 there were 81 disabled community members, 10 households without breadwinners, one single elderly person, eight low-income households, 11 households with large numbers of dependent children and one household with single parent as provided by the khokimiyat. There are three pre-school educational facilities with 240 pupils and one local school with 935 pupils.

Aynakul community is located approximately 5.7 km away from the north-eastern part of the Project site. The total area of Aynakul community is 101.19 ha, and the total number of households amounts to 1,102. According to the information provided by the khokimiyat, the total number of residents is 5,677. The ethnic breakdown of the community is Uzbeks (5,675), Tajiks (two). Regarding the vulnerable groups, in 2022 there were 112 disabled community members, nine households without breadwinners, two single elderly people, five low-income households, 86 households with large numbers of dependent children and 18 households with single parent as provided by the khokimiyat. There is one local school with 1,412 pupils.

5.5.6 Access to services

This information has only been collected by the khokimiyats at a district level. Community level information will be collected during the planned socioeconomic surveys for the ESIA.

5.5.7 Transport

The M39 road is located approximately 1.3 km to the north part of Project site. This road is used by residents of both districts Kamashi and Guzar. The M39 road is directly connected to Tashkent and the nearest logistics hubs are Karshi and Samarkand. The distance from the Project Site to Samarkand along the M 39 road is approximately 130 km and to Karshi is approximately 50 km. The town of Shahrisabaz is not considered as a logistics hub due to the fact that it is difficult for large vehicles to cross the Takhta Karacha Pass. The distance from the site to Shahrisabz by road is approximately 55 km and about 10 km to Guzar. In order to get to the Project site, a short access road from the M39 will be constructed. The smallest distance from the M39 to the project site is about 300m. The M39 is a double lane road between the two cities of Shahrisabz and Yakkabog, and a single lane between Yakkabog and Guzar. Viaducts and overhead lines/structures and some road damage (potholes, cut-outs, etc.) were observed on the road from Shahrisabz to the site.

5.5.8 Cultural heritage, tourism and recreation

There are no recreation facilities or cultural features of national and international importance in the AoI. Significant objects of cultural heritage or international reputation are located in Bukhara and Samarkand regions.

Consultation with the Academy of Sciences of the Republic of Uzbekistan, National Archaeological Center[41] provided information on the general area. In the area of the Project or its vicinity, particularly in the central part of Kashkadarya and Karshi area, the initial archaeological exploration of archaeological sites was carried out by S.K. Kabanov in 1946-1948. In 1946, as part of the Amu darya Expedition, A.I. Terenokhin and L.I. Albaum conducted the search and examination of archaeological sites in the Qarshi region. During that period, researchers identified approximately 30 archaeological sites of three different types in the region. During this period, many archaeological sites from the pre-Arab invasion period were discovered in a small territory.

The overall study of the Karshi area, re-registration of archaeological sites, mapping, and documentation work were undertaken under the leadership of R.X. Suleymonov, an employee of the Institute of Archaeology of the Uzbekistan Academy of Sciences, in 1973 (Suleymanov, 1973). Subsequent research in 1981, conducted by R.H. Suleymanov and N.Yu. Nefedov, focused on archaeological sites in the Kashkadarya region, including the Dehqonobod flood control structure[42], as well as the construction of water reservoirs in Qizilsuv, Oyoqchisoy, and Toshloqsoy (Suleymanov, Nefedov, 1981). In the following years, various archaeological scholars, including R. Suleymanov, M. Isomiddinov, M. Xasanov, and A. Raimqulov, conducted research in these areas. Between 2020 and 2022, archaeological studies were organized in the Karshi area and Guzar district by A. Raimqulov and S. Suyunov. Presently, S. Suyunov has compiled a catalogue of archaeological sites in the Guzar district of the Kashkadarya region, which has been published (Suyunov, 2022). Notable sites in this region include Xo'ja Buzurgtepa, Vahmtepa, Nomsiz Tepa No. 16, Handalaktepa, Yarg'unchatepa, Nomsiztepa, and several others. Xo'ja Buzurgtepa, considered a historical city of Eskifagan, covers an area of no less than 200 hectares and was one of the major cities in Southern Sagdiana during the Middle Ages, reflecting the center of the ancient Huzor. Xo'ja Buzurgtepa is situated approximately 2 km from the designated area, surrounded by various fortresses and rural settlements. In 2023, archaeological research is being conducted in the Guzar district of the Kashkadarya region by S. Suyunov, a scholar at the Archaeology Institute of the Samarkand Branch of the Republic of Uzbekistan's Cultural Heritage Agency. But no archaeological investigations have been conducted in or near the project area at this time.

Within the vicinity of the area where the project is planned, there may currently exist a memorial defensive wall, which is poorly preserved but has the shape of the letter "L" (Figure 47). However, a final conclusion on this can only be made only after an archaeological study of this preserved hill. To the west of the PV Site boundary there are the remains of another unspecified monument with preserved western and southern walls. It will only be possible to clarify whether this hill is an archaeological monument or a cultural heritage site only after archaeological investigations. As part of the ESIA process, further archaeological survey works shall be carried out to determine the presence of a cultural heritage object or archaeological monuments within the project boundary and to gain further clarity on the features near the PV Site

Field surveys by archaeological search and excavation methods were conducted on the planned construction site.

As a result of the research, number of stone artifacts belonging to the historical period from the Middle Paleolithic to the Neolithic period were collected on the ground level of the specified area. The causes and factors of the spread of archeological finds on the ground level of the project site in this area were

41 Preliminary archaeological survey work and research was conducted by A. Mukhamadiev of the National Center of Archaeology.

42 special hydrotechnical facility built to drain flood waters and protect an area from flood disasters

clarified. Several Stone Age workshops and settlements were found 8-10 km south of the designated area, around the spring called Butabulok.

No archaeological objects and monuments with a cultural layer were identified in the area of the project site.

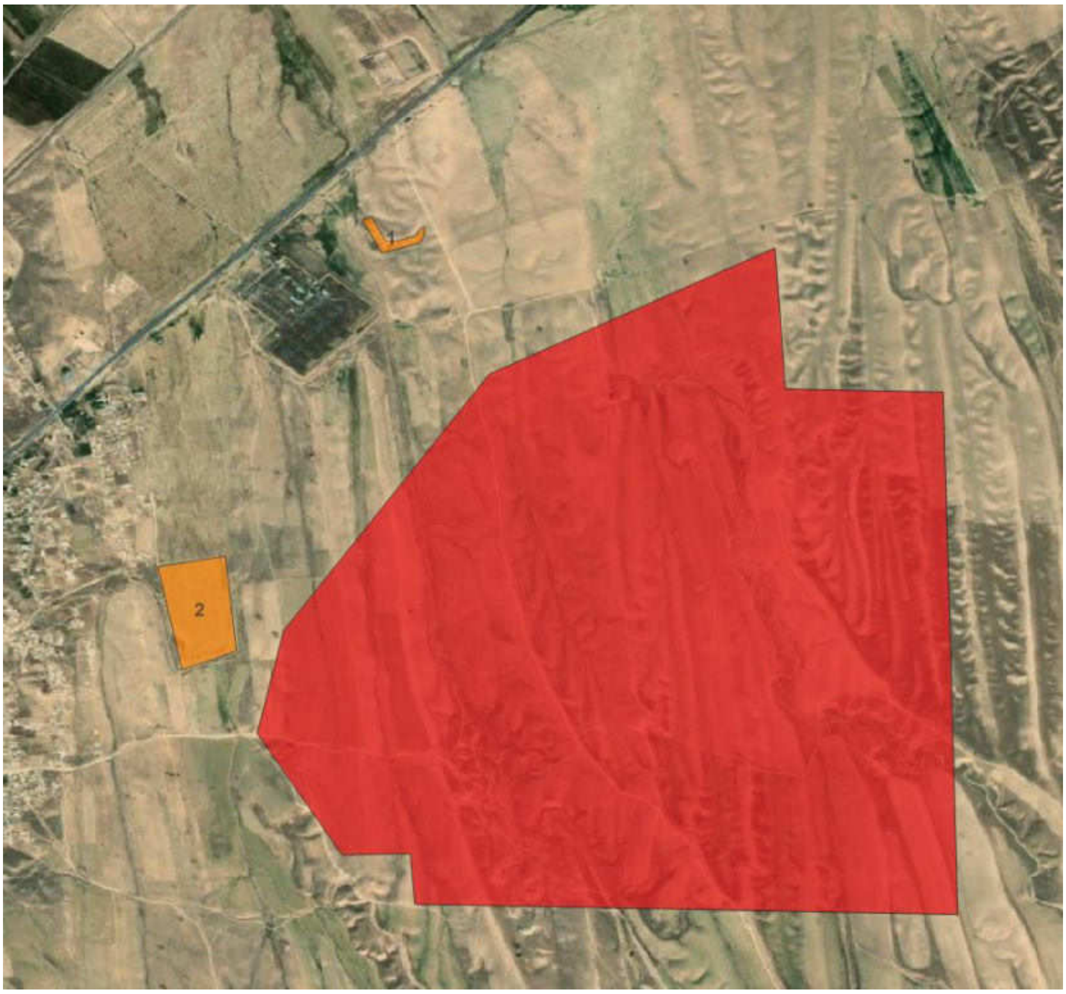


Figure 46 – (1) “L” shaped memorial defensive wall and (2) remains of other unspecified monument whose western and southern walls have been preserved

The communities of Yangiabad and Khalqabad each have one cemetery and Aynakul community has three cemeteries, these are not expected to be impacted by the Project.

5.5.9 Indigenous Peoples

IFC PS8 defines Indigenous peoples (IPs) as a distinct social and cultural group possessing the following characteristics in varying degrees:

- Self-identification as members of a distinct indigenous social and cultural group and recognition of this identity by others.
- Collective attachment to geographically distinct habitats, ancestral territories, or areas of seasonal use or occupation, as well as to the natural resources in these areas.
- Customary cultural, economic, social, or political institutions that are distinct or separate from those of the mainstream society or culture.

- A distinct language or dialect, often different from the official language or languages of the country or region in which they reside.

No IPs were observed during the site visit or identified during communications with the nearest communities. IPs are not considered present in the AOI.

5.6 Biodiversity overview

5.6.1 Protected areas and internationally recognized Key Biodiversity Areas

Three areas with international protected status (Key Biodiversity Areas[43]) and zero areas with nationally/legally protected status were identified in the 50km AoI (Figure 47). Further information about these areas is provided in Table 17.

Table 17: Protected areas and KBAs

Name	National site (IUCN Management Category)	Internati onal site	Area (ha)	Distance to project site	Organisation	Purpose
Gissar State Nature Reserve	N/A	IBA UZ042	549,577	20 km	Uzbekistan Society for the Protection of Birds	Mountains host breeding of various raptors, vultures, and other montane birds
Chimkurgan Reservoir	N/A	IBA UZ041	4,124	20 km	Uzbekistan Society for the Protection of Birds	Reservoir hosts large concentrations of waterfowl and other waterbirds during winter and migration
South-west Gizzar Foothills	N/A	-IBA UZ043	19,635	30 km	Uzbekistan Society for the Protection of Birds	Spring migration bottleneck area for Demoiselle Cranes and various raptors

43 Key Biodiversity Areas (KBAs) are the most important places in the world for species and their habitats. Faced with a global environmental crisis we need to focus our collective efforts on conserving the places that matter most. The KBA Programme supports the identification, mapping, monitoring and conservation of KBAs to help safeguard the most critical sites for nature on our planet – from rainforests to reefs, mountains to marshes, deserts to grasslands and to the deepest parts of the oceans (www. https://www.keybiodiversityareas.org/).

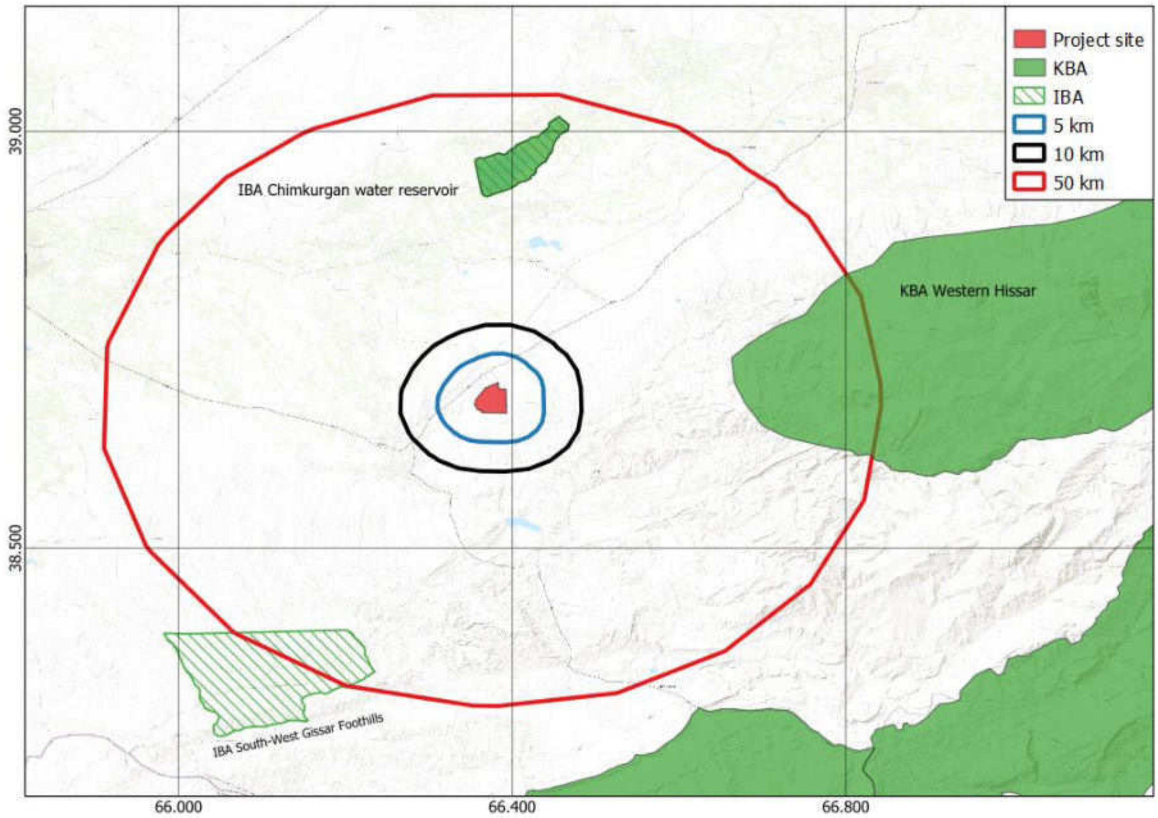


Figure 47 Nur Kashkadarya Solar Project AoI[44]

(Dark gray polygon in center) buffered by 1km, 10km, and 50km (concentric circles around project area), showing Project location in relation to internationally recognized Key Biodiversity Areas (KBA, pink areas) within 50 km. All three of the KBA depicted in this map are Important Bird Areas (IBA), as designated by Uzbekistan's BirdLife International partner organization, the Uzbekistan Society for the Protection of Birds. This figure is reproduced from a report produced with the International Biodiversity Assessment Tool (IBAT) by Caleb Gordon on 04 October, 2023, under license 30569-49620. This report also indicated that there are no nationally/legally protected areas within 50 km of the Project area.

5.6.2 Desk based review and critical habitat/PBF screening

A biodiversity screening exercise was performed based on a review of the Suntrace Report[45] and Juru review of an IBAT report[46], and publicly available information relevant to the area, including information from the IUCN Red List of Threatened Species database[47], and eBird[48], to determine species potentially triggering a Critical Habitat (CH) or Priority Biodiversity Feature (PBF) determination for the Project under either IFC PS6 or EBRD PR6. This exercise identified a preliminary list of nine species that are provisionally classified as PBF based on the threatened species criterion (Table 18). No species were identified in the desktop review as potential triggers of CH or PBF based on the restricted range or migratory/congregatory species criteria of either IFC or EBRD, and no likely CH triggers were identified

44 At the time of the survey, an earlier version of the Project layout was considered. The findings of this report are not changed by the final Project layout as depicted in Figure 2 32 and are still considered representative of the final layout.

45 Suntrace, 2022. Guzar Solar PV Critical Habitat Assessment. Prepared for Uzbekistan Ministry of Energy and ADB. January, 2022.

46 IBAT PS6 & ESS6 Report. Generated under license 30569-49620 from the Integrated Biodiversity Assessment Tool on 04 October 2023 (GMT). www.ibat-alliance.org

47 IUCN Red List of Threatened Species. http://www.iucnredlist.org . Accessed 13 October, 2023

48 https://ebird.org/home, accessed 13 October, 2023

during the screening. Based on the level of anthropogenic disturbance and land use history of the Project site, the area is not considered likely to trigger a CH or PBF determination under any of the CH/PBF criteria related to threatened/unique ecosystems, key evolutionary processes, or ecosystem services.

Table 18: Species Identified during the desktop review as potential Priority Biodiversity Features or Critical Habitat triggers for the Nur Kashkadarya PV solar project

Common Name	Scientific Name	IUCN Category	Uzbekistan Red data book Category	Identified in IBAT report	Identified in Sun trace report
Sociable Lapwing	*Vanellus gregarius*	CR	VU	X	Possible in transit, not likely to depend on site
Saker Falcon	*Falco cherrug*	EN	EN	X	Potentially nesting in Project area, but unlikely to depend on area
Steppe Eagle	*Aquila nipalensis*	EN	VU	X	Possible in transit, unlikely to depend on area
Imperial Eagle	*Aquila heliaca*	VU	VU		Possible rarity in transit
Greater Spotted Eagle	*Clanga clanga*	VU	VU		Possible rarity in transit
Egyptian Vulture	*Neophron percnopterus*	EN	VU	X	Confirmed present in Project area, but unlikely to depend on area
Great Bustard	*Otis tarda*	VU	CR		Possible rarity in transit, but habitat is unsuitable for wintering
Macqueen's (Asian Houbara) Bustard	*Chlamydotis macqueenii*	VU	VU		Unlikely
Marbled Polecat	*Vormela peregusna*	VU	VU		Potentially present but rare

5.6.3 Vegetation

The project area is situated in the clayed foothills on the western slopes of the Hissar mountain range. The primary land use in this area is traditional rain-fed agriculture, focused on seasonal crops such as chickpea, winter wheat, barley, and flax during the winter season (December - April or February - April).

A preliminary habitat assessment, based on a scoping site visit, revealed that approximately the majority of the PV site is covered by cultivated fields or has been plowed in the past with small areas of dry grasslands of loamy foothills and dry ravines.

Figure 48: Plowed rain fed fields

Figure 49: Overgrazed unplowed clayed hilly plain

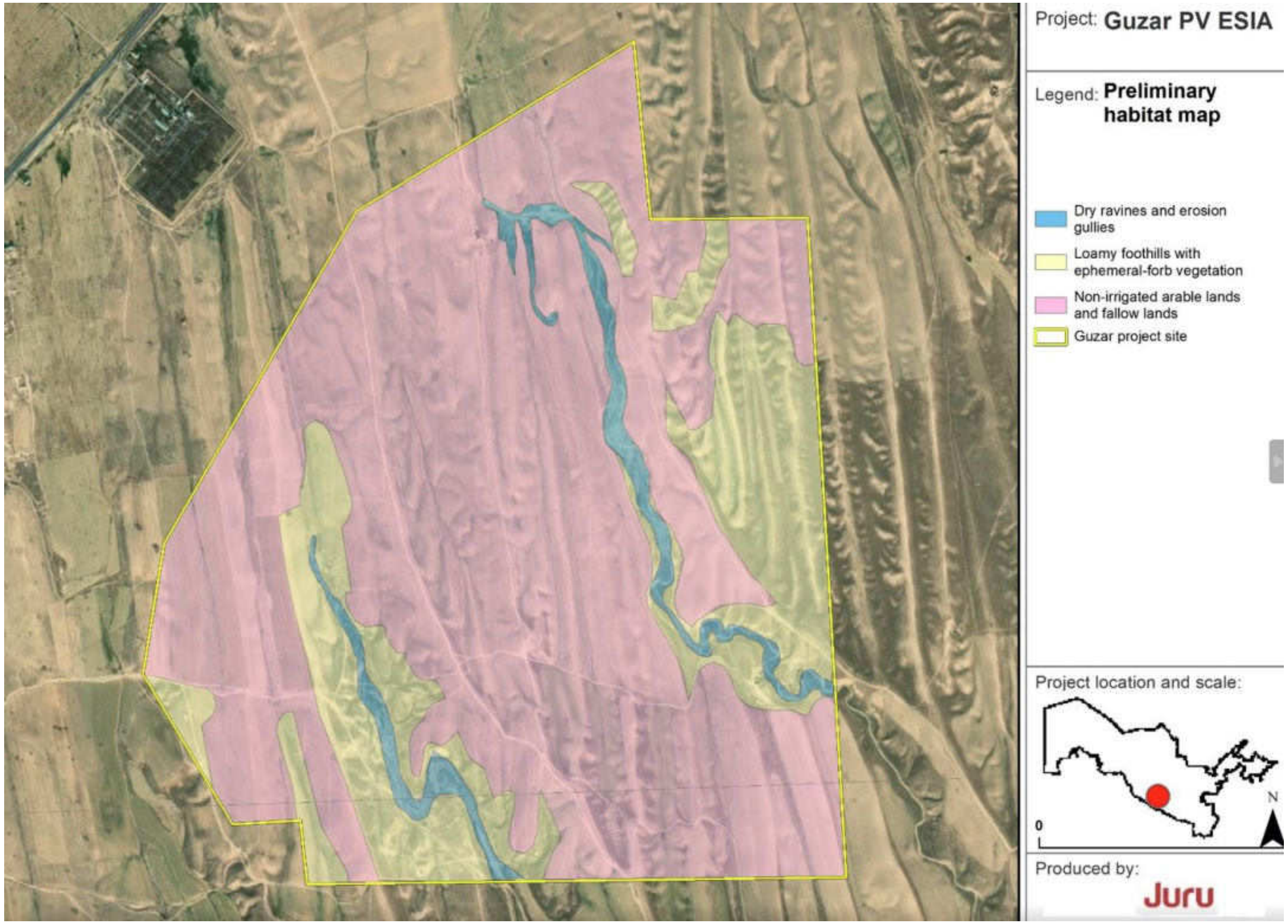


Figure 50: Preliminary habitat

5.6.4 Ornithofauna

The avifauna of Uzbekistan consists of approximately 477 species, with 48 of these having various conservation statuses, including DD, NT, VU, EN, and CR. The project area doesn't have an exact list of bird species, but data analysis showed the potential presence of 62 bird species (Table 19).

The Scoping survey showed that territory is mainly covered with non-irrigated arable lands, but in dry grasslands of loamy foothills the habitat is suitable for breeding Isabelline Wheatear *Oenanthe isabelline* and Crested lark *Galerida cristata.* In dry ravines and erosion gullies the habitat is suitable for breeding Little owl *Athene noctua*, sand martins *Riparia riparia*, Roller *Coracias garrulus*, sparrows and et al. It is unsuitable for waterbirds also, although it is possible that ducks like Common Teal *Anas crecca* and small waders could visit the small wetlands formed from spills from waterpipe, which is located outside the PV site to the north of the site. Also, geese and ducks could visit territory for foraging during winter, when the area planted by wheat and other plants. Although near small waterbody we observed White wagtail *Motacilla alba*, Common sandpiper *Actitis hypoleucos* on Scoping in 2023. We also observed Common quail *Coturnix coturnix* and a few migrating birds during the scoping visit – Long legged buzzard *Buteo rufinus*,. The groups of migrating flocks of Lesser short-toed lark *Calandrella rufescens* are common here. Various raptors could potentially visit this area for resting on poles of TL and foraging, as rodent holes, potentially indicating the presence of raptor prey, were observed during the scoping visit, but only in the unplowed parts. May more rodent holes were observed outside of the project site to the east. Several synanthropic bird species were observed on the site – Barn swallow *Hirundo rustica*, Rock dove *Columba livia*, Common Myna *Acridotheres tristis*.

Table 19: List of potential bird species occurring at the site

No.	Common name	IUCN Red list	Uzbekistan Red Data Book	Note
1	Common Quail *Coturnix coturnix*			Was recorded on the eastern border September 11, 2023.
2	Greater White-fronted Goose *Anser albifrons*			Abundant species on Talimarjan (Lampila et al. 2018 a, b), potential use of the fields on the territory as foraging grounds
3	Lesser White-fronted Goose *Anser erythropus*	VU	VU:R	Recorded in 2018 on Talimarjan and the Amu Darya (Lampila et al. 2018 a, b), potential passage over the territory
4	Greylag Goose *Anser anser*			Abundant species on Talimarjan and the Chimkurgan reservoir (Lampila et al. 2018 a, b), potential use of the fields on the territory as foraging grounds
5	Ruddy Shelduck *Tadorna ferruginea*			Breeds nearby, can forage in winter fields in winter
6	Eurasian Wigeon *Anas penelope*			Potential migration through the territory (Lampila et al. 2018 a, b), potential use of the fields on the territory as foraging grounds
7	Common Teal *Anas crecca*			Was recorded in 2021
8	Mallard *Anas platyrhynchos*			Potential foraging and migration through the territory (Lampila et al. 2018 a, b)

No.	Common name	IUCN Red list	Uzbekistan Red Data Book	Note
9	Black Stork *Ciconia nigra*		VU:R	Potential migration through the territory (Lampila et al. 2018 a, b)
10	White Stork *Ciconia ciconia*		NT	Potential migration through the territory (Lampila et al. 2018 a, b)
11	Lesser Kestrel *Falco naumanni*		NT	Potential foraging and migration through the territory
12	Common Kestrel *Falco tinnunculus*			Potential foraging and migration through the territory
13	Merlin *Falco columbarius*			Potential wintering on this territory
14	Saker Falcon *Falco cherrug*	EN	EN	This species breeds in Hissar mountains
15	Peregrine Falcon *Falco peregrinus*		VU:R	Potential wintering on this territory
16	Barbary Falcon *Falco pelegrinoides*		VU:R	This species breeds in Hissar mountains
17	Osprey *Pandion haliaetus*		VU:R	Potential migration through this territory
18	European Honey-buzzard *Pernis apivorus*			Potential migration through this territory
19	Oriental Honey-buzzard *Pernis ptilorhynchus*			Potential migration through this territory
20	Black Kite *Milvus migrans*			Potential migration through this territory
21	White-tailed Sea-eagle *Haliaeetus albicilla*		VU:R	Potential migration through this territory
22	Himalayan Griffon *Gyps himalayensis*	NT	VU:R	Short-distance winter migrations to the plain
23	Griffon Vulture *Gyps fulvus*		VU:D	Short-distance winter migrations to the plain
24	Cinereous Vulture *Aegypius monachus*	NT	NT	Short-distance winter migrations to the plain, was recorded in Nov, 2021
25	Bearded Vulture *Gypaetus barbatus*	NT	VU:R	Short-distance winter migrations to the plain
26	Egyptian Vulture *Neophron percnopterus*	EN	VU:D	Breeds near the Pachkamar reservoir in 11 km from project site (2018, 2021) could visit territory during migration and foraging.
27	Short-toed Snake-eagle *Circaetus gallicus*		VU:D	There is a possibility of nesting near the territory
28	Western Marsh Harrier *Circus aeruginosus*			Potential foraging and migration through the territory

No.	Common name	IUCN Red list	Uzbekistan Red Data Book	Note
29	Hen Harrier *Circus cyaneus*			Potential foraging and migration through the territory
30	Pallid Harrier *Circus macrourus*	NT	NT	Potential foraging and migration through the territory
31	Montagu's Harrier *Circus pygargus*			Potential foraging and migration through the territory
32	Shikra *Accipiter badius*			Potential foraging and migration through the territory
33	Eurasian Sparrowhawk *Accipiter nisus*			Potential foraging and migration through the territory
34	Northern Goshawk *Accipiter gentilis*			Potential foraging and migration through the territory
35	Eurasian Buzzard *Buteo buteo*			Potential wintering on this territory
36	Long-legged Buzzard *Buteo rufinus*			Breeds nearby, can stay for the winter. Was recorded in September 2023
37	Upland Buzzard *Buteo hemilasius*			Potential wintering on this territory
38	Rough-legged Buzzard *Buteo lagopus*			Potential wintering on this territory
39	Greater Spotted Eagle *Aquila clanga*	VU	VU:R	Potential foraging and migration through the territory
40	Steppe Eagle *Aquila nipalensis*	EN	VU:D	Potential foraging and migration through the territory
41	Eastern Imperial Eagle *Aquila heliaca*	VU	VU:D	Potential foraging and migration through the territory
42	Golden Eagle *Aquila chrysaetos*		VU:R	Potential foraging and migration through the territory
43	Booted Eagle *Hieraaetus pennatus*		VU:D	Potential migration through this territory
44	Demoiselle Crane *Anthropoides virgo*			Potential migration and roosting through this territory
45	Common Crane *Grus grus*			Potential migration and roosting through this territory
-	Great Bustard *Otis tarda*	VU	CR	This species is known as wintering in Kashkadarya region, but the survey conducted in 2020-2021, showed that there is no stable wintering location in this region. Such place was indicated only in Djizzak region. The oral interviews showed that local people don’t know Great bustard.

No.	Common name	IUCN Red list	Uzbekistan Red Data Book	Note
-	Asian Houbara *Chlamydotis macqueenii*	VU	VU:D	The habitat is not suitable for breeding and migrating.
46	Little Bustard *Tetrax tetrax*	NT	VU:D	Potential wintering on and migration through this territory
47	Northern Lapwing *Vanellus vanellus*	NT		Potential wintering on and migration through this territory
48	Sociable Lapwing *Vanellus gregarius*	CR	VU:R	It can use this territory during migration (Donald et al 2016, Azimov 2018). The survey will be conducted in
49	Common sandpiper Actitis hypoleucos			Migrates through this territory (September 2023)
50	Rock Pigeon *Columba livia*			Resident synanthropic species (July 2021, September 2023)
51	European Turtle-dove *Streptopelia turtur*	VU	VU:D	Potential migration through this territory
52	Pallid Scops-owl *Otus brucei*			Potential migration through this territory
53	Little Owl *Athene noctua*			Nesting (July 2021)
54	European Roller *Coracias garrulus*			Nesting (July 2021)
55	European Bee-eater *Merops apiaster*			Nesting (July 2021)
56	Common Hoopoe *Upupa epops*			Nesting (July 2021)
57	Crested Lark *Galerida cristata*			Resident species (July 2021, September 2023)
58	Lesser short-toed lark *Calandrella rufescens*			Migrating flocks were recorded on site (September 2023)
59	Collared Sand Martin *Riparia riparia*			Nesting (July 2021)
60	Barn swallow *Hirundo rustica*			Resident synanthropic species (September 2023)
61	Isabelline Wheatear *Oenanthe isabellina*			Nesting (July 2021), Resident (September 2023)
62	Common Myna *Acridotheres tristis*			Resident synanthropic species (July 2021, September 2023)
		17	26	

5.6.5 Reptiles

The preliminary list of reptiles living on the project territory in the western foothills of the Hissar mountain range, consists of 13 species, including 4 rare species, 2 synanthropic species (Table 20). Among them the important is a Central Asian tortoise *Testudo horsfieldii*. During the scoping site visit, the oral review of local herders showed the presence of tortoises and Desert monitor, also 8-10 potential holes were observed on the site. The planned Spring survey 2024 will confirm the presence of these species (see section 8, ESIA TOR).

Table 20: List of reptile species inhabiting the project area

Species name	Uzbekistan Red data book	IUCN Red list	Endemism	Scoping results
Central Asian tortoise *estudo horsfieldii*	VU	VU		Holes, and oral data from local herders
Turkestan thin-toed gecko *Tenuidactylus fedtschenkoi*			UZ, TJ, TM	Was recorded in 2021
Steppe Agama *Trapelus sanguinolentus*				Secondary data
Glass lizard *Pseudopus apodus*				Secondary data
Asian snake-eyed skink *Ablepharus pannonicus*				Secondary data
Gold or Schneider's skink *Eumeces schneideri*				Secondary data
Rapid Lizard *Eremias velox*				Was recorded in 2021
Caspian Monitor *Varanus griseus caspius*	VU:D			Rare lizard recorded not far from fields
Tatary sand boa *Eryx tataricus*	NT			Secondary data
Spotted whip snake *Hemorrhois ravergieri*				Secondary data
Spotted desert racer *Platyceps karelinii*				Secondary data
Diadem snake *Spalerosophis diadema*				Secondary data
Central Asian cobra *Naja oxiana*	NT	DD		Secondary data

Figure 51 Potential tortoise hole

5.6.6 Mammals

According to available data and habitat condition, about 15 species of mammals belonging to 6 orders potentially habit on the project site and adjacent areas:

1. **Long-eared Hedgehog** (***Hemiechinus auritus*** **Gmelin, 1770)** is widely distributed throughout the Kashkadarya basin (Mecklenburtsev, 1958).
2. **Buchara Horseshoe Bat (*Rhinolophus bocharicus* Kastschenko et Akimov, 1917** – the diaries of A. P. Kuzyakin indicate that this species was caught by R. N. Meklenburtsev in the Kashkadarya valley 30 km downstream of Shakhrisyabz (Bogdanov, 1953).
3. **Common Pipistrelle (*Pipistrellus pipistrellus* Schreber, 1774)** is one of the most common species of the country, living in populated areas. It was recorded in the town of Guzar (Bogdanov, 1953; Volozheninov, 1986; Mecklenburtsev, 1958)
4. **Serotine Bat (*Eptesicus serotinus* Schreber, 1774)** - was recorded in Guzar in 1935 (Mecklenburtsev, 1958)
5. **Tolai hare (*Lepus tolai* Pallas, 1778)** - the western border of the tolai's range runs along the eastern shore of the Caspian Sea, going further to the south, to Iran and Afghanistan (Sokolov et al., 1994).
6. **Small five-toed Jerboa (*Allactaga elater* Lichtenstein, 1825)** – the southern boundary of the range enters Kazakhstan from the east, from China, runs along the southern edge of the ili depression to the west along the foothills of the Tian Shan, skirts the Karatau Range from the north, includes the Fergana valley, goes around the spurs of the Pamir-Alai Mountains from the north and west and runs in their southern foothills to enter the South-Tajik depression, where it encompasses the Vakhsh valley, crosses the lower reaches of the Panj and goes to the south, to Afghanistan (Shenbrot et al., 1995). It is worth noting that the small five-toed jerboa was found 45 km north of Karshi, in a biotope consisting of small sedge, sagebrush, camel thorn and saltworts (Meklenburtsev, 1958)
7. **Large Souslik (*Spermophilus fulvus* Lichtenstein, 1832)** - widely distributed throughout the Kashkadarya basin (Mecklenburtsev, 1958)

8. **Severtsov's jerboa (*Allactaga severtzovi* Vinogradov, 1925)** - it was recorded 45 km north of Karshi, in a biotope consisting of small sedge, sagebrush, camel thorn and saltworts (Mecklenburtsev, 1958). It is locally extinct in the Karshi steppe as a result of the development of virgin area and fragmentation of populations (Schoenbrot et al., 1995).
9. **Red Fox (*Vulpes vulpes* Linnaeus, 1758)** - it is widely distributed throughout the Kashkadarya basin (Mecklenburtsev, 1958; Ishunin, 1961)
10. **Corsak fox (*Vulpes corsac* Linnaeus, 1758).** it is widely distributed throughout the Kashkadarya basin (Mecklenburtsev, 1958; Ishunin, 1961)
11. **Golden Jackal (*Canis aureus* Linnaeus, 1758)** - recorded between Karshi and Shakhrizyabs, generally common in the Kashkadarya valley (Mecklenburtsev, 1958; Ishunin, 1961)
12. **The wolf (*Canis lupus* Linnaeus, 1758)** is extremely rare in the desert territories of Kashkadarya region (Mecklenburtsev, 1958; Ishunin, 1961), but this predator is common in the Hissar Range (Ishunin, 1961)
13. **Steppe Polecat (*Mustela eversmanii* Lesson, 1827)** - was recorded in the vicinity of Guzar (Ishunin, 1961)
14. **Marbled Polecat (*Vormela peregusna* Guldenstaedt, 1770)** - it was common in Guzar district, according to H. Salikhbayev (1939). 109 skins were received by procurement centres in 1934, 106 – in 1935, and 44 in the first half of 1936 (Ishunin, 1961)
15. **Steppe Cat (*Felis silvestris ornata* Gray, 1832)** - inhabits desert and foothills throughout (Heptner and Sludsky, 1972)

Goitered Gazelle (***Gazella subgutturosa* Guldenstaedt, 1780)** – was recorded in large numbers in the flat and hilly parts of Guzar and Dehkanabad districts. Specifically for the project area, there is information that in 1935 it was found in the foothills east of the town of Guzar (Salikhbayev, 1939). The habitat is not suitable for this species, because the territory is modified and degraded with high stress of human disturbance (grazing et al.). The Goitered Gazelle that had formerly inhabited the territory was completely exterminated.

Table 21: Rare species of mammals potentially inhabiting the project area

No.	Species	Uzbekistan Red Data Book	IUCN status	CITES	CMS	Endemism
1	Bukhara Horseshoe Bat	-	LC	-	-	AF, IR, KZ, KR, TM, TJ, UZ
2	Severetsov's Jerboa	-	LC	-	-	KZ, TM, TJ, UZ
3	Marbled Polecat	2(VU:D)	VU	-	-	-
4	Steppe Polecat	2(VU:D)	LC	-	-	-
5	Corsak fox	2(VU:D)	LC	-	-	-

The Scoping survey 11.09.2023 indicates several holes of rodents (Large souslik, jerboas, and foxes (Red Fox or Corsak). We didn't observe any mammal species on the territory. The number of holes to the east of the project site was much higher. It showed that territory is not suitable for habitation because of regular plowing. The quality of this habitat is very low, probably due to the drought, and high grazing pressure.

Figure 52 Fox hole

The mammal survey has been conducted on the project site in Autumn 2023.

6 Identification of Potential E&S Impacts

Table 22 provides the outcomes of the scoping review for potential E&S issues and impacts for the Project's construction, operation, and decommissioning phases. The scoping assessment considers the potential source of impact, relevant nearby receptors (as determined during the scoping baseline review) and provides a scoping evaluation to identify those E&S issues that may give rise to significant impacts and require further assessment in the ESIA. Issues and impacts have been scoped in or out based on current understanding of the Project, the baseline environment and with reference to other projects in the region.

Where aspects are scoped in for further assessment, an outline of the requirements for further baseline studies and impact assessment are summarised A detailed scope is provided in the ESIA TOR in chapter 8. For those issues scoped out, no further assessment is considered necessary in the ESIA. If new information becomes apparent during the ESIA process, issues scoped out may be scoped back in.

Table 22: ESIA Scoping

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
Air quality (Construction/Decommissioning) (C/D)	There is little traffic and no industrial facilities in the site's immediate vicinity; therefore, current air quality (AQ) is considered good. The site may be subject to sandstorms and high winds at certain times of the year. The largest source of AQ emissions will be fugitive dust from vehicle movements and fugitive emission (NOx, SOx) from construction traffic on unpaved public and private roads and fugitive dust emissions from site preparation and general construction works. The nearest potential receptors are workers at the site and adjacent substation, nearby communities, herders, and residents in the worker accommodation.	✓		The ESIA will include a high-level qualitative assessment of construction phase AQ impacts (traffic and dust within 250m of the proposed Project to understand impacts on potential receptors and define GIIP for managing construction dust and fugitive emissions in a framework ESMP. To support any future construction air quality benchmarking, spot check monitoring at two locations for the following parameters: Nitrogen dioxide (NO_2); Sulphur dioxide (SO_2); Carbon monoxide (CO), TSP (total suspended particles); PM2.5 and PM10 will also be performed.
Air quality, dust (Operation) (O)	No operational emissions are anticipated from the Project's operation (PV/BESS/OHTL). Project-related fugitive emissions will be intermittent and slight.		✗	This topic is scoped out for further assessment in the ESIA.
Noise and Vibration (C/D)	There is potential for noise impact on workers and residents of the communities along the transportation routes to the site and nearby households to the east of the site. Workers at the substation and the house immediately adjacent to the substation are also considered noise sensitive receptors. Noise impact will arise due to the use of plant and machinery and construction-related traffic due to works at the substations and along the OHTL route. The nearest noise-sensitive human and ecological receptors are between 270 m and 1 km from the site boundary and the 250m either side of the four communities along the transportation route to the site. There is unlikely to be any blasting requirement for the works; however, this is also a potential intermittent noise source.	✓		The ESIA will consider potential noise-generating sources and impacts and perform a qualitative assessment of potential impacts to define GIP for construction noise management in a framework ESMP. The existing noise baseline at the nearest sensitive receptors locations will be ascertained to support future noise benchmarking.
Noise and Vibration (O)	Operational noise impacts mainly arise from maintenance works during operation. Receptors are not considered to be close enough for any residual noise from the OHLT (if selected as the preferred option), the PV plant or the BESS.		✗	This topic is scoped out for further assessment in the ESIA.

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
	Maintenance works will be short-term and intermittent. Abnormal or emergency noise events will be short-term and temporary. Given the proximity of noise-sensitive receptors, operational noise impacts are not expected to be significant.			
Landscape and visual impact (C/D)	Construction activity may constitute a visible activity (presence of construction traffic, compound, plant and equipment and localised light pollution) that may modify the landscape in which they are set. These activities will be temporary, short-term/transient, reversible activities. Potential landscape and visual impacts from these sources are therefore regarded as negligible. This, combined with the general absence of receptors in the project zone of visible influence, means the impact is likely to be insignificant.		✗	This topic is scoped out for further assessment in the ESIA.
Landscape and visual impact (O) (including glint and glare)	Generally, the height of the permanent installation of solar PV panels and BESS containers will be low (less ten 2.5m above ground level). There are no anticipated visual impacts associated with the electrical interconnection being underground and within an existing substation. In all cases, the project proposes to use standard/sensitive material choices for the structures on site. The permanent installation of solar panels can result in glint and glare impacts on airfields, road users and residential receptors south of the PV site. The absence of receptors south of the site means that the significance of the expected impact will be insignificant.		✗	This topic is scoped out for further assessment in the ESIA.
Surface water quality (C/D)	The scoping site visit and review of existing documentation has identified seasonal surface water features within the Project site. The site design has considered a set back from these streams to avoid or minimise impact on drainage and water quality, however this must be assessed further. There is no plan to discharge wastewater to into adjacent water bodies. There is a minimal risk that construction works may result in an unforeseen release into the surface water features. Flood risk associated with the seasonal streams may can also be a risk.	✓		The ESIA will consider in detail the impact of the construction and operation on the seasonal surface water features considering their importance in the wider catchment area. The ESIA will consider the need for project specific mitigation measure and also define GIIP construction management methods for controlling spills and unforeseen discharges in the ESMP. A flood risk assessment will be performed to conform risk of flooding. A specific impact assessment on water quality will not be performed.

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
Hydrogeology (C/D)	Although foundation works are shallow, the groundwater levels at the site are not confirmed. There is the potential to impact groundwater due to the foundation works.	✓		The EISA will qualitatively assess the potential to impact groundwater based on further geotechnical works to be performed for the Project. The ESIA will outline management and mitigation measures (GIIP and project-specific) in the framework ESMP.
Hydrogeology use (O)	No groundwater will be used during the operation phase.		✗	This topic is scoped out for further assessment in the ESIA.
Waste (C/D)	Construction-related waste will be a mixture of inert material (soils), general/domestic construction waste and hazardous wastes (e.g. oils, paints, greases etc.) and some biomass resulting from the clearance of the site, site levelling, civil works, equipment installation (packaging, metals, paints, coatings), electrical cut-offs and domestic waste. The baseline review indicates that waste management options in the Project area for the transportation and disposal of waste in line with GIIP may be limited.	✓		The ESIA will confirm the waste generated during the Project construction phase and collect further baseline data on the availability of waste infrastructure that aligns with GIIP that may be used by the Project. The ESMP will define the requirements for a Site Waste Management Plan aligned with GIIP, including requirements for waste segregation, recycling options, and duty of care obligations,
Waste (O)	The operational phase will include general hazardous and non-hazardous waste connected with maintenance works. One key waste stream will be waste from electric and electronic equipment (WEEE), including PV panels and batteries. It is anticipated that any wastes will be removed from the site area by the engineers for offsite disposal, storage or decommissioning (in the case of old/obsolete equipment), but the final disposal route is not confirmed now, and options within Uzbekistan may be limited.	✓		The ESIA will identify requirements for operational waste management (general waste, hazardous waste and WEEE) and define management requirements for an operational WMP plan, including a framework for supplier policies and buyback schemes. Additional baseline information on suitable options for disposing of electrical equipment and other hazardous waste in line with GIIP will be identified and outlined in the ESIA.
GHG emissions (all phases)	The Project will not generate Scope 1 and Scope 2 emissions in excess of 100,000 tonnes of CO_2 equivalent annually.		✗	This topic is scoped out for further assessment in the ESIA.
Climate (all phases)	The Project's location is in an arid area susceptible to physical climate change risks, including (among other things) extremes of temperature, high winds, and sand storms, flash flooding/ & landslides which in the long term may have adverse consequences on the Project that must be considered in the design and for the management or worker welfare during construction and operation.	✓		The ESIA will include a Climate Change Risk Assessment of relevant physical risks (following the methodology outlined by the TCFD). Suitable design and other mitigation in the final design specification will be outlined in the ESMP.

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In	Scoped Out	Requirements for ESIA /baseline data collection
Soils and land quality (erosion potential/ land contamination) (C/D)	The project area is characterized by Grey-brown soils. These soils have formed over time on the skeletal-fine grained alluvial sediments of the subaerial deltas of the Kashkadarya river. They are brownish gray at the surface characterized by a light topsoil layer low in humus content. These soils are usually free of vegetation and thus are exposed to increased degradation and deflation. Grey-brown soils dominate the northern and eastern parts of the region. Takyr soils, interlocked with grey-brown and sandy soils. They are formed in shallow depressions with high clay contents, which collect water. Given the susceptibility to erosion and degradation of the minimal humus layers, there is potential for the Project to exacerbate this during clearance and construction works. Soil contamination is a risk at all construction sites and industrial facilities where hazardous substances are handled and stored. Exposure to hazardous substances in the workplace puts workers at risk. Hazardous substance spills at the project site could adversely impact soils, surface water, groundwater/well water, community health and biodiversity.	✓		The ESIA will consider the erosion susceptibility of the soils in the Project area to assess the soil structure (topsoil) to identify mitigation or enhancement measures to manage soil degradation. Specific soil and erosion management requirements and requirements for site clearance and rehabilitation/reinstatement of land aligned with GIIP will be outlined in the ESMP. Conduct soil sampling at six locations within site corresponding to representative land-use types to confirm contamination levels and potential risks to workers. Define GIIP for managing construction soil contamination risks in the ESMP.
Soils and land quality (erosion potential/land contamination) (O)	There will be no impact on the soil during the operation phase. The potential for contamination may occur during abnormal operating scenarios.	✓		GIIP for managing operational contamination risks will be outlined in the ESMP. A specific operational impact assessment is scoped out from the ESIA.
Water resource use (C/D)	Water availability in the AOI area is assumed to be scarce. Water requirements during construction are moderate (including cement batching). Water will be tankered in from offsite sources. Given the water scarcity in the local area, water suppliers may have limited capacity for even small additional water use needs resulting in a potentially significant impact on competing water users.	✓		The ESIA will include a water use assessment for construction water use impacts on other water users and will define requirements for water minimisation aligned with GIP in the ESMP.
Water resource use (O)	Water availability in the AOI area is assumed to be scarce. Potable water needs are low based on the limited number of on-site personnel. Operational water use requirements are expected to be negligible, given that water will not be used for panel cleaning.		✗	This topic is scoped out for further assessment in the ESIA.
Socio-economic (all phases)	There are four communities within 10km of the Project site and small numbers of community members in scattered individual households located within 5km of the Project that	✓		A socio-economic survey will be undertaken as part of the ESIA to identify Project impacts and benefits and any groups vulnerable to project impacts (e.g. grazing herders).

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
	may be impacted by the Project. Some herders also unofficially use the site for grazing livestock during most of the year. These people are the most likely to have contact with the Project and its workforce. Other community members are located at a distance that they are unlikely to be impacted by construction nuisance. Community members may have contact with Project workers should workers be accommodated in the nearby communities (which may be allowed, at least before construction camps have been built), travel through the communities (traffic and transportation impacts are discussed further below) or visit the communities during their rest periods. However, community members may benefit from the Project through employment or by providing goods and services to the Project, or its workforce.			The ESIA will also include an assessment of accommodation options in local communities. The ESIA will include consultations and focus group discussions with local communities and their representatives to identify possible risks and inform the ESIA. Outputs will be outlined in the ESMP, including preparing a worker code of conduct.
Labour and working conditions (including labour risks in the supply chain) (C/D)	While there is evidence of child labour in Uzbekistan, it is mainly restricted to the agricultural industry, particularly cotton picking. There have been allegations of forced labour in the solar panel supply chains, particularly in the mining of polysilicon. Additionally, there are general concerns with labour and working conditions in the construction industry, particularly for migrant workers. Some areas that need to be addressed are retention of passports, payment of workers (particularly overtime payments), provision of adequate accommodation (where relevant)	✓		The ESIA will determine the risk of forced labour in the supply chain and child labour in the construction phase (secondary data and consultations). The ESIA will identify requirements for managing and monitoring workers on-site, in the supply chain and accommodation facilities. A Human Rights Due Diligence will be prepared to support the outcomes of the ESIA.
Labour and working conditions (O)	Labour and working conditions are less of a concern during operations, given that workers are usually higher skilled and better paid. However, there continue to be risks, particularly with the most vulnerable workers, such as security guards (who are often overlooked in labour monitoring) and cleaning staff.	✓		The ESIA will identify labour and working condition risks and requirements for the operation phase and outline management and monitoring requirements in the ESMP.
Occupational Health and Safety (all phases)	Occupational health and safety is a risk for all Projects; some possible occupational health and safety concerns related to the Project include slips, trips and falls; electrocution; falls from heights' working in enclosed spaces; and weather extremes. Given the distance from the project site to the	✓		The ESIA will consider OHS risks and the provision of medical care and outline management and monitoring requirements in the ESMP.

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
	nearest medical facilities, access to medical care in the case of an emergency will need to be carefully discussed.			
Electromagnetic field (EMF) and electrostatic impacts (operation)	The interconnection is not finalised at this time and may be above ground or below ground or a combination of both. There is potential for EMF connected with overhead line and although exposure periods are not expected to be exceed safe levels, considering the prevalence of small farms in the vicinity of the OHTL ROW there may be requirements for some mitigation to ensure farm workers return to work under the OHLT and do so safely.		✗	The ESIA will qualitatively consider EMF risks in relation to farm workers working with the OHLT ROW. .
Community, heath, safety and security (communicable diseases, public access, GBV/H.)	The distance between the local communities and the Project site minimises the risk of impacts on the local community. There is a possibility that workers may be housed either in temporary accommodation at the project site or existing accommodation within nearby communities, the latter of which represents a greater community health and safety risk. The potential for incidents of GBV/H was raised during the scoping phase, but insufficient information was received; therefore, this will be more fully addressed during future consultations. There is not expected to be a large influx of workers, but this will be further assessed in the ESIA, and the need for an influx management plan will be determined.	✓		The ESIA will identify project requirements for accommodation and assess the risk concerning local communities. The assessment will also determine whether worker influx is a concern and if an influx management plan will be required. During the socioeconomic survey work, consultation with local communities about community health and safety risks, including GBV/H, will be performed to inform the socio-economic risk assessment and definition of mitigation measures in the ESIA.
Traffic and Transportation (C/D)	A main road that runs past the north of the Project site will likely be used to transport project components to (or from) the site during construction and decommissioning. The route will pass through three of the four nearby local communities . It is not expected that significant impacts will occur due to the size of the Project components and the fact that the road is a main road and communities will be accustomed to traffic. However, a further review of possible vulnerable receptors will be required to confirm this assumption. There will also need to develop a traffic management plan to reduce the risk of accidents on site from transporting workers to and from the site.	✓		The ESIA will identify potential transportation routes and determine if there are any vulnerable receptors that the transportation of Project components might impact. The ESIA will also assess safety risks related to increased road traffic and define requirements for management and monitoring in a traffic management plan.
Traffic and Transportation (O)	Traffic during the operations phase will be limited to no more than ten workers and maintenance vehicles. It is unlikely that		✗	This topic is scoped out for further assessment in the ESIA.

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
	there will be significant impacts on the local communities from operational traffic and transportation activities.			
Security (C/D)	The Project site will be fenced to stop local community members from accessing the site. Nevertheless, security guards are expected to be used during the Project's construction phase. The distance between local communities and the site reduces the risk of security concerns for the Project or between local community members and security personnel. However, there is a small risk should herders bring their livestock near the Project, or that there are conflicts between the security guards and the houses located near to the Guzar substation, but this is considered manageable with the use of training, vetting of guards and consultation with herders and nearby households. This potential impact is still scoped in as measures must be identified to reduce these minor security risks.	✓		The ESIA will identify measures to reduce security risks on site and outline management and monitoring requirements for developing a security management plan.
Security (operation)	During the operation phase, it is expected that security provision will be remote (no permanent security presence); nevertheless, a security management strategy will be required.	✓		The ESIA will outline an operational security management strategy as part of the operational ESMP.
Emergency preparedness and response (all phases)	Possible natural disasters and climate-related risks relevant to the project location include earthquakes, wildfires, dust storms, drought, flash floods, landslides and heat waves. There may also be project related emergences related to spills, release of chemicals or other accidents. Given the distance from the project site to local communities and the type of technologies used, it is unlikely that emergencies on-site will impact local communities, but workers could be affected. Access to medical care in the case of an emergency will need to be carefully considered in determining the risk. The consideration of additional risks from the BESS will also be required.	✓		The ESIA will review potential emergencies through a review of secondary data and outline the requirements of an emergency preparedness and response plan.
Land acquisition, land use and livelihoods (including ecosystem services) (all phases)	There are two landowners for the Project site and, seven leaseholders. The SWID Committee and the Kamashi State Forestry Department are the two government landowners, and they will likely transfer their rights to use the land to Masdar (through the Ministry of Energy or NEGU).Compensation will need to be paid for the seven leaseholders. Herders from the	✓		The ESIA process will include consultations, focus groups and socioeconomic surveys, including specific discussions with the landowners, as well as nearby land user and herders. A Resettlement Plan, or Land Acquisition and Livelihood Restoration Plan (LALRP) including a Grievance Mechanism (GM) will be required to outline provisions for land acquisition,

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
	nearest communities also stated that they use the Project site during certain seasons to graze livestock.It is understood that alternative grazing land may need to be found for these herders. Following the scoping site visit, it does not appear that there will be any physical relocation of household's permanent residences, required. However, there are some structures on the site that will need to be removed and compensated. Further information will be sought to identify how often and for what purpose the structures are used.			compensation and livelihood restoration aligned with EBRD PR5.
Protected Areas and Internationally Recognized Key Biodiversity Areas (Priority Biodiversity Features)	No national protected areas were identified in AOI. Three areas with international status (KBA) were identified in AOI: ● KBA Western Hissar ● IBA Chimkurgan Reservoir ● IBA Southwest Gizzar Foothills		✗	Considering the baseline review, the Project is not expected to have any direct or indirect impacts on PAs and KBAs. Impact on protected areas is scoped out from further assessment in the ESIA, this information will be presented in the ESIA for completeness.
Invasive Species	No invasive plant or animal species were identified as significant ecological threats for the Project	✓		AIS surveys as part of the botanical surveys will be conducted to confirm the scoping review for the PV site and OHTL in spring 2024. Where relevant, the ESIA will address the management of invasive species.
Threatened Habitats	The entire Project site consists of Modified Habitat per IFC PS6, and specifically cultivated crop fields.		✗	Considering that the entire Project is sited in Modified Habitats, there is no potential for impacting Natural Habitat, per IFC PS6, and impacts to Natural, or threatened habitats are scoped out from further assessment in the ESIA. Valued biodiversity species and priority biodiversity features, will be considered separately, see below.
Reptiles	The presence of Steppe tortoise Testudo horsfieldi (IUCN VU, UzRDB VU[49]) was confirmed through a desktop study and oral interviews with local herders. Furthermore, during the scoping	✓		Tortoise surveys will be completed between March and the end of April / early May 2024 when tortoises are likely to be

49 UzRDB – Uzbekistan Red data book. Vol.2. Animals. Tashkent 2019.

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
	phase, potential tortoise burrows were discovered in unplowed lands, indicating their habitat within the project area.			most active and most obvious. However peak periods of activity are dependent on temperature and prevailing weather conditions. Determination of requirements for no-net loss and species relocation plan will be defined as part of the ESIA and ESMP.
Plants	During the scoping stage, no threatened species were identified, and no trees were observed on the site, with the exception of planted trees near one herder's camp.	✓		To ensure the exclusion of potential threatened species, detailed botanical surveys will be conducted to confirm the scoping review for the project site in October 2023 and for the Project site and OHTL in spring 2024 to feed into the ESIA biodiversity impact assessment and management plans as necessary.
Fish	There are no permanent surface water features on the site.		✗	Small ponds formed by spills from water pipes are unsuitable for fish habitation.
Birds	The project site is not suitable to the breeding of threatened species, primarily because a significant portion of the territory has been modified. Additionally, the project site is situated along a bird migration route. Nevertheless, there is potential for bird species to pass through the area and impacted by the potential OHTL.	✓		Vantage point bird counts will be conducted during Autumn 2023 and Spring 2024 to feed into the ESIA biodiversity impact assessment. This will include the OHTL route
Mammals	There is potential habitation of several threatened species - Marbled Polecat (IUCN VU, UzRDB VU), Steppe Polecat (UzRDB VU) and Corsac Fox (UzRDB VU).	✓		Additional mammal surveys will be conducted to confirm presence or absence of these species, and to confirm the findings from the Critical Habitat Screening and inform any mitigation and monitoring plans to be developed as part of the ESIA.
Indigenous peoples	It is not expected that the Project will impact indigenous peoples. No indigenous peoples were identified during the scoping site visit. There may be people from other ethnic minorities, such as Turkmen or Kazak living in the communities surrounding the project site. There is a possibility that they will be considered vulnerable people, but they would not be regarded as indigenous peoples.		✗	This topic is scoped out for further assessment in the ESIA [NB - Numbers of ethnic minority peoples and their vulnerability will be identified during the socioeconomic survey collection. This will determine if special measures are required].
Cultural heritage (C)	There are no areas of UNESCO cultural heritage significance near the Project site. Consultation with the Institute of Archaeology under the Academy of Sciences of the Republic of Uzbekistan to determine if they have surveyed the Project location and the response has indicated a potential	✓		The ESIA will undertake further consultation and studies to understand the presence and importance of this feature and identify appropriate mitigation measures. The institute Archaeology (IoA) will be engaged to perform a survey to confirm the potential sensitivity of the military wall and to

E&S Aspect (Construction (C), Decommissioning (D) / Operation (O))	Potential source of impact, receptors and scoping evaluation	Scoped In/Out		Requirements for ESIA /baseline data collection
	archaeological site outside the Project boundary to the west and north of the site. The prevalence of some features near the Project site, could indicate the potential presence of unknown cultural heritage features in the project area.			identify any other archaeological or cultural heritage receptors within the site. The scope of this assessment will be defined by the IoA. In addition, a desktop review of secondary data combined with consultations with the local community will be performed to confirm any potential intangible cultural heritage features. The ESIA will include an impact assessment to determine the potential significance of archaeological and cultural heritage risks.
Cultural Heritage (O/D)	There will not be any additional earthworks associated with the operation or decommissioning of the Project. Therefore, the impacts on cultural heritage during these phases are highly unlikely and insignificant.		✗	This topic is scoped out for further assessment in the ESIA.
Cumulative impacts (C/D)	The scoping assessment has not identified any potential cumulative impacts.	✓		The ESIA will not include a cumulative impact assessment.
Cumulative impacts (O)	During operation, cumulative effects are not deemed relevant.		✗	This topic is scoped out for further assessment in the ESIA.
Transboundary Impacts	The Project site is located more than 50 km from the border with Turkmenistan and is considered outside the direct and indirect AOI for the Project. There will be no water abstraction from transboundary water courses and there are no transboundary air quality considerations.		✗	This topic is scoped out for further assessment in the ESIA.

7 Scoping Stakeholder Engagement

7.1 Stakeholder engagement approach

A Stakeholder Engagement Plan (SEP) has been prepared as part of the scoping phase that outlines legal and lender obligations, detailed stakeholder mapping, an overview of engagement performed to date (also summarised below), stakeholder engagement principles, the requirements for the ESIA phase and beyond and the Project Grievance Mechanism.

7.2 Scoping engagement

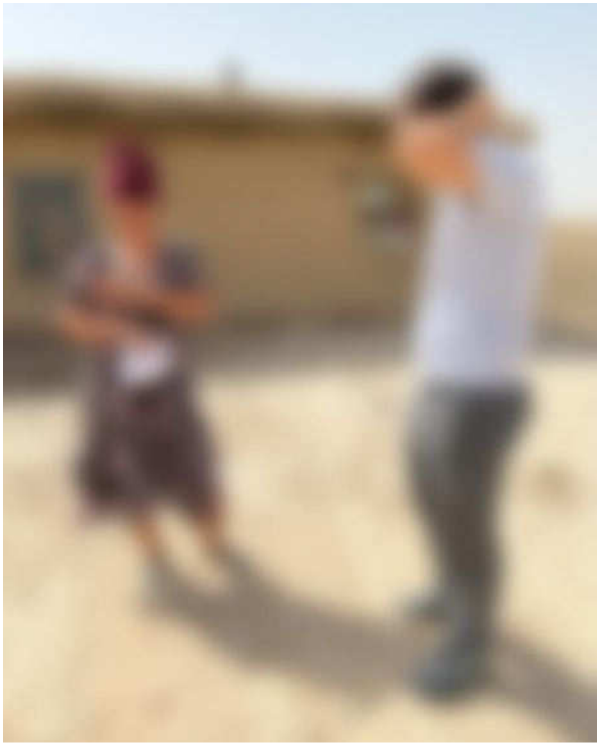
During the site visit on 11 and 12 September 2023 and 25 January 2024 that was undertaken to inform the preparation of the Scoping report, meetings with following stakeholders were held:

- Guzar District Municipality;
- Kamashi District Municipality;
- Guzar 500kv substation;
- Guzar District Power Grids Department;
- Guzar District Cadastral Department;
- Guzar District Construction Department;
- Kamashi District Cadastral Department;
- Kamashi District Agricultural Department;

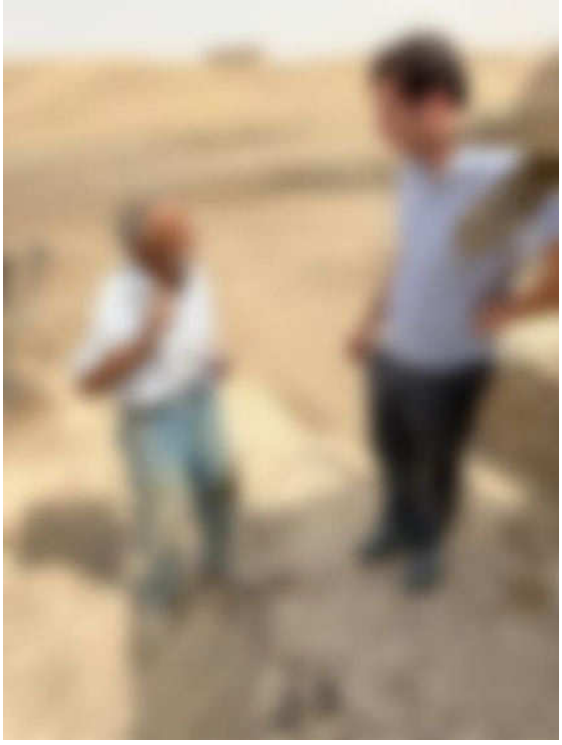
During the meetings, participants were provided with leaflets that provided the key information about the planned Project as well as contact details of the ESIA Consultant. A sample of the leaflet is provided in Annex A. The summary of meetings with local social receptors is summarised in Table 23 below.

Figure 53 : Scoping Engagement

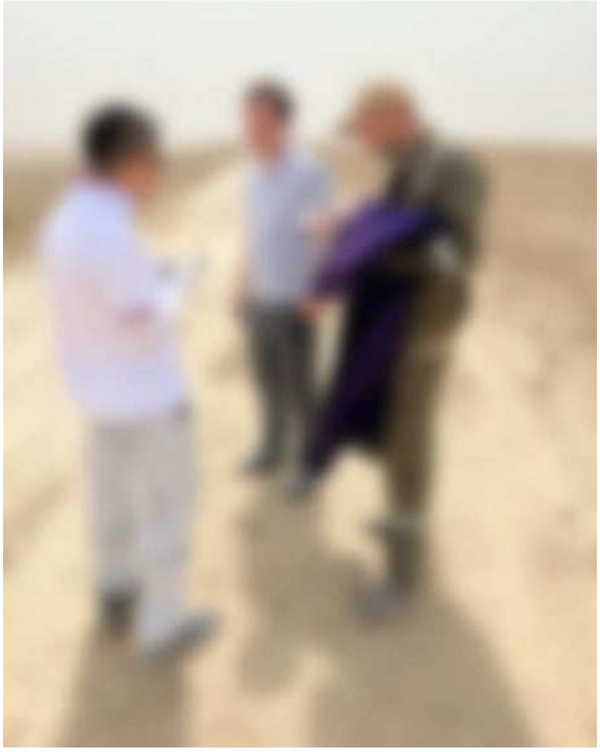
a. Herder on the site



b. Household outside the project side (north-east)

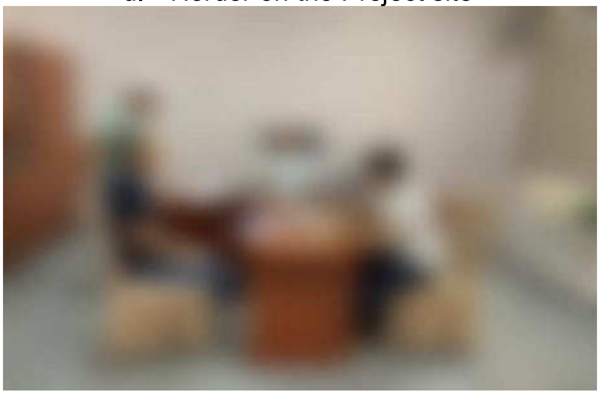


c. Herder outside the site



d. Herder on the Project site

e. Guzar municipality, Deputy mayor and representatives of relevant departments (September 2023)

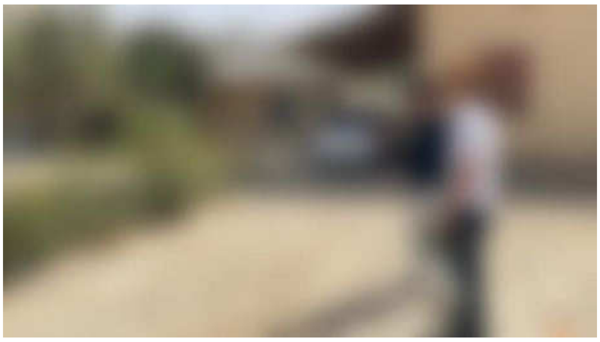


f. Kamashi municipality, Deputy mayor and representatives of relevant departments (September 2023)

g. Nearest community office, community head and representatives (September 2023)

h. Nearest community office, community head and representatives (September 2023)

i. Nearest community office, community head and representatives (September 2023)



j. Houses near Guzar 500 substation (September 2023)

k. Worker of Guzar 500 substation (September 2023)

l. Community member (September 2023)

During the meetings, participants were provided with leaflets with key information about the planned project as well as contact details of the ESIA Consultant. A sample of the leaflet is provided in Annex A. A summary of meetings with local social receptors is provided in Table 23 below:

Table 23: Summary of the scoping stage meetings

Name of receptor	Date and place of meeting	Person met	Summary of discussion	Concerns raised/ comments	Responses provided	How concerns were addressed/ follow up required	Information disclosed
Guzar District Municipality	11 September 2023	Mayor and Deputy Mayor of Construction	Provided information on the Project, shared Project leaflet. Discussed Project site use matters.	n/a	n/a	n/a	Project leaflet
Kamashi District Municipality	11 September 2023	Mayor and Deputy Mayor of Construction	Provided information on the Project, shared Project leaflet.	n/a	n/a	n/a	Project leaflet
Guzar District Power Grids Department	11 September 2023	Head of the department	Provided information on the Project, shared Project leaflet.	n/a	n/a	n/a	Project leaflet
Guzar District Power Grids Department	11 September 2023	Head of Guzar District Power Grids Department	Provided information on the Project.	n/a	n/a	n/a	Project leaflet
Guzar District Cadastral Department	11 September 2023	Head of Guzar District Cadastral Department	Provided information on the Project. Consultation regarding land use was conducted.	n/a	n/a	n/a	Project leaflet
Guzar District Construction Department	11 September 2023	Head of Guzar District Construction Department	Provided information on the Project.	n/a	n/a	n/a	Project leaflet
Kamashi District Cadastral Department	11 September 2023	Head of Guzar District Cadastral Department	Provided information on the Project. Consultation regarding land use was conducted.	n/a	n/a	n/a	Project leaflet
Kamashi District	11 September 2023	Representative of	Provided	n/a	n/a	n/a	Project leaflet

Name of receptor	Date and place of meeting	Person met	Summary of discussion	Concerns raised/ comments	Responses provided	How concerns were addressed/ follow up required	Information disclosed
Agricultural Department		Kamashi District Agricultural Department	information on the Project.				
Guzar 500kv Substation	12 September 2023	Worker and guard of the Guzar 500 Substation	Consultation regarding land use was conducted	n/a	n/a	n/a	Project leaflet
Nearest living communities' heads: Oynakul, Batosh, Khalqabad, Yangiabad;	12 September 2023	Head of living community	Information on Project was provided and consultations regarding Project site use was conducted.	n/a	n/a	n/a	n/a
Committee of the Republic of Uzbekistan for the Development of Sericulture and Wool Industry	Letter sent on 27.09.2023	n/a	The letter with request of information on Project site use.	n/a	No response received yet	Phone consultations is ongoing	Information on the Project was provided in the letters
Sanitary-epidemiological welfare and public health committee	Letter sent on 27.09.2023	n/a	The letter with request of information on relevant health protection and buffer zones related to the Project .	n/a	No response received yet	Phone consultations is ongoing	Information on the Project was provided in the letters
Ministry of Mining Industry and Geology of the Republic of Uzbekistan	Letter sent on 27.09.2023	n/a	The letter with request of information on Project site use.	n/a	No response received yet	Phone consultations is ongoing	Information on the Project was provided in the letters
Kashkadarya regional Cadastral department	Letter sent on 27.09.2023	n/a	The letter with request of information on Project site use.	n/a	Information on Project site use and details of land users (registered on	n/a	Information on the Project was provided in the letter

Name of receptor	Date and place of meeting	Person met	Summary of discussion	Concerns raised/ comments	Responses provided	How concerns were addressed/ follow up required	Information disclosed
					official database of Cadastral department) was provided.		
Institute of Archaeology under Academy Sciences of the Republic of Uzbekistan	Letter sent on 27.09.2023	n/a	The letter with request of information on archaeological matters related to the Project.	n/a	No response received yet	Phone consultations is ongoing	Information on the Project was provided in the letters
Guzar District Municipality	25 January 2024	Mayor and Deputy Mayor of Cadastre and Agriculture departments	Provided information on the Project new layout, Discussed Project site use matters.	n/a	n/a	n/a	Map of new Project layout
Kamashi District Municipality	25 January 2024	Mayor and Deputy Mayor of Cadastre Agriculture departments	Provided information on the Project new layout, Discussed Project site use matters.	n/a	n/a	n/a	Map of new Project layout

8 ESIA Terms of Reference

The proposed terms of reference (TOR) for the ESIA are outlined in the following subsections:

8.1 Objectives

The ESIA aims to achieve the following objectives:

- Review applicable local environmental and social rules and regulations, and establish environmental and social requirements to be met by the Project
- Establish an environmental and social baseline for the Project and the Area of Impact (AoI) based on the information available
- Identify and assess potential environmental and social impacts of the Project
- Conduct an impact assessment and determination of significance for Project environmental, social, health, and safety impacts during construction, operation, and decommissioning phases and describe mitigation measures
- Develop a stakeholder engagement plan (SEP) and grievance mechanism aligned with national requirements and EBRD PR10
- Develop an Environmental and Social Management Plan (ESMP)

8.2 Structure of the ESIA

The ESIA report will contain the following volumes:

- Volume I: Non-Technical Summary
- Volume II: ESIA Main Report
- Volume III: ESIA Technical Appendices and Supporting Documents
- Volume IV: Environmental and Social Management Plan (ESMP)
- Volume V: Resettlement Plan (RP)
- Volume VI: Stakeholder Engagement Plan (SEP)

8.3 Non-technical summary

A non-technical summary (NTS) will be prepared in English, Russian and Uzbek. This will provide a high-level overview identifying the scope and nature of the Project and predicted environmental and social impacts. The NTS will be used as a tool to aid consultation and information disclosure.

8.4 ESIA technical scope

The ESIA process will investigate relevant environmental and social aspects, as defined by this Scoping Report and summarised in Table 25..

The outcomes will the documented in an ESIA report as set out in Table 24.

Table 24 Proposed structure of ESIA report

Chapter	Description of Content
Introduction	Presents a brief overview of the Project, a description of the developer and a brief outline of contents of the report, etc.
Project Description	Describes the Project, including its main elements and activities for construction and operation.

Project Need and Analysis of Alternatives	Presents the purpose and rationale of the Project and examines alternatives to the proposed project site and generation technology including the no project alternative.
Policy, Legislative and Institutional Context	Description of legislative and institutional context in Uzbekistan including: • Institutional Framework and National Regulators • National EIA process • Relevant laws, regulations and applicable guidance for each environmental and social topic contained in the ESIA • International conventions The chapter will also refer to the applicable EBRD standards and guidelines.
ESIA methodology	Sets out the stages of the ESIA process. Robust criteria for determination of the significance of the anticipated impacts will be developed. Definitions of significance will be clearly defined and make reference as applicable to the magnitude, geographic extent, duration and frequency, irreversibility and ecological, social, and economic context.
Information Disclosure, Consultation and Participation	Provides an overview of the consultation processes as defined in the stakeholder engagement process and summarises results including specific reference to comments made during consultation and how they were addressed in the ESIA.
Topic-specific impact assessment	With reference to this Scoping Report and the activities required to be undertaken as part of the construction and operation of the Plant. Each impact assessment chapter will contain the following sub-headings: • Introduction • Methodology and assessment criteria (specific legislation, a summary of relevant consultation comments, criteria for determining significance and associated limitations • Baseline summary • Impact identification and assessment • Cumulative and transboundary impacts • Mitigation and enhancement measures • Residual impact summary
Conclusions	Conclusions
References	List of references cited in the report.

In accordance with international lender requirements for environmental and social impact assessment, the scope of works for this ESIA will consider the following scope:

- Environmental, social, labour, gender, health, safety, risks and impacts
- Project and related and associated facilities (where relevant)
- Risks and impacts that may arise for each project activity
- Role and capacity of the relevant parties, including government, contractors and suppliers
- Potential third-party impacts including supply chain considerations

This ESIA will identify beneficial and adverse, direct and indirect impacts of the Project related to the biophysical and socio-economic environment. Cumulative and transboundary impacts are scoped out.

The following topics are scoped in and out of the ESIA

Table 25 Topics scoped in and out of the ESIA

Environment and Health	Social	Labour
• Air quality (C/D); • Noise and vibration (C/D) • Waste (including hazardous waste) (C/O/D) • Climate resilience (O) • Soil and (C/D) • Water resources (C/D); • Hydrogeology (C/D) • Biodiversity (habitat loss, impact on PBF) (C, O)	• Community health and safety (C/O/D) • Traffic and Transportation (C/D) • Security (C/D) • Livelihood and land use (including ecosystem services) (C) • Cultural heritage (C) • Electromagnetic interference, Radio and TV interference (O)	• Occupational Health and Safety (C/O/D) • Emergency preparedness and response (C/O/D) • Labour rights (C/O/D) • Employment (positive) (C/D) • Gender-Based Violence and Harassment (GBVH) (C/O/D) • Human rights (including gender) (C/O/D) • Procurement/supply chain (C/O/D)
Scoped out: Air quality (O) Noise (O) Soils (O) Landscape and visual impact (C, D, O) (including glint and glare) Traffic and transportation (O) Greenhouse gases (C, O) Cultural heritage (O, D) Indigenous Peoples (C, O, D) Cumulative impacts (C, O) Transboundary impacts (C, O) Security (O)		
Note: C = Construction, O = Operations, D = Decommissioning		

For the topics scoped into the ESIA, the impact assessment process will consider the magnitude of the impact and the sensitivity of the receiving environment to evaluate the overall significance of the impact. A framework for assigning magnitude, sensitivity and impact significance is described below. For each E&S topic the proposed mitigation and management measures are considered to give an overall residual impact significance conclusion.

Sensitivity criteria for receptors are categorised into high, medium, or low. Generic criteria used to determine the receptor sensitivity are provided in Table 26. Each topic-specific chapter of the ESIA will define the relevant receptors and assign a receptor sensitivity based on topic-specific criteria.

Table 26: Generic criteria for the allocation of Receptor sensitivity – criteria for allocation

Sensitivity	Physical Receptor	Human Receptor	Biodiversity Receptor	Climate (physical)
High	Little or no capacity to absorb proposed changes and has national or international value, e.g. receptors where people or operations are particularly susceptible to noise or air quality changes)	Receptors with high vulnerability and permanent presence within the direct or indirect AOI (e.g. school, poor or vulnerable household, hospital). No capacity to absorb project changes or no opportunity for mitigation.	Substantial loss of ecological functionality	Climate variability will threaten the sustainability of the project (e.g. work may be precluded from taking place during certain months of the year).
Medium	Moderate capacity to absorb proposed changes, e.g. where it may cause some discomfort, distraction or disturbance	Receptors with moderate to high vulnerability and or somewhat affected by project impacts. Limited capacity to absorb changes. Potential opportunities for mitigation	Moderate but sustainable change which stabilises under the constant presence of impact source, with ecological functionality maintained	Management actions can address potential impacts (e.g. design, implementation management).
Low	Good capacity to absorb proposed changes and not protected or has low value, e.g. receptors where the disturbance is minimal.	Receptors with low to moderate vulnerability are infrequently located in the AOI. Good capacity to absorb changes with no lasting effects, or good access to mitigation measures.	Species or communities unaffected or marginally affected	Potential impact does not affect the sustainability of the Project.

The magnitude of the potential impact is determined based on the professional judgement of the specialist undertaking the assessment or by quantitative means where possible (e.g. modelling and comparison against legislative standards) considering the five criteria provided in Table 27.

Table 27: Determination of magnitude – example criteria for allocation

Magnitude	Intensity / Compliance	Duration	Spatial extent	Reversibility	Likelihood/Frequency
High	High intensity / non-compliant / large numbers of people affected/ very disruptive/noncompliant with legal standards	Beyond the constructio n phase or permanent change	Direct AOI & Indirect AOI	Permanent impact	Continuous
Medium	Medium intensity/ actions need to be taken to become fully compliant/medium disruption or disruption to vulnerable groups or sectors of the	> 3 months up to the completion of the constructio n phase	Indirect AOI	Reversible but requires mitigation and/or compensation	Intermittent

	community or workforce / Quality of life diminished due to change in character / borderline compliant with legal standards.				
Low	Low intensity /legally compliant/affects small numbers of people / non-intrusive or does not cause changes in quality of life	One-off event or occurs for Three months or less	Direct AOI	Reversible following the end of the phase under consideration	Infrequent/one-off event

The temporal influence of the Project will be assessed by comparing the existing baseline conditions (environmental, socio-economic and biological) over the expected duration of the Project activities.

Based on impact magnitude and receptor sensitivity as defined above, the significance of the impact is classed as insignificant, minor, moderate, major or critical, as presented in Table 28. Those that are deemed moderate, major or critically significant are considered the main focus of the management and implementation framework going forward based on the following considerations:

- **Critical**: These effects represent key factors in the decision-making process. They are generally, but not exclusively, associated with impacts where mitigation is impractical or ineffective.
- **Major**: These effects are likely to be important considerations but where mitigation may be effectively employed such that resultant adverse effects are likely to have a Moderate or Slight significance.
- **Moderate**: These effects, if adverse, while important, are not likely to be key decision-making issues and can be managed effectively with appropriate plans and monitoring.
- **Minor**: These effects may be raised but are unlikely to be important in the decision-making process and can be managed effectively with appropriate plans and monitoring.
- **Neutral**: No effect, not significant; need not be considered as a determining factor in the decision-making process.

Impacts are typically considered adverse, but it is also possible for positive impacts to be realised. Where positive impacts are identified, these are assigned a degree of positive impact based on the sustainability (duration) and scale (number of receptors) of the positive outcomes.

Table 28 Significance evaluation

Significance		**Magnitude**					
		Negative			Positive		
		Low	Medium	High	Low	Medium	High
Receptor Sensitivity	Low	Insignificant	Minor	Moderate	Insignificant	Minor	Moderate
	Medium	Minor	Moderate	Major	Minor	Moderate	Major
	High	Moderate	Major	Critical	Moderate	Major	Critical

Baseline data collection to inform the impact assessment shall be generated through a combination of approaches for all specialist areas and include primary and secondary source information as defined in the scoping report. The baseline data collection approach shall be underpinned by stakeholder consultation consisting of public meetings with affected communities and interviews and focus groups with key informants such as representatives from local authorities and the local community.

Baseline studies and assessment locations are outlined below and in Figure 54.

- Air quality – 24 hours continuous monitoring at 5 locations: one near the residential area, another at the herder` building for the following parameters: Nitrogen dioxide (NO_2); Sulphur dioxide (SO_2); Carbon monoxide (CO), TSP (total suspended particles); PM2.5 and PM10.
- Noise - day and night-time noise measurements for 5 representing the nearest sensitive receptors: one near the residential area, another at the herder` building for 24 hours continuously each location for i) A-weighted equivalent continuous noise level in decibels - L_{Aeq} dB(A); ii) minimum and maximum A-weighted sound pressure level in decibels (Lmax (A), Lmin (A).
- Soil sampling (at 5 locations) to confirm the composition of the topsoil and the presence of any elevated contaminants.
- Water sampling (at 1 locations) from the temporary water body (originated from the leakages from the water pipeline)
- Socio-economic surveys of up to 350 households.
- Flora, avifauna, mammal and reptile surveys - the biological terms of reference and work plan are provided in Annex C.

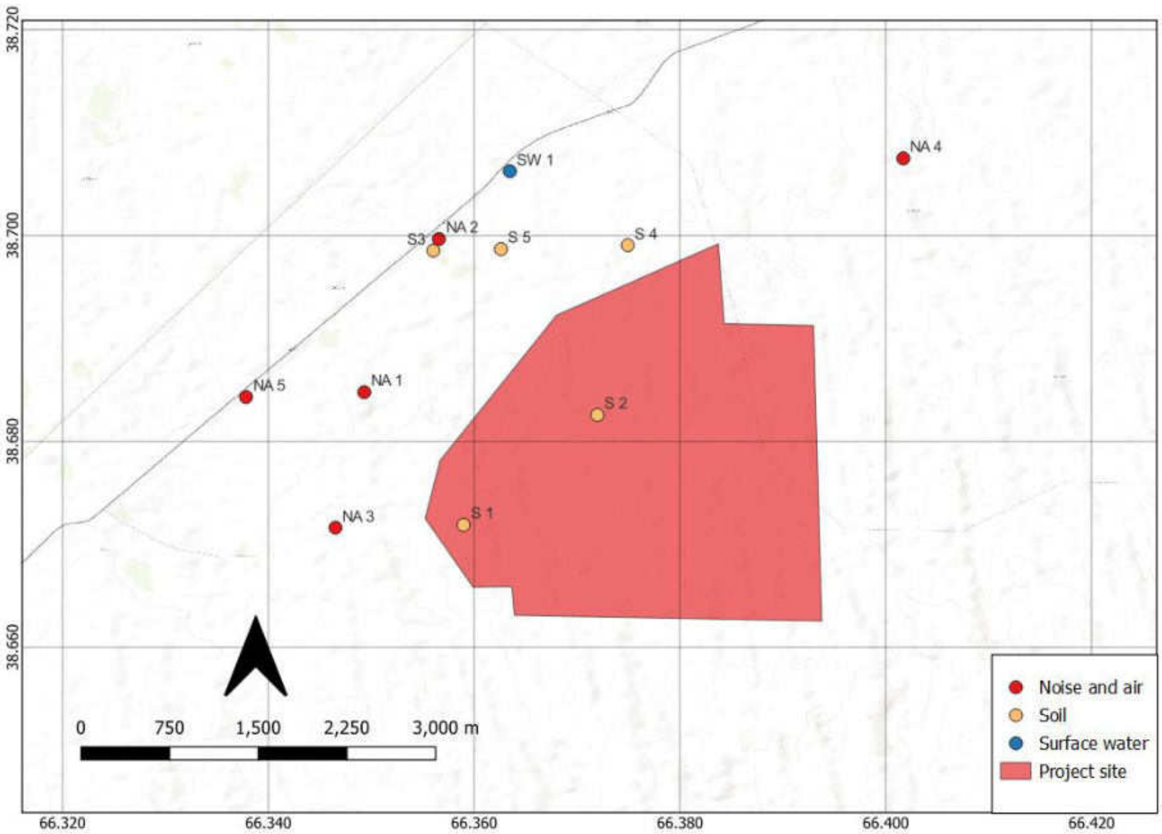


Figure 54: Preliminary locations for the baseline surveys

The significance of impacts will be discussed before and after mitigation (i.e., residual impact) for each aspect. Based on the above approach, major, moderate or critical impacts will be classified as significant.

For each significant impact, the ESIA will define the mitigation and management actions in the form of a framework Environmental and Social Management Plan (ESMP). In general, the following hierarchy of mitigation measures will be applied to reduce, where possible, the significance of impacts to acceptable levels:

- Avoidance and reduction through design (embedded mitigation)
- Abate impacts at source or receptor
- Repair, restore or reinstate to address temporary construction effects
- Compensation for loss or damage, such as replacement planting elsewhere
- Once the application of mitigation and management measures has been defined, the residual significance will be determined.

Significant residual impacts are those impacts that remain acceptable after applying mitigation and enhancement measures.

Any uncertainties associated with impact prediction or the sensitivity of receptors due to the absence of data or other limitations will be considered and articulated in the final report.

The ESIA will make commitments concerning monitoring measures that should be implemented in the ESMP.

8.5 Critical Habitat Assessment (including PBF determination)

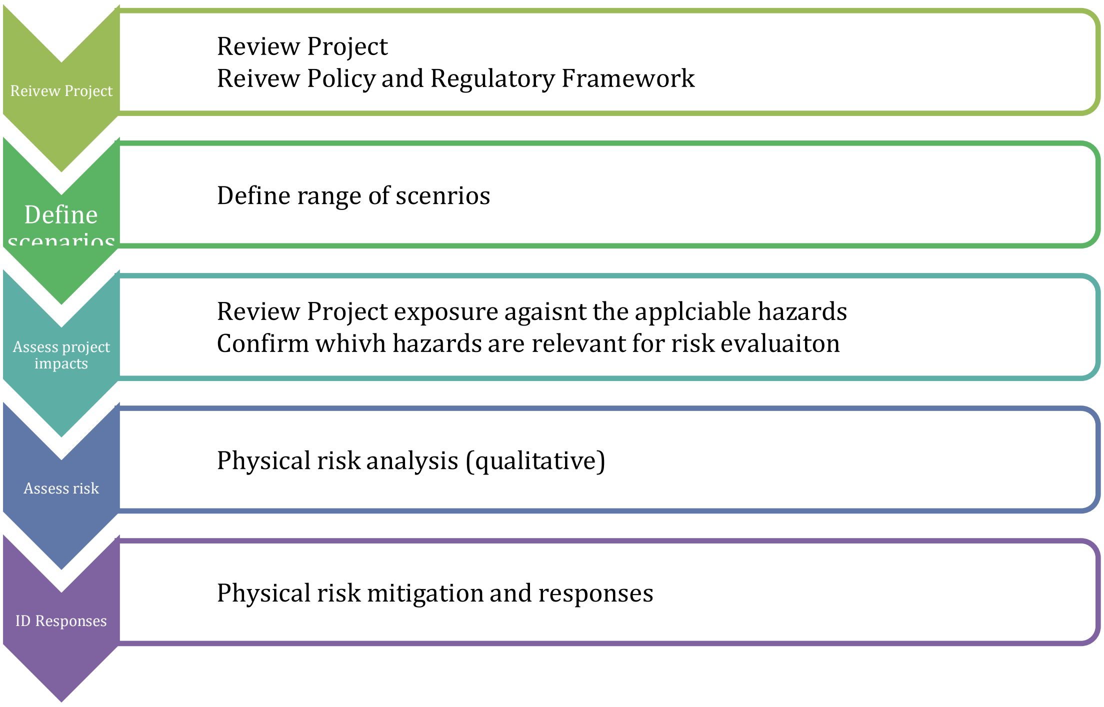
A Critical Habitat Assessment (CHA) will be performed which will be based both of review of desktop-level information, as well as the results of the biodiversity baseline surveys, and which will be incorporated into the biodiversity baseline section of the ESIA. This CHA will be conducted in a manner that is consistent with both IFC PS6 and EBRD PR6. While these two lender policy documents are largely aligned with respect to CHA methodologies, and criteria for triggering a CH determination, they are divergent with respect to the “second tier” of sensitive biodiversity receptors that trigger a “no net loss where feasible” mitigation requirement (Natural Habitat in the case of IFC; Priority Biodiversity Features in the case of EBRD). This is important for the CHA because our recent experience working with EBRD in Uzbekistan, including the development of imminent nationwide guidance for solar project development, has shown that EBRD regards the identification of PBF as an essential component of CHA. This means that in addition to identifying biodiversity features that trigger a CH determination, the CHA and ESIA will also be expected to identify the biodiversity features that trigger a PBF determination, under the EBRD’s separate criteria for such (there is no analogue in IFC PS6), and while CH features will be subject to a “net positive gain” mitigation standard and other requirements (e.g. preparation of a BAP), PBF will be subject to a “no net loss where feasible” mitigation standard. By contrast, the IFC ascribes the “no net loss where feasible” mitigation standard to “Natural Habitat,” which is defined in IFC in a manner for which there is no analogue in EBRD PR6. Even though the concepts of PBF and Natural Habitat are not shared between IFC PS6 and PR6, they both trigger the “no net loss where feasible” mitigation requirement, according to the policies of one or the other of these lenders. Therefore, in order to be bankable to both of these lenders, the Project’s CHA will need to identify both CH features and PBFs, and the ESIA will need to identify the presence of both PBF and Natural Habitat, in addition to Critical Habitat features, and to include proposed mitigation and management measures for these resources, accordingly.

If Critical Habitat (CH) is triggered for any of the species present within the Project site, a Biodiversity Action Plan (BAP) in line with IFC PS6, EBRD PR6 and associated guidance notes will be required, in order to demonstrate a net gain strategy for each of the receptors for which CH has been triggered.

8.6 Climate change risk assessment

The ESIA will include a Climate Change Risk Assessment (CCRA) following requirements set out in Equator Principles 4 to identify, appraise and demonstrate physical climate risks,[50] vulnerabilities and opportunities for the primary elements of the proposed development and its long-term climate resilience and the impact of the Project on wider receptor vulnerability and capacity to adapt to climate change. The main steps undertaken to complete the assessment for physical risks will be aligned with Taskforce for Climate Related Financial Disclosures (TCFD) process for applying scenario analysis of climate-related risks and opportunities.[51] We will follow the procedure outlined in Figure 55.

Figure 55: Process for climate change risk assessment



The CCRA will facilitate the integration of climate resilience considerations into decision-making early on in project design and prioritization of resilience needs and budget. Inputs from the engineering design team will inform the CCRA to determine levels of climate risk and vulnerability (low, medium or high) and identify appropriate adaptation responses for integration within the proposed development design, the ESMP and the SEP. The CCRA will be a stand-alone annexe to the ESIA with relevant referencing in the main body of the report.

8.7 Human rights and gender

Human Rights requirements are described in international standards as securing dignity and equality for all. Every human being is entitled to enjoy their human rights without discrimination. The HRRA will be prepared to guide the project to respect human rights within its area of influence and monitor that third party companies are respecting human rights too.

50 For all projects, in all locations, when combined Scope 1 (direct) and Scope 2 (indirect electricity) emissions are expected to be more than 100,000 tonnes of CO2 equivalent annually. For these projects the CCRA is to include consideration of climate-related ‘Transition Risks’ (as defined by the TCFD). Theis project will not trigger an exceedance of this threshold.

51 June 2017, TCFD recommendations report

The United Nations Guiding Principles (UNGPs) are the key framework for assessment of human rights and to set management systems. The UNGPs state that the responsibility to respect human rights is a global standard of expected conduct for all business enterprises wherever they operate.

Addressing adverse human rights impacts requires taking adequate measures for their prevention, mitigation and, where appropriate, remediation. The Guide to Human Rights Impact Assessment and Management (HRIAM) states that the scope of a human rights risks and impact assessment should consider, at the very minimum:

- The key human rights risks associated with the country of operation
- The human rights risks of key business relationships, including associated facilities and third-party organisations
- The human rights risks and impacts relating to the business activity itself
- The range of stakeholders (potential and actual) that are directly or indirectly affected by the business activity
- The nature and level of the risks and impacts, at different key stages of the project's lifecycle

The HRRA will identify rights holders and duty bearers and the possible risks the project will have on them. Contrary to an ESIA where significance of a risk or impact is identified, UNGP Principle 14 requires that the severity of human rights impacts are identified. As per Principle 14 'severity of impacts will be judged by their scale, scope and irremediable character"[52]. The HRRA will rank risks per their significance and identify possible mitigation measures. Many of these will align with mitigation measures identified elsewhere in the ESIA. The HRRA will be a stand-alone annex to the ESIA with relevant referencing to and from the main body of the report.

A gender risk assessment will be integrated into the ESIA. The gender risk assessment will consider as a minimum women participation in the labour market and in business, the impact of social norms on participation, gender-based violence and harassment (GBVH), labour rights, equality and discrimination in the workplace.

8.8 Volume III: Technical Appendices

Supporting baseline studies and other documentation will be provided in Volume III.

8.9 Volume IV: Framework ESMP

The primary aim of the ESMP is to safeguard the environment, site staff and the local population against site activity which may cause harm or nuisance. The ESIA will include an ESMP including the following content:

- Project Description – overview of the Project description.
- Applicable Regulatory Standards and Guidelines – legal and other relevant standards and guidelines for the Project.
- Environmental and Social Management – provides the environmental and social aspects and impacts along with proposed outline mitigation measures for the construction and operational phases.
- Environmental and Social Monitoring – this section will outline the physical environmental and social monitoring and measurement activities and indicators for the construction and operational phases.

52 guidingprinciplesbusinesshr_en.pdf (ohchr.org)

8.10 Volume V: Resettlement Plan

A Resettlement Plan (RP) will be developed based on the following guiding principles:

- Land acquisition and resettlement will be minimized or avoided where possible;
- All land acquisition and livelihood restoration activities will be managed through the RP and implementation will be documented and monitored;
- All Project affected persons (PAPs) will be meaningfully consulted and be active participants throughout the design and implementation of the RP;
- PAPs will be assisted in their efforts to improve their livelihoods and standards of living, or at least to restore them to pre-Project levels; and
- All compensation will be paid prior to the commencement of civil works in affected areas.

8.11 Volume VI: SEP

A stakeholder engagement plan (SEP) will be developed to coordinate the public participation process following national and IFC PS1. The SEP will guide the following activities:

- Identify all affected people (i.e. by construction activities or during operation) and facilitate the spread of information to relevant authorities and interested and affected parties (IAPs).
- Consult with relevant NGOs and government departments and agencies that may have a stake in the Project and its impacts.
- Prepare a stakeholder consultation plan that provides an opportunity for relevant authorities and IAPs to voice concerns and issues related to the project and allow for the identification of additional alternatives or recommendations. The stakeholder consultation plan will also describe a schedule for public consultation with the relevant groups including the frequency and method of communication (i.e. media announcements, town hall meetings, questionnaires, etc.). Results of the public consultation process will be summarized in an appendix to the ESIA.

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Annex A: Scoping Notification Leaflet

ENVIRONMENTAL AND SOCIAL IMPACT ASSESSMENT (ESIA)

The Project is currently in the scoping phase of the ESIA process. Next steps include collecting physical, biological and socio-economic baseline data within the Project site and nearby communities. Potential positive and negative impacts for the construction, operation and decommissioning phases will then be assessed for significance and management and mitigation measures identified to reduce risk to acceptable levels. The ESIA process will:

- Identify actions that can be taken to eliminate, or at least reduce any negative impacts as a result of the Project, to acceptable levels, and enhance Project benefits.
- Confirm that costs are not levied on the public or individuals that are greater than the benefits they will receive.
- Sequence of Tasks for ESIA Study

Scoping
↓
Baseline Data Collection
↓
Identification and Assessment of Impacts
↓
Mitigation and Benefit Enhancement
↓
Management and Monitoring
↓
Environmental and Social Auditing

NUR-KASHKADARYA SOLAR PV INTRODUCTION

Abu Dhabi Future Energy Company PJSC ("Masdar") has been awarded by the Ministry of Energy, Government of Uzbekistan to design, build, finance, construct, commission and operate & maintain the Nur-Kashkadarya Solar photovoltaic (PV) project with a capacity of 300 MW_{AC} and 75 MW Battery Energy Storage System ("**Project**"). The Project will be implemented through a long term i.e., 25 years power purchase agreement (a "**PPA**") with JSC National Electric Grid of Uzbekistan ("NEGU"). The Project will be designed to meet national regulations and international standards.

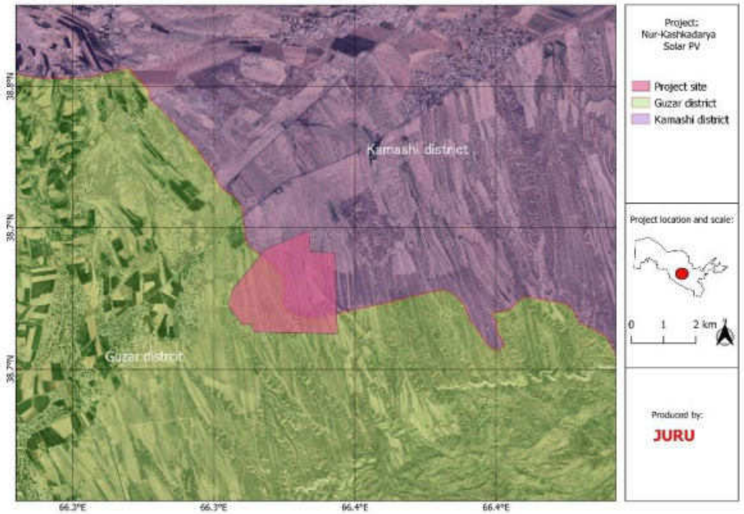
The Project will support Uzbekistan to:

- Reduce energy dependence on carbon-based fuels.
- Meet renewable energy targets.
- Reduce greenhouse gas emission rates.

This leaflet has been produced to provide information about the basic characteristics of the Project and its surroundings, and how the environmental and social impacts will be assessed and managed.

PROJECT DESCRIPTION AND LOCATION

The proposed site covers approximately 802 Ha area of land and is spread across two districts - the Kamashi and Guzar Districts of Kashkadarya region, of the Republic of Uzbekistan. It is located 8 km northeast of Guzar city. The site is accessible from the M39 roadway. The site has an undulating topography; there are no permanent surface water bodies or rivers on the site.



Construction of the Project will consist of the following basic infrastructure components:

- Foundations for the mounting structures for PV panels, and installation of the panels, cabling and hydraulic fittings;
- An inverter and transformer station and substation;
- Construction of internal access roads and maintenance paths, as well as storm water drainage system;
- Construction of temporary construction camp/offices and construction lay-down area;
- The Battery System including the batteries themselves and a control and monitoring system (BMS);
- Operation and maintenance rooms, offices and parking areas;
- Security fence, lightning and accommodation for workers (including water and sanitation).
- The Project will include a new Plant substation and a 0.75 km underground cable to connect the new substation to the grid at the Guzar 220kV substation.

Figure 2. PV Plot Boundary – Nur-Kashkadarya Solar PV Project

The preliminary layout is presented in Figure 2. Stakeholder engagement shall be undertaken during the preparation of the scoping report and ESIA in accordance with national regulations and good practice. Stakeholder engagement activities will include Project Affected Persons ("PAP") and communities concerned by the Project e.g., local and traditional leaders, representatives of the communities, land users, potential vulnerable groups such as youth and women.

CONTACT DETAILS

All complaints, comments or queries relating to the ESIA for the Nur-Kashkadarya Solar PV Project should be sent to:

JURU CONSULTING **Name**: Viktoriya Filatova, Asad Nabiev **Address**: 10A, Chust Str., Tashkent, Uzbekistan, 100077 **Email**: guzar_SolarPV_ESIA@juru.org **Phone:** +998 91 101-27-79	**MASDAR Clean Energy** **Name**: Khurshid Karamatov **E-mail**: kkaramatov@masdar.ae **Phone:** +998 93 522 00 70

Annex B: Project Grievance Form

Ref №		
1	Name (indicate if complainant preferred to be anonymous)	Full name (if applicable): Gender: Age: Address (if applicable): Occupation (if applicable): I wish my identity not to be disclosed: ____
2	Contact information (need to specify the way to get back to complainant)	Mob phone: Fax: Email: Other (specify):
3	How compliant/ feedback/request was received and by whom	Phone call: Text/WhatsApp applications: Verbal communication: Letter/Email: Receiver's name:
4	Purpose of contact	Make a compliant: Give feedback: Request information: Other (specify):
	Date application was received	Date: Time:
5	Text of applicant's message	
6	Response message (after receipt of application)	Dear _______________ We confirm that we have received your application. We would like to inform you that your application is under review. You will receive the response within two weeks of submission of the application. We also would like to inform you that you will get written response for the issues you have raised in your request. We will keep you updated. Thank you for your understanding. *This message was delivered to the applicant by_____________ on ________ at ___ via _________*
7	Summary of the response provided to the applicant	
8	Follow up actions required	
9	Date the application was closed	Date:

The message was addressed by ____________________________
Date/Month/Year _______________________________________
The response was delivered by ___________________________
Date/Month/Year _______________________________________
Signature and stamp ____________________________________

Annex C: Biodiversity Survey Work Plan

As part of the ESIA, biodiversity baseline surveys are needed to validate assumed status and fill any information gaps.

Stage 1 – Desktop Assessment

The ESIA will include a comprehensive desktop assessment, which consists of a review of existing information from platforms including but not limited to, International Union for Conservation of Nature, Birdlife International, World Database on Protected Areas, Global Critical Habitat Screening Layer, Integrated Biodiversity Assessment Tool, Global Biodiversity Information Facility, World Database of Key Biodiversity Areas, Global Invasive Species Database, grey literature and published research articles, national Biodiversity Strategy and Action Plan(s), regional Red List(s), citizen science reports, and any other relevant and verifiable documents.

The outcomes of the Desktop Assessment will include:

- Identification of the boundaries of the baseline study area, taking into account the principles of Area of Influence (AoI) as well as Ecologically Appropriate Area of Analysis (EAAA);
- Preparation of a Land Cover/Land Use map covering the baseline study area boundaries, identifying preliminary habitats and sensitive receptors using remote sensing techniques; and
- A consolidated narrative compiling the findings of data review and consultation, inclusive of:
    - mapping spatially relevant information, such as the boundaries of IBAs, KBAs, protected areas, and/or known species distributions;
    - listing relevant biodiversity values that may or do exist within the study area, such as habitats, species, and unique ecosystems, focusing on those that could constitute Critical Habitat features or Priority Biodiversity Features, per EBRD PR6;
    - provision of an initial assessment of biodiversity status of the study area, with focus on habitats and species that could trigger a Critical Habitat determination, per EBRD PR6;
    - identification of any information gaps or weaknesses that should be addressed through field surveys; and
    - proposed detailed field surveying methodology tailored and updated as per the findings of the desktop assessment.

Stage 2 – Field Surveying

The following surveys are planned to capture information ranging across biodiversity values such as habitat, flora, and additional terrestrial fauna species (birds, reptiles and fishes).

The field surveying methodology may be further tailored once the comprehensive desktop assessment has been undertaken. However, the below field surveying methodology has been proposed given the general lack of detailed information present throughout the project site and surrounding area.

The below sub-sections outline the completed and proposed field survey methodology.

Habitat and Flora Surveying

A specialist will undertake a walkover in early October to broadly categorise the habitat types of the project site and compile flora species lists, additional supplementary Spring botanical survey will be performed in April 2024.

A report will be prepared detailing the methods and results, including details of habitat types present, basic habitat maps, and any status of particular protected plant species (IUCN and Uzbekistan Red Data Book) will be required.

The report will, in part, compare the results of the 2023 survey to the results and assessment of any previous surveys undertaken (where applicable) to note changes in habitats and species recorded.

The main tasks of expert-botanist are following:

- field botanical survey and processing of field data (Autumn 2023 and Spring 2024);
- analysis of any previous botanical surveys and other available data (publications, reports, etc.) compared with the results of the 2023 survey;
- detailed description and GIS-based mapping of habitat types present within the project site;
- compilation of the check-list of plant species recorded within the project site (in particular, threatened species included in the Rad Data Book of Uzbekistan and/or the IUCN Red List);
- Botanist will also assess the habitat condition/quality using the scoring method for assessing natural habitat no net loss impacts.

The field research on the project area will be conducted using the traditional methods of botanical survey commonly used for sampling and mapping of native non-forest vegetation, recognition of floristic composition and spatial patterns of plant communities.

Vegetation structure and species composition will be described from 50x50 m geobotanical sample plots (squares) chosen in an area with homogeneous vegetation. Sample plots (squares) will be located away from roads and boundaries between different vegetation communities (coordinates of these boundaries observed during the survey were recorded separately). For each square, photographs of the landscape and vegetation were taken using a digital camera, and following data will be recorded: location and physical environment (including GPS coordinates, elevation, topography, and soil), state of vegetation and disturbance factors (grazing, etc.), plant association, canopy cover (%), canopy height, all plant species present at the plot, their cover and abundance, phenological stage and height. Microcomplexes (e.g., along dry riverbeds) will be described separately. Coordinates of populations of endemic, red listed or alien species, number of individuals and area occupied by population also will be recorded.

Species cover and abundance will be determined using the Braun-Blanquet cover-abundance scale (1965) widely used in geobotanical and ecological studies as rapid visual assessment technique.

Esri ArcGIS 10.1 software will be used for vegetation mapping. The vegetation map will be compiled in ArcGIS by visual interpretation of the satellite image using the field data, a topographical map (1:100.000) and a soil map of the region.

Results will be presented as per the tables provided below:

Botanical survey results for every survey location

Plant species	Life form	Height, cm	Abundance	Phenol. stage	Conservation status

Summary of botanical survey results

Plant species	Life form	Family	Abundance	Habitat type	Conservation status

Herpetological survey

The aim of the field herpetological survey is to assess the status of reptiles in the study area of the project territory (specification of the species and quantitative composition, territorial distribution, including places of concentration, the state of habitats).

The main tasks of the herpetological survey are the following:

- Field herpetological survey and processing of field data in early October 2023;
- Supplementary Central Asian tortoise survey will be conducted to confirm presence of this species in Spring 2024;
- Analysis of any previous herpetological surveys and other available data (publications, reports, etc.) compared with the results of the 2023 survey;
- Compilation of the checklist of species recorded within the project site (in particular, threatened species included in the Red Data Book of Uzbekistan and/or the IUCN Red List).

The research will combine static and transect survey methods. Locations for stationary surveys and transects will be selected following different types of habitats and will be determined during preparatory work based on the maps and literature data.

The quantitative assessment of reptiles and amphibians will be mainly based on the transect survey. The transect method consists in counting individuals along a fixed long line (transect), on both sides of it, with the duration of the survey determined by the known distance, which is selected depending on the type of reptile and the area, but does not exceed 1 km in one way. In this case, all individuals encountered on the transect are registered, regardless of the distance they are identified at. The perpendicular distance is measured between the transect axis and each individual. The results obtained are used to calculate the density of recorded reptiles.

The list of reptile species inhabiting the project territory will be presented in the following form:

The list of reptile species

No	Species name	Species name acc. to literary sources	Author's earlier personal data	April, June 2023 field expedition data	Species abundance	Endemism	Nature conservation status		
							UzRDB	IUCN	CITES

Bird survey

The aim of the field ornithological survey is to assess the status of bird in the study area of the project territory (specification of the species and quantitative composition, territorial distribution).

It is planned to conduct Autumn and Spring bird monitoring surveys in the project area, which is located along the main flyway of threatened species, including Steppe Eagle (IUCN EN and UzRDB VU), Greater Spotted Eagle (IUCN and UzRDB VU), Imperial Eagle (IUCN and UzRDB VU), as well as other threatened species. The survey will be a combination of VP and point count survey. This will provide a more complete characterization of songbird, shorebird, and other small bird use of the site, to complement the VP surveys.

The survey will be conducted 4 days per season and those 4 days should be divided between VP and point counts, so on each day, the observer will do 4 point counts in the morning, 20 min each, and then two x two hour VP surveys. The surveyor will use the same 4 point counts and two VP locations each time, which should be sufficient to cover the whole site. That will give 4 rounds or days of survey per migratory season, including 8x 2 hour VP surveys and 16 x 20 minute point counts per season.

The Autumn survey will be conducted in the period between September to November 2023 and the Spring survey between March and May 2024.

The main tasks of the survey are the following:

- VP and point counts Autumn and Spring bird monitoring surveys;
- Analysis of any previous ornithological surveys and other available data (publications, reports, etc.) compared with the results of the 2023 survey; and
- Compilation of the checklist of species recorded within the project site (in particular, threatened species included in the Red Data Book of Uzbekistan and/or the IUCN Red List).

The list of bird species inhabiting the project territory will be presented in the following form:

The list of bird species

No	Species name	Data	Number	Nature conservation status		
				UzRDB	IUCN	CITES

Mammals Survey

The survey for terrestrial mammals will aim to establish the presence and absence of mammal species within the project territory and its surrounding areas. To accomplish this, the following steps will be taken:

- Conduct a ground survey of non-flying mammal species in Autumn 2023 throughout the study area (including broader Area of Influence, or Ecologically Appropriate Area of Analysis);
- Install 4 camera traps in the most suitable locations for a period of two months and collecting data from camera traps to get information about mammal species presence, distribution, seasonal dynamics and behaviour;

- Collect questionnaire data from local people on mammal species presence/absence, status and threats;
- Analyse the preliminary field data including species richness, composition, and distribution; and
- Compile a mammal species list based on field data, questionnaire data and data from literature sources, including endangered and non-endangered species.

The ground survey will be conducted in early October, 2023, using a transect-based approach. The ground survey will also include nocturnal survey. The data collected by the phototraps will be analysed and included in the final mammal baseline report.

During the field research, a non-invasive methodology will be used, not related to the capture and killing of wild animals, including:

- Visual observation of mammals both by eye and using 10x binoculars;
- Registration of tracks of the vital activity of wild mammals, including animal tracks (paw footprints on the ground), faeces, digging, burrows, dead animals, etc.
- Taking photos of the animals, their tracks and traces of their vital activity, typical habitats.

Characteristics of mammals found in the project area will be presented in the table provided below:

Mammal species

№	Species	Status of threat IUCN / Uzbekistan RDB	Sources

Annex D: Preliminary Permit Register

Permit / Required Activity	Permit Title	Issuing Authority	Implementing Law	Responsible Party for Obtaining License
Pre-construction				
Construction activities	Construction Permit	Khokimiyats of Project region	• Resolution of the Oliy Majlis of the Republic of Uzbekistan “On the list of activities for which a license is required” No. 222-II of 12.05.2001; • Resolution of the Cabinet of the Republic of Uzbekistan No. 54 of 02/25/2013. Appendix 1 “Regulations on the procedure for granting land plots in populated areas for the implementation of urban planning activities of design and registration of construction objects, as well as the acceptance into operation of objects”	EPC
Construction activities	The Positive Conclusion of SEE for the national EIA report (Stage I and/or Stage II)	MNR	• Law «On Nature Protection» (1992); • Law of the Republic of Uzbekistan “On Ecological Expertise” (2000); and • Regulations "On the State Environmental Expertise" (SEE), approved by the Resolution of Cabinet of Ministers No. 541 “On further improvement of the environmental impact assessment mechanism’ (2020).	Masdar
Construction activities	Cultural Heritage Clearance	Ministry of Culture of Uzbekistan	• Law on the Protection and Use of Cultural Heritage Objects (2001)	Masdar
Pre-construction activities	Approval of the Protocol of pre-construction surveys and relocation by MNR	MNR	• Law «On Nature Protection» (1992); • Law "On the Protection and use of flora” No. 409 dated 21.09.2016 • Law "On the Protection and use of wildlife” No. 408 dated 19.09.2016	Masdar
Pre-commissioning				

Permit / Required Activity	Permit Title	Issuing Authority	Implementing Law	Responsible Party for Obtaining License
Construction activities	The Positive Conclusion of SEE for the national EIA report (Stage III)	MNR	- Law «On Nature Protection» (1992); - Law of the Republic of Uzbekistan “On Ecological Expertise” (2000); and - Regulations "On the State Environmental Expertise" (SEE), approved by the Resolution of Cabinet of Ministers No. 541 “On further improvement of the environmental impact assessment mechanism’ (2020).	SPV / PC

Annex E: Information about protected areas in the scoping AOI

IBA "Chimkurgan Reservoir" [53]

The reservoir locates in 24 km to the north from project area. Site has only IBA status, no state protection.

The reservoir is of the channel storage type and is located in the Kashkadarya river basin. The area of the reservoir is 44.4 km2, with a length of 17.5 km, maximum width of 5.5 km, average depth of 17.2 m, and a maximum depth of 30m. Its water is used for irrigation. The area has an acutely continental climate with hot and long summers, and short winters with little snow. This site is important for wintering waterbirds, over 20,000 waterbirds winter here in good years.

172 bird species of 15 orders have been recorded in the Kashkadarya river valley (Meklenburtsev, 1958). According to the literature (Meklenburtsev, 1958) Crex crex was recorded once during autumn migration near Chirakchi and Aythya nyroca once on the Kashkadarya river. Coracias garrulus is common and numerous breeding bird in the steep loess banks of the Kashkadarya (Meklenburtsev, 1958).

Non-bird biodiversity: R.N. Meklenburtsev (1958) published data on the vertebrate animals in the Kashkadarya river valley. From insectivores, Hemiechinus auritus is common and H.hypomelas was found once. There are 5 species of Chiroptera but they are not numerous and are only found sporadically. Predators include Vulpes vulpes (widespread) and Canis aureus, C.lupus and Felis lybica are rare. Sus scrofa was recorded between Shahrisabs and Chirakchi. Of Rodents, Spermophilus fulvus, Rhombomis opinus and Ellobius tancrei are widespread. Reptiles are represented by 28 species: Testudines – 1 species (Agrionemis horsfieldi), lizards – 16 species, with Cyrtopodion caspius, Trapelus sanguinolentus and Phrynocephalus interckapularis being common; Varanus griseus is more rare. There are 11 species of snakes with Natrix tessellata, Coluber karelini and Psammophis lineolatum being common.

KBA "Western Hissar" [54]

The site locates in 25 km to the East from project area. Site has only KBA status, no state protection. It was determined in frame of CEPF project "Mountains of Central Asian ecosystem"

This is high mountain habitat, which has A1, B1 Global KBA criteria.

Site is important for habitation next species:

Plants: *Acantholimon annae, Acantholimon gontscharovii, Acantholimon hissaricum, Acantholimon taschkurganicum, Acantholimon vvedenskyi, Allium brevidentiforme, Allium dolichomischum, Allium hexaceras, Allium majus, Allium tytthanthum, Amygdalus bucharica, Astomaea galiocarpa, Astragalus bobrovii, Astragalus butkovii, Astragalus komarovii, Astragalus massagetowii, Astragalus pseudanthylloides, Astragalus schutensis, Astragalus stipulosus, Astragalus terrae-rubrae, Astragalus tupalangi, Bergenia hissarica, Cephalopodum hissaricum, Cicer incanum, Cousinia allolepis, Cousinia campyloraphis, Cousinia subcandicans, Cousinia vvedenskyi, Dianthus uzbekistanicus, Dimorphosciadium gayoides, Dionysia hissarica, Dracocephalum formosum, Eremurus aitchisonii, Eremurus iae, Eremurus pubescens, Erysimum nabijevii, Euphorbia kudrjaschevii, Ferula*

[53] BirdLife International (2023) Important Bird Area factsheet: Chimkurgan Reservoir. Downloaded from http://datazone.birdlife.org/site/factsheet/chimkurgan-reservoir-iba-uzbekistan on 11/10/2023.

[54] https://www.cepf.net/sites/default/files/mountains-central-asia-ecosystem-profile-english.pdf

fedtschenkoana, Ferula pratovii, Ferula sumbul, Hedysarum bucharicum, Hedysarum kudrjaschevii, Hedysarum magnificum, Iskandera hissarica, Jurinea asperifolia, Jurinea pjataevae, Jurinea sangardensis, Lepidium minor, Ostrowskia magnifica, Oxytropis lasiocarpa, Oxytropis microcarpa, Oxytropis tyttantha, Parrya pjataevae, Pedicularis grandis, Rhus coriaria, Ribes malvifolium, Saponaria gypsacea, Scutellaria guttata, Scutellaria holosericea, Scutellaria villosissima, Seseli merkulowiczii, Silene michelsonii, Sphaerosciadium denaense, Tanacetopsis botschantzevii, Thesium ramosissimum, Tulipa carinata, Tulipa ingens, Tulipa lanata, Tulipa orythioides, Tulipa tubergeniana, Ungernia victoris, Vvedenskia pinnatifolia, Zeravschania regeliana, Xylanthemum rupestre, *Malus sieversii.*

Birds: *Falco cherrug, Neophron percnopterus*

Mammals: *Panthera uncia*

IBA "South-western Gizzar" [55]

The reservoir locates in 32 km to the south-west from project area.

The IBA covers the vast foothills zone of the north-western base of the Baysuntau and Kugitang ridges of the western part of the Pamiro-Alay mountain system and consists of well-vegetated hills and small plains. This site is remote from large settlements, though there are a few cattle-breeders' huts, abandoned in winter. The site is a migratory bottleneck, especially for *Grus virgo*. There are many farm tracks, often impassable in spring because of rain, and vehicles are rare. There is a haymaking in May and June when the number of people increases considerably. There is a recent, progressive development of the foothills for dry crop cultivation which is reducing the amount of land available for cranes, but the majority of the area is still virgin land.

This site is important for spring migration when the high mountains to the east are still covered with snow. The list of migrants is about 240-250 species. Apart from Passeriformes, the only recorded breeding species are *Falco naumanni*, *Circaetus gallicus*, *Aquila chrysaetos* and *Buteo rufinus* (common). Vultures are resident.

Non-bird biodiversity: The foothills are rich in wildlife in spring. There are many *Testudo horsfieldi*, *Ophisaurus apoda*, *Naja naja* and *Vipera libetina* are found regularly. Rodents: in some years there are very high numbers of *Meriones lybica* and *Rhombomis opimus* in the lower areas. Predators (*Vulpes vulpes*, *V. corsac* and *Felis lybica*) are common. *Gazella subgutturosa* used to be common but has now been exterminated.

[55] BirdLife International (2023) Important Bird Area factsheet: South-west Gizzar Foothills. Downloaded from http://datazone.birdlife.org/site/factsheet/south-west-gizzar-foothills-iba-uzbekistan on 11/10/2023.